Perfect play in a finite nonlocal game requires infinite entanglement entropy

Seth Douglas Nidhal Mghirbi

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Abstract

How much infinity can a finite nonlocal game require? We construct a finite two-player game with classical questions, classical answers and a Boolean winning rule. It has a perfect strategy on a tensor product of separable Hilbert spaces. Every perfect normal spatial strategy contains, under local isometries, the same pure entangled state with infinite entropy. For mixed strategies, both marginal entropies are infinite and the pure state remains a tensor factor. Thus infinite dimension can be strengthened to infinite entropy without losing spatial attainment.

The proof constructs a rational Bell functional whose perfect strategies contain a state with Catalan Schmidt multiplicities, and compiles it into Boolean tests. For either construction, the minimum marginal entropy at loss ε is $\Theta(\log(1/\varepsilon))$. The lower bound holds for every normal spatial strategy, including mixed states and POVMs. It follows from polynomial decoding on the actual state and a finite-strip entropy estimate. Height and size cutoffs give the matching upper order. The sharp leading constant remains open.

Changes in version 1.1.0 are described at the start of Section 9.

AI systems assisted with this work. The authors take responsibility for its claims; see "How this work was done." This preprint has not been peer reviewed.

Contents

1 Introduction **4**
1.1 Prior work and scope of the contribution . . . 5
1.2 Proof route . . . 6

2 Normal states and support calculus **6**

3 The Catalan relations and the pure tensor factor **7**
3.1 Five bipartite equations . . . 8

4 The normal flagged magic-square bridge **9**
4.1 Questions, calibration and exact bindings . . . 9
4.2 Normal bridge theorem . . . 10
4.3 A shared question fixes transported labels . . . 11

5 Rational projection equations that certify products **11**
5.1 Graphs and triangles . . . 12
5.2 Binding prescribed parameters . . . 12

6 A common Catalan table from two bridges **13**
6.1 Nine corners and the small polynomial input . . . 13
6.2 Extraction and mirrors . . . 14
6.3 Simultaneous normal realization . . . 14

7 The rational Bell functional **15**
7.1 Literal coefficient convention . . . 16

8 Rational rows as Boolean winning conditions **16**
8.1 A certified scalar matrix algebra . . . 16
8.2 Signed reflections and column graphs . . . 17
8.3 The explicit game and proof of Theorem C . . . 18

9 Entropy near the optimum **18**
9.1 Divergence at bounded entropy . . . 19
9.2 Height cutoffs and the logarithmic upper bound . . . 20
9.3 A spectral taper and its exact loss . . . 22
9.4 Restricted lower bound for all spectral filters . . . 24

10 Discussion **25**

How this work was done **25**

A Reproducibility **26**

B Polynomial decoding on the actual state **27**
B.1 Statement, conventions, and scope . . . 27
B.2 A calculus that supplies every suffix mirror . . . 28
B.3 Raw graphs, triangles, words, and prescribed parameters . . . 29
B.3.1 Graphs . . . 29
B.3.2 Triangles and placements . . . 29
B.3.3 The parameters are the specified physical events . . . 30
B.4 Normal row–column magic-square rigidity . . . 31
B.5 One actual pair of isometries and the two comparison blocks . . . 32
B.5.1 Mirrors before compression . . . 32
B.5.2 Round only three coarse sectors . . . 33
B.5.3 Compress, dilate, and extract each magic sector . . . 33
B.6 Calibrated signs and shared labels without coefficient inversion . . . 34
B.6.1 Compression and sharpness are linear in vector error . . . 35

B.6.2 A bounded raw-sign mirror precedes label transport . . . 35
B.6.3 The same opposite questions fix every refinement . . . 35
B.6.4 Calibrate and repair the same two transports . . . 36
B.7 Transport the raw labels; leave the frame in the root . . . 36
B.8 Bounded root generators and all five mirrors . . . 37
B.8.1 Exact domain restrictions without an exponent loss . . . 38
B.8.2 The root mirrors . . . 38
B.9 Extend the root table to the actual full state . . . 39
B.10 The Boolean compiler has a linear loss reduction . . . 39
B.10.1 Raw scalar matrix units and physical slot events . . . 39
B.10.2 Identify the column projections in the tested orientation . . . 40
B.10.3 The anchored vector identity . . . 41
B.11 Dependency, exponent, and suffix ledgers . . . 41
B.12 Consequences and the sharp ceiling on a decoder exponent . . . 42

C State-weighted Catalan rounding **43**
C.1 Statement and conventions . . . 43
C.2 Spectral banding with explicit constants . . . 44
C.2.1 Mirrors needed before any localization . . . 44
C.2.2 The common random partition . . . 45
C.2.3 Localization without a new square root . . . 45
C.2.4 The exact input to the incoming-row repair lemma . . . 46
C.2.5 The unused bottom block . . . 47
C.2.6 A cumulative-tail recurrence with kernel mass four . . . 47
C.3 Repair of incoming rows . . . 48
C.4 Finite-strip renewal and its entropy . . . 50
C.5 Assembly and explicit logarithmic remainder . . . 53
C.6 A full exact extension, with a simpler top boundary . . . 55
C.7 Compatibility with the decoder and the game corollary . . . 57
C.8 Dependency order and boundary conventions . . . 58
C.9 What is proved about possible improvements . . . 59
C.10 The single finite-strip lemma supporting the rounding . . . 59

D Exact local algebras and extraction limits **60**
D.1 The sharp exact-algebra law . . . 60
D.2 A limit on full-state extraction . . . 62

1 Introduction

How "inherently infinite" can a finite quantum correlation be? Coladangelo and Stark posed this question and exhibited a finite correlation that requires infinite dimension [8]. Dimension does not measure the amount of entanglement: a state can have infinitely many nonzero Schmidt coefficients and finite entropy. The game Γ gives this distinction in a perfect-winning setting: it has a perfect spatial strategy, no finite-dimensional perfect strategy, and a finite-entropy perfect strategy [11, Theorems 1.1 and 13.1]. A different rung is provided by Douglas and Mghirbi: a finite 0/1 game has perfect normal commuting play in type III and none when either ambient algebra is semifinite [10, Theorem 1.1]. We ask whether a finite game can force infinite entropy at every perfect spatial strategy.

We answer this question affirmatively. The optimum is attained by a normal state on a tensor product. The conclusion therefore concerns an actual spatial strategy, rather than a limit of strategies or a commuting-operator optimum outside the tensor-product model. The game has a perfect normal spatial strategy, and perfect play extracts the same pure state with infinite entropy from every such strategy, including mixed states. The rational-row compiler gives a separate way to convert synchronized additive identities into Boolean tests.

A *normal spatial strategy* consists of local POVMs on Hilbert spaces $\mathcal{H}_A, \mathcal{H}_B$ and a positive trace-class operator ρ of trace one on $\mathcal{H}_A \otimes \mathcal{H}_B$. Its probabilities are

$$p(a, b \mid x, y) = \operatorname{Tr}\bigl(\rho(E_{a|x} \otimes F_{b|y})\bigr).$$

We use entropy in bits, $S(h) = -\operatorname{Tr}(h \log_2 h)$, allowing the value $+\infty$. For pure bipartite states this is entanglement entropy. For mixed states our conclusion is infinite marginal entropy together with a pure entangled tensor factor.

Let $\mathcal{W}$ be the words generated from the empty word by appending A at any height and B only at height at least two. The letters respectively raise and lower the height by one. Write $n(w)$ for the number of A letters. There are $\mathrm{Cat}_n = (n+1)^{-1}\binom{2n}{n}$ words of size n. Define

$$\psi_{\mathrm{Cat}} = \frac{1}{\sqrt{2}} \sum_{w \in \mathcal{W}} 2^{-n(w)} |w, w\rangle. \tag{1}$$

The Catalan generating function normalizes this vector. Its reduced density has infinite entropy.

Theorem C (the game). *The explicitly specified game G has* 169 460 *questions per party, at most* 256 *answers per question, the uniform distribution on all question pairs, and a $\{0,1\}$ winning predicate. It has a perfect normal spatial strategy on $\ell^2 \otimes \ell^2$. For every perfect normal spatial strategy there are local isometries V_A, V_B and a normal state τ such that*

$$(V_A \otimes V_B)\rho(V_A \otimes V_B)^* = |\psi_{\mathrm{Cat}}\rangle\langle\psi_{\mathrm{Cat}}| \otimes \tau. \tag{2}$$

Both original marginal entropies are infinite. No finite-dimensional strategy is perfect, and finite-dimensional winning probabilities approach one.

Here and below local isometries are asserted on the state supports and extended to unused space by enlarging the target auxiliary spaces. The finite specification of G is in `game.zip`. Sections 5–8 also give finite rules generating every constraint. This is a large construction; the size is an explicit bound, not an optimality claim.

The core is the following Bell equality. The functional uses the ordinary conditional probabilities without an additional question-distribution normalization.

Theorem A (the Bell functional). *The explicitly tabulated rational Bell functional B has* 5 625 *questions per party, at most* 12 *answers, and* 73 769 *nonzero coefficients with common denominator* 2048. *For every normal spatial strategy, $B \le 0$. A normal spatial strategy attains zero, and every such equality state satisfies* (2). *No finite-dimensional strategy attains zero; finite-dimensional values approach zero. If H is a lower-bounded local Hamiltonian with $\operatorname{Tr} e^{-\beta H} < \infty$ for some $\beta > 0$, its mean energy in either equality marginal is infinite.*

Section 8 states and proves Theorem B, which compiles finitely many rational linear identities among synchronized projections into Boolean tests. Section 9 proves Theorem D: the minimum marginal entropy at loss ε is $\Theta(\log(1/\varepsilon))$. Height and size cutoffs give the upper bound. Polynomial decoding and state-weighted rounding give the lower bound for every normal spatial strategy. The radial power $1/5$ remains optimal within the scalar-filter class.

1.1 Prior work and scope of the contribution

Coladangelo–Stark concern infinite-dimensional spatial correlations [8]. The game Γ supplies an earlier Boolean game with spatial perfection, no finite-dimensional perfect strategy, and finite-entropy attainment [11, Theorems 1.1 and 13.1]; its displayed pure perfect strategy attains the minimum entropy $\log_2 24 + 7/9$ bits. Neither dimension separation alone nor an infinite-rank self-test implies the every-optimum entropy assertion here. Pauwels proves finite-dimensional nonattainment for I_{3322} [29]. Coladangelo's I_{3322} theorem [6] is another dimension separation. Its attaining state has finite entropy by the tail argument in Section 10.

There is a different route to infinity outside the tensor-product model. Paddock and Schmidt self-test exact embezzlement with a commuting-operator correlation that has no tensor-product realization [27]. Harris characterizes exact embezzlement through Cuntz isometries and a quasi-free state, with a type III conclusion [17]. Beside these results, Douglas and Mghirbi give a finite 0/1 game with 243 uniformly sampled pairs that is perfect on $R_{1/2}$ and has no perfect strategy with a normal marginal on a semifinite ambient algebra of either party [10, Theorem 1.1]. This is a type-III-only requirement for normal play, rather than a spatial entropy requirement. The structural relation between embezzlement and type III algebras is developed in [33, 34]; algebraic embezzlement is studied in [22]. Exact embezzlement is impossible in tensor products even with infinite entropy [4]. Our state is spatial and is not an exact embezzler. These are different levels of the question, not priority claims about infinite dimension, type III algebras, or embezzlement.

Limit-only resource requirements occur in the three-player game of Ji, Leung and Vidick [19] and the two-player construction of Coladangelo [5]. They differ from normally attained perfection. Mančinska's flagged magic-square game supplies the finite nonmaximally entangled bridge used here [23]. We prove the needed exact normal-state extension. Chen gives a robust finite-dimensional self-test of a related five-dimensional state [2]. Its squared Schmidt weights $(1, 4, 4, 4, 4)/17$ agree with our scaffold conditioned on $e_0 + M_2$, not with one calibrated bridge, whose weights are $(1, 2, 2, 2, 2)/9$. Chen uses extraction channels for the quantitative state assertion; that result does not directly provide the operator mirrors needed here. The quantitative argument draws on Coladangelo–Stark's rigidity framework and Vidick's state-dependent stability lemma, presented in [7, Lemma 4.7], together with his state-dependent Gowers–Hatami method [35, 15], and the spectral-cut approach of Slofstra–Vidick [32]. We prove the required normal-state magic-square lemma directly; we do not use Coladangelo–Stark's Theorem 6.9 as an arbitrary-dimension statement. The entropy comparison uses Alicki–Fannes and Winter and the infinite-dimensional formulation of Kuznetsova [1, 36, 21].

Projection partitions and synchronous game algebras are standard [13, 18, 28]. Known reductions encode multiplicative relations and relations between game algebras [31, 14, 16]. Sums of projections also underlie the correlation self-tests of Mančinska, Prakash and Schafhauser

[24] and the nonclosure constructions of Musat and Rørdam [26]. Those correlation statements do not supply the Boolean row tests below. Chen and Zhao encode real-coefficient unitary presentations by graph and column projections in synchronous correlations [3, Section 4]. Their result uses prescribed tracial correlation values. Our Boolean compiler instead enforces rational additive identities among the retained synchronized projections by zero-probability tests after finite maximally entangled amplification. To our knowledge, this extension is new as stated. The graph and Hadamard-column arguments below give the additional identification proofs.

To our knowledge, Theorem C is the first finite two-player Boolean game with a normally attained perfect strategy for which every perfect normal spatial strategy has infinite marginal entropy, and Theorem D pins its near-perfect price to $\Theta(\log(1/\varepsilon))$. Our earlier paper *The uncomputable price of precision* (E2, version 4) already gives logarithmic entanglement laws for games whose perfection is not attained [12]; we claim no priority for logarithmic laws in general. This bibliographic statement rests on the cited comparisons and targeted literature searches. It is not an exhaustive priority certificate. In particular, we claim no priority for nontracial perfect games. The known synchronous obstruction of Kim, Paulsen and Schafhauser [20, Theorem 3.10 and Corollary 3.11] explains why the nonsynchronous bridge questions are essential.

1.2 Proof route

The operator relations in Section 3 force the Catalan spectrum. Five bipartite equations force its pure tensor factor. Two calibrated flagged magic-square bridges supply density ratios. Shared opposite questions fix the gauge of duplicated bridge measurements. Graph projections and triangles then encode the required products as rational linear projection identities. Their Bell penalties prove Theorem A. The row compiler replaces those penalties by Boolean tests and proves Theorem C. Exact finite computations check the exported objects; the infinite statements follow from the written proofs.

2 Normal states and support calculus

We collect the support facts used throughout the proof. They are elementary operator facts, not additional novelty claims. A trace-class density has separable support. Thus even if the ambient spaces are nonseparable, all state-bearing arguments take place on separable spaces. We purify mixed states on $\mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_E$; no queried measurement acts on E.

Perfect matching. For opposite-party effects E, F, zero mismatch implies

$$0 = \langle E + F - 2EF \rangle = \langle (E - F)^2 + (E - E^2) + (F - F^2) \rangle. \tag{3}$$

Each positive term annihilates the purification Ω. In coefficient-operator coordinates for the cut $A \mid BE$, the first identity is $EK = K(F \otimes I_E)^T$. The range of K is dense in Alice's state support. Hence E preserves that support, and self-adjointness makes it reducing. Repeat on the $B \mid AE$ cut. On these supports the positive sharpness defects vanish: if h is injective there and $D \geq 0$, then $\mathrm{Tr}(hD) = 0$ gives $D^{1/2}h^{1/2} = 0$, and dense range gives $D = 0$. Thus matched effects restrict to projections. Multiplying the intertwining equation and its adjoint gives $[E, h_A] = 0$, and similarly on Bob. This order of reasoning establishes reduction before using projectivity.

Projective compression. If P projects onto a state support, $0 \leq E \leq I$, and $PEP = Q$ is a projection on $P\mathcal{H}$, then E reduces $P\mathcal{H}$. Indeed $\langle v, (I - E)v \rangle = 0$ for $v \in \operatorname{ran} Q$ forces

$Ev = v$. For $v \in \ker Q \cap P\mathcal{H}$, positivity forces $Ev = 0$. Self-adjointness finishes the reduction argument.

Sharp marginals. A POVM with sharp coarse marginals is their commuting joint spectral measurement. Each compressed fine effect lies below the corresponding marginal projections. If $0 \le q \le Z$ and Z is a projection, then $q = ZqZ$. Summing over the other outcomes shows that the marginals commute and that their joint products equal the fine effects (or zero for omitted outcome patterns). If this argument starts with compressed effects, projective compression is used only after sharpness has been proved.

Forbidden events. Suppose D_A, K_A, K_B are support projections and $(K_A - K_B)\Omega = 0$. A zero event for D_A and K_B gives $D_A K_B \Omega = 0$, hence $D_A K_A \Omega = 0$. Dense Schmidt range gives $D_A K_A = 0$ on the support, and taking adjoints gives the reverse product. This proof needs no density commutation in the orthogonality step.

A full synchronized question has zero probability of unequal answers. Coarse-graining gives (3) for every answer subset. Consequently all its support effects are projections commuting with the local marginal. We always mean these support-restricted projections when using local polynomial identities.

3 The Catalan relations and the pure tensor factor

Let A, B, C be bounded operators and P, Q projections. The local relations are

$$A^*A = I, \qquad AA^* + BB^* = I - P, \qquad B^*B = I - P - Q, \tag{4}$$

$$CP = P, \qquad CQ = Q/2, \qquad 0 \le C \le I, \tag{5}$$

$$CA = AC/2, \qquad CB = 2BC. \tag{6}$$

For an injective normal density h, the accompanying modular relations are

$$hA = Ah/4, \qquad [h, B] = [h, C] = 0. \tag{7}$$

The canonical representation on $\ell^2(\mathcal{W})$ appends the legal letter, with B zero where it is not allowed. It has $C|w\rangle = 2^{-\mathrm{ht}(w)}|w\rangle$, $P = |\varnothing\rangle\langle\varnothing|$ and Q the height-one projection.

Lemma 3.1 (Normal Catalan classification)**.** *Every normal density satisfying (4)–(7) has, on its support, the form*

$$\mathcal{H} = \ell^2(\mathcal{W}) \otimes \mathcal{J}, \qquad h = h_{\mathrm{Cat}} \otimes \sigma, \qquad h_{\mathrm{Cat}} = \frac{1}{2} \sum_{w \in \mathcal{W}} 4^{-n(w)} |w\rangle\langle w|,$$

where σ is a normal density and the five operators are the canonical operators tensored with identity.

Proof. The equations $CP = P$ and $CQ = Q/2$ make P, Q reducing spectral subprojections of C and imply $PQ = 0$. The operator B is a partial isometry. The three range projections AA^*, BB^*, P are orthogonal and sum to identity. Covariance makes these range projections commute with C.

Put $E_k = 1_{\{2^{-k}\}}(C)$ and $E_\infty = 1_{\{0\}}(C)$. There is no spectrum in $(1/2, 1)$: there B is isometric and would send it above one. The same argument on $E_0 - P$ gives $E_0 = P$. On $E_1 - Q$, B would isometrically map into P, orthogonally to its range, so $E_1 = Q$. Iteration excludes every open interval $(2^{-k-1}, 2^{-k})$. Bounded functional calculus for $CB = 2BC$ justifies these spectral transports. Thus $I = E_\infty + \sum_{k \ge 0} E_k$.

The zero-counter space reduces A, B, C, h. There $P = Q = 0$ and A, B are isometries with orthogonal ranges summing to identity. If $t = \mathrm{Tr}(hE_\infty)$, trace cyclicity and (7) give range masses $t/4$ and t. Their sum is t, so $t = 0$. There is no zero-counter state mass.

Starting with $P\mathcal{H}$, apply legal words in chronological order; write W_w for the resulting isometry from the root space. Different sectors $W_w P\mathcal{H}$ are orthogonal. Different final letters have orthogonal ranges, and a nonempty sector is orthogonal to P. Matching final letters can be canceled using $A^*A = I$ or $B^*B = I - P - Q$, which is identity on each legal predecessor for B. Repeating proves the assertion. On sector w, $C = 2^{-\mathrm{ht}(w)} I$ and

$$hW_w = 4^{-n(w)} W_w h.$$

The root projection commutes with h, so each word-sector projection does too. Its density block is $4^{-n(w)} W_w h_0 W_w^*$, where $h_0 = PhP$.

The closed span of the root sectors reduces all the operators and h. Suppose its orthogonal complement carried positive mass. The compact density on that complement has a largest positive eigenvalue λ, with a nonzero finite-dimensional eigenspace L. The adjoint modular equation is $hA^* = 4A^*h$, so $A^*L = 0$. Since $P = 0$ on the complement, $BB^* = I$ on L. The space L reduces B; its restriction is therefore a unitary. It also reduces C. Multiply $CB = 2BC$ by the unitary B^* on the right and trace on L. This gives $\mathrm{Tr}_L C = 2\,\mathrm{Tr}_L C$. Positivity forces $C = 0$ there, contradicting the absence of zero-counter mass. Hence the word sectors exhaust the support.

For $n \geq 1$ and $0 \leq m \leq n - 1$, reflection of the first bad prefix counts the words with n letters A and m letters B as

$$\binom{n+m-1}{m} - \binom{n+m-1}{m-1}.$$

Summing gives Cat_n; the empty word gives $\mathrm{Cat}_0 = 1$. The generating function $C(z) = 1 + zC(z)^2$ gives $\sum_n \mathrm{Cat}_n\, 4^{-n} = 2$. Thus $\mathrm{Tr}\, h_0 = 1/2$, and $\sigma = 2h_0$ gives the claimed identification. □

Its entropy is

$$S(h_{\mathrm{Cat}}) = \sum_{n \geq 0} \frac{\mathrm{Cat}_n}{2\,4^n}(1 + 2n) = \infty, \qquad \frac{\mathrm{Cat}_n}{4^n} \sim \frac{1}{\sqrt{\pi} n^{3/2}}. \tag{8}$$

Infinite rank alone would not give this conclusion. Tensoring with any normal density preserves the divergence.

3.1 Five bipartite equations

On a purification Ω require each party's local relations and

$$(A_A/2)\Omega = A_B^*\Omega, \qquad A_A^*\Omega = (A_B/2)\Omega, \tag{9}$$

$$B_A\Omega = B_B^*\Omega, \qquad B_A^*\Omega = B_B\Omega, \tag{10}$$

$$C_A\Omega = C_B\Omega. \tag{11}$$

Tensor identities and the identity on the environment are suppressed. Both adjoint equations and both factors 1/2 matter.

Lemma 3.2 (Pure extraction)**.** *The local relations and* (9)–(11) *imply* (2)*.*

Proof. For a pure vector with coefficient operator K, the two A equations read $A_A K = 2K\overline{A_B}$ and $A_A^* K = K A_B^T/2$. Combining either with the adjoint of the other gives $h_A A_A = A_A h_A/4$. The B equations and the C equation give the remaining commutators. The same holds on Bob. For a purification, use the two original local cuts. Paired equations ensure support reduction, without inverting K.

Lemma 3.1 identifies the word sectors on both sides. The C equation gives $P_A\Omega = P_B\Omega$ by functional calculus. Put $\Omega_0 = P_A\Omega$. Transferring adjoint letters in their actual order across the vector gives

$$(W_w^{A*} \otimes I)\Omega = 2^{-n(w)}(I \otimes W_w^B)\Omega.$$

The root projection on the transferred side selects the same word sector. Consequently

$$(P_w^A \otimes I)\Omega = 2^{-n(w)}(W_w^A \otimes W_w^B)\Omega_0.$$

The sectors exhaust the state and $\|\Omega_0\|^2 = 1/2$. Their sum is $\psi_{\mathrm{Cat}} \otimes \eta$, where $\eta = \sqrt{2}\Omega_0$ lies in the root multiplicities and the environment. Tracing out the environment proves the mixed-state assertion. □

4 The normal flagged magic-square bridge

The magic-square relations originate with Mermin and Peres [25, 30]. We use Mančinska's flagged magic-square (FMS) geometry [23]. We spell out the predicate and prove its normal extension because the published finite-dimensional statement does not suffice here. All indices start at zero.

4.1 Questions, calibration and exact bindings

Alice has R_0, R_1, R_2, T_0, T_1 and Bob has C_0, C_1, C_2, U_0, U_1. A row question has a flag answer f or a triple of product $+1$. A column question has f or a triple of product $+1, +1, -1$ according to its column. The magic square is

$$\begin{pmatrix} XI & IX & XX \\ IZ & ZI & ZZ \\ XZ & ZX & YY \end{pmatrix}. \tag{12}$$

Here the displayed matrices describe a canonical realization; the predicate only checks the corresponding signs. Select column-zero outcomes $c_0 = (+++)$, $c_1 = (+--)$ and row-zero outcomes $r_0 = (+++)$, $r_1 = (+--)$. Question T_x has answers $+, -$ and the three column-zero triples other than c_x; U_y analogously retains all row-zero triples other than r_y.

On R_i, C_j, reject flag/nonflag mismatch and otherwise check the intersecting cell; two flags are accepted. On T_x, C_0, a sign is allowed only against f or c_x; a retained triple must equal the opposite triple. On R_0, U_y use the analogous rule with r_y. On T_x, U_y, reject two signs $a, b \in \{+1, -1\}$ unless $ab = (-1)^{xy}$; all pairs with a nonsign answer are accepted. Other question pairs are accepted. For a conditional instance add an inactive answer $\perp$, accept two inactive answers, and reject exactly one inactive answer. Bind each noninactive event to its synchronized active projection and all row/column flags to the synchronized flag.

Let M be the nonflag magic support. The four column-zero projections $m_{00}, m_{01}, m_{10}, m_{11}$ correspond respectively to $(+++), (+--), (-+-), (--+)$. On Alice, m_{00} is read from the retained $(+++)$ answer of T_1 and the other three from T_0. On Bob all four come from C_0. Each of these chosen occurrences is matched to its opposite synchronized binary question. The positive nonflag ZI event is read from Alice's R_1 component 1 and Bob's C_1 component 1. The positive IX event comes from Alice's R_0 component 1 and Bob's C_1 component 0. Put

$$x = I - 2M + 2E_{ZI,+}, \quad z = I - 2M + 2E_{IX,+}, \quad t_{ab} = x^a z^b m_{00}.$$

These are reflections extended by identity outside M, and $t_{ab}^* t_{ab} = m_{00}$, $t_{ab} t_{ab}^* = m_{ab}$.

4.2 Normal bridge theorem

Lemma 4.1. *A perfect normal conditional FMS strategy has on its active supports the decomposition*

$$\mathcal{H}_{\text{active}} = \mathcal{H}_f \oplus (\mathbb{C}^4 \otimes \mathcal{J}), \qquad h = h_f \oplus (I_4 \otimes h_m).$$

The calibrated transports w_A, w_B from the flag to the selected c_0 corner satisfy $w^* w = f$, $w w^* = m_{00}$ and $hw = 2wh$. On a purification their matched components satisfy

$$\Omega_{c_0 c_0} = \sqrt{2}(w_A \otimes w_B \otimes I_E)\Omega_{ff}. \tag{13}$$

The other three magic directions obey the transported versions of this equation.

Proof. First apply Section 2 to the activity, flags and checked cells. They reduce the original state supports and are sharp there. On the compressed nonflag space each compressed fine effect lies below its assigned sharp cell marginals. Thus it is their joint spectral projection. The projective-compression fact then makes the original fine effects reduce the support. Alice's row relations and Bob's column relations transfer through the dense coefficient range, in reversed product order; within a context the products commute.

For completeness, write $a = M_{00}, b = M_{01}, c = M_{10}, d = M_{11}$ for the four corner reflections of the square. Context relations give

$$[a, b] = [c, d] = [a, c] = [b, d] = 0, \qquad (ac)(bd) = -(ab)(cd).$$

Cancel the outside a, d to obtain $cb = -bc$. Commutation of ac and bd then gives $da = -ad$. The pairs (a, d) and (b, c) are anticommuting Pauli pairs commuting across pairs. Each pair exchanges its two eigenspaces unitarily. Hence they generate $M_4 \otimes I_{\mathcal{J}}$. Marginal commutation gives the stated density form. The synchronized magic units identify its coefficient block as $I_4 \otimes D$, with D positive, injective and dense-range. Write the flag coefficient block as F with the same properties.

Calibration makes Alice's sign support $S_x^A = f + \Pi_{c_x}$ and Bob's $S_y^B = f + \Pi_{r_y}$. The selected directions have overlap $\langle c_x, r_y \rangle = \sigma_{xy}/\sqrt{2}$, where $\sigma_{xy} = (-1)^{xy}$. Their compressed coefficient operator is

$$W_{xy} = \operatorname{diag}(F, \sigma_{xy} D/\sqrt{2}).$$

For compressed sign effects E_a, B_b, a forbidden pair gives $E_a^{1/2} W_{xy} (B_b^T)^{1/2} = 0$, and therefore $E_a W_{xy} B_b^T = 0$. Summing the two complete binary effects gives

$$E_a W_{xy} = W_{xy} B_{\sigma_{xy} a}^T = E_a W_{xy} B_{\sigma_{xy} a}^T.$$

Injectivity and dense range imply $B_b^2 = B_b$ and $E_a^2 = E_a$ on the sign supports. Their zero extensions give projective compressions; thus the original sign effects reduce the state supports. Their differences X_x, Y_y obey

$$X_x W_{xy} = \sigma_{xy} W_{xy} Y_y^T. \tag{14}$$

Identify the magic directions using the actual matrix units. Set $H = \operatorname{diag}(F, D/\sqrt{2})$ and $J = \operatorname{diag}(I, -I)$. The three positive equations give a common X, Y with $XH = HY^T$. The remaining one gives $XJH = -JHY^T$, hence $XJ = -JX$. Thus

$$X = \begin{pmatrix} 0 & \beta^* \\ \beta & 0 \end{pmatrix}, \qquad \beta : \mathcal{H}_f \longrightarrow \mathcal{J} \text{ unitary.}$$

Self-adjointness and $XH = HY^T$ imply $[X, H^2] = 0$. Its off-diagonal block gives $h_m\beta = 2\beta h_f$. The corresponding corner transport has the required ratio.

The actual Alice extraction is $w_A = \Pi_{c_0} X_0 f$. Bob's r_0 projection is the retained $(+++)$ answer of U_1, equivalently the sign support of U_0 minus its flag. His extraction to the same c_0 direction is

$$w_B = \sqrt{2}\Pi_{c_0}\Pi_{r_0} Y_0 f.$$

The coefficient $\sqrt{2}$ occurs in extraction, not in the predicate. To see the phase in (13) directly, put $\Omega' = (S_0^A \otimes S_0^B)\Omega$. The zero sign events give $X_0\Omega' = Y_0\Omega'$. Project onto $c_0 \otimes f$. The left side is $(w_A \otimes I)\Omega_{ff}$; the right side is

$$2^{-1/2}(I \otimes fY_0|r_0\rangle\langle c_0|)\Omega_{c_0c_0}.$$

Apply Bob's flag-to-r_0 block and then $|c_0\rangle\langle r_0| = \sqrt{2}\Pi_{c_0}\Pi_{r_0}$. This gives (13).

For a mixed state, first apply the pure-vector calculation on the cut $A \mid BE$. The synchronized opposite effects reduce the Schmidt support, and the sign effects do so after their compressed sharpness is proved. Thus every compressed product used above is the restriction of an original Bob-only product tensored with I_E. The resulting Alice identities hold on her original marginal support. Repeat on $B \mid AE$ for Bob's identities. The component calculation then gives Eq. (13) on the original purification, using $w_A \otimes w_B \otimes I_E$. Tracing E preserves the claimed pure factor. □

4.3 A shared question fixes transported labels

Lemma 4.2 (Label refinement)**.** *Retain the original FMS questions. For a synchronized projection Z, add Alice's $T_0[Z]$ with answers (a, z). Ignore z against the unchanged U_0, U_1, C_0 and use the original predicates; match z to Bob's binary Z. Add the symmetric $U_0[Z]$ against unchanged T_0, T_1, R_0. Also retain activity matching. In every perfect normal strategy the coarse signs equal the original signs and commute with Z on their state supports. Conversely commuting labels give these refinements. Distinct labels require no mutual commutation.*

Proof. Write P_A for the state support and R_A for the sign support. For an arbitrary new fine effect $Q_{a,z}$ first form $q_{a,z} = P_A Q_{a,z} P_A$. Calibration identifies its compressed retained effects and the coarse sign sum R_A. Let X' be its compressed sign difference. The unchanged opposite sign gives $X'W_{0y} = W_{0y}Y_y^T$, with exactly the same dense-range W_{0y} as for X_0. Hence $X' = X_0$. Its coarse effects are $(R_A \pm X_0)/2$, hence projections. Projective compression now makes the original coarse effects reduce P_A.

Label matching gives $P_A(\sum_a Q_{a,1})P_A = Z_A$. Positivity gives $0 \le q_{a,1} \le Z_A$ and $0 \le q_{a,0} \le I - Z_A$. Thus the compressed coarse effects commute with Z_A, and so does X_0. The symmetric argument applies on Bob. Conversely use $E_a Z, E_a(I - Z)$ when the original coarse effects commute with Z. A separate duplicated question for each label always returns the same original sign. This proves the final assertion. □

If a label is block diagonal with flag block Z_f and identical magic blocks Z_m, the commutator says $\beta Z_f = Z_m\beta$. In particular, for a graph projection of a block unitary it certifies transport of that same unitary by the calibrated w. It does not supply a fresh bridge with an independent gauge.

5 Rational projection equations that certify products

All projections in this section are synchronized. We give the finite rules used in the core presentation. The projection-partition viewpoint is standard [13]; the formulas here provide the needed actual multiplication identities. See also the graph-projection encoding in [3, Section 4].

5.1 Graphs and triangles

Let F_i, F_j be orthogonal, $S = F_i + F_j$, and introduce projections G, H, K_G, K_H with

$$G + K_G = S, \quad H + K_H = S, \quad 5H + S = 3F_i + 4G. \tag{15}$$

On S, put $Z = F_i - F_j$, $X = 2G - S$. These are reflections, and $2H - S = (3Z + 4X)/5$. Squaring forces $ZX + XZ = 0$. Thus

$$X = \begin{pmatrix} 0 & u_{ij} \\ u_{ij}^* & 0 \end{pmatrix}, \qquad u_{ij}^* u_{ij} = F_j, \quad u_{ij} u_{ij}^* = F_i.$$

The graph blocks are certified, rather than assumed. Conversely any such unitary corner map gives the four projections in (15).

For a partition $F_1 + F_2 + F_3 = I$ impose the graph tests on all three pairs and add a projection D with

$$3D = 2(G_{12} + G_{23} + G_{13}) - I. \tag{16}$$

Its block matrix has diagonal $I/3$ and off-diagonal $u_{ij}/3$. The $(1,3)$ block of $D^2 = D$ gives $2u_{13} + u_{12}u_{23} = 3u_{13}$, hence $u_{13} = u_{12}u_{23}$. Conversely this product identity makes the entire matrix a projection.

Start with a reference triangle for identity. Its scalar matrix algebra is called the *frame*. It gives scalar matrix units and $\mathcal{H} = \mathbb{C}^3 \otimes \mathcal{K}$. For every unitary label U, use graphs in all three positions. Triangles $(U, 1, U)$ and $(1, U, U)$ identify the placements. A triangle (U, V, W) then certifies $W = UV$; a triangle with product identity certifies inverse labels. Build each word from its proper prefix and final token. Finally a real rational equation $\sum_w q_w U_w = 0$ is precisely

$$2\sum_w q_w G_{12}(U_w) - \Big(\sum_w q_w\Big)(F_1 + F_2) = 0. \tag{17}$$

Its two off-diagonal blocks are the equation and its adjoint. All products therefore refer to actual common operators. Synchronization makes the extracted unitaries commute with the base marginal density.

5.2 Binding prescribed parameters

A prescribed synchronized projection E is linked to the frame by projections EF_i, NF_i with

$$EF_i + NF_i = F_i, \quad \sum_i EF_i = E, \quad \sum_i NF_i = I - E.$$

These symbols are names for projections before the equations identify their products. For every pair ij, introduce $J_{11}, J_{10}, J_{01}, J_{00}$ and impose

$$\begin{aligned} J_{11} + J_{10} &= EF_i + EF_j, & J_{00} + J_{01} &= NF_i + NF_j, \\ J_{11} + J_{01} &= G_{ij}(1), & G_{ij}(s_E) &= J_{11} + J_{00}. \end{aligned}$$

Projection sums give the indicated joint partitions. They force E to commute with the frame and identify the extracted s_E with $2e - I$, where $E = I_3 \otimes e$. In particular these parameters are the specified FMS events, not independent replacements. There are thirteen such parameters below.

These rules are a deterministic finite compiler for the finite word list. Cancellation of adjacent inverse tokens is performed before taking proper prefixes. All coefficients stay rational. Conversely a centralizer unitary assignment with the prescribed parameter projections gives every graph, complement, H, triangle and joint projection simultaneously by these formulas.

6 A common Catalan table from two bridges

6.1 Nine corners and the small polynomial input

Take two conditional FMS instances, indexed $g = 1, 2$. Use nine orthogonal base corners

$$e_0;\ m_{1,00}, m_{1,01}, m_{1,10}, m_{1,11};\ m_{2,00}, m_{2,01}, m_{2,10}, m_{2,11}.$$

Put $M_g = \sum_{ab} m_{g,ab}$, $e_1 = m_{1,00}$, $e_2 = m_{2,00}$, and impose $e_0 + M_1 + M_2 = I$. Bridge g has flag e_{g-1}, magic support M_g, and activity $e_{g-1} + M_g$. Every event binding is a zero-mismatch test in both party orientations. In particular Bob's flag/four-magic/inactive C_0 partition certifies the activity decomposition.

Let $w_g : e_{g-1} \to e_g$ be the calibrated transports. Then

$$v = w_2 w_1, \quad v^* v = e_0, \quad vv^* = e_2, \quad hv = 4vh. \tag{18}$$

The bridge masses and cover give root mass $1/25$, with weights 1, four copies of 2, and four copies of 4. Thus the root cannot vanish.

Use the thirteen parameters e_0, the eight $m_{g,ab}$, and the four positive ZI, IX events. Bind them to the scalar frame by Section 5. Introduce four global unitary axis labels equal to x_g, z_g of Section 4. For $r = 0, 1, 2$ and $k \in \{b_+, b_-, p, q, c\}$ use unitary labels $u_{r,k}$ and impose

$$(u_{r,k} - I)(I - e_r) = 0. \tag{19}$$

A unitary fixing $e_r^\perp$ reduces e_r, so $e_r u = e_r u e_r$. Define the actual polynomials

$$\begin{aligned} b_r &= e_r(u_{r,b_+} + u_{r,b_-})/2, & p_r &= e_r(I + u_{r,p})/2, \\ q_r &= e_r(I + u_{r,q})/2, & c_r &= e_r(2I + u_{r,c} + u_{r,c}^*)/4. \end{aligned} \tag{20}$$

Require $u_{0,p}^2 = u_{0,q}^2 = I$. Hence p_0, q_0 are projections, b_r are contractions, and $0 \le c_r \le e_r$.

For each g, k introduce one unitary $v_{g,k}$ and impose, with $R_{ab} = x_g^a z_g^b$,

$$v_{g,k} = I - (e_{g-1} + M_g) + e_{g-1} u_{g-1,k} + \sum_{ab} R_{ab} e_g u_{g,k} R_{ab}^*. \tag{21}$$

For each bridge, refine both FMS questions as in Lemma 4.2 for the six frame projections $F_1, F_2, F_3, G_{12}(1), G_{13}(1), G_{23}(1)$ and the five projections $G_{12}(v_{g,k})$. This means eleven labels per bridge, twenty-two refined questions per party. Frame preservation comes first. It makes $w_g = I_3 \otimes \widetilde{w}_g$ because the selected magic parameters generate the full M_4 algebra and already commute with the frame. The graph-label refinements then give

$$w_g u_{g-1,k} = u_{g,k} w_g. \tag{22}$$

Thus the repeated operators are transported by the same bridge.

Take two more global unitaries a_+, a_- and $a = (a_+ + a_-)/2$. The remaining nine equations are

$$a = e_2 a e_0, \qquad a^* a = e_0, \qquad aa^* + b_2 b_2^* + p_2 = e_2, \tag{23}$$

$$b_0^* b_0 = e_0 - p_0 - q_0, \qquad p_0 q_0 = 0, \qquad c_0 p_0 = p_0, \tag{24}$$

$$c_0 q_0 = q_0/2, \qquad c_2 a = a c_0/2, \qquad c_0 b_0 = 2 b_0 c_0. \tag{25}$$

Together there are four axis equations, fifteen support equations, ten block equations, nine equations above, two root reflection equations, and one cover: forty-one polynomial inputs.

Expand every parameter $e = (I + s_e)/2$ and apply the prefix/graph rules. They give 358 words and 4 529 rational linear projection rows, fully listed in `core.json`.

For clarity, four reductions from a longer presentation are exact. The axis right sides are reflections because their positive events lie under M_g. The omitted source/range equations for t_{ab} follow from the actual Pauli cells and the stated outcome bindings. Equation (19) removes the final e_r from every sandwich in (20). Finally $t_{ab} = R_{ab}e_g$ and (19) turn $t_{ab}u_{g,k}t^*_{ab}$ into the summand in (21). No unproved equation deletion enters the smaller list.

6.2 Extraction and mirrors

After factoring the synchronized scalar frame, work on the root corner and define

$$A = v^*a, \quad B = b_0, \quad C = c_0, \quad P = p_0, \quad Q = q_0. \tag{26}$$

These are fixed bounded corner expressions in the original effects. From (18)–(25), $A^*A = e_0$, $AA^* = e_0 - P - BB^*$, $B^*B = e_0 - P - Q$. The two counter boundary equations are explicit. Moreover

$$CA = c_0v^*a = v^*c_2a = v^*ac_0/2 = AC/2, \qquad CB = 2BC.$$

All centralizer words commute with the full density. Since $hv = 4vh$, the normalized root marginal satisfies $h_0A = Ah_0/4$. Thus every local relation of Section 3 holds for one common table.

The synchronized scalar M_3 extracts a maximally entangled qutrit. For real-coefficient base words, graph matching gives $T_A\Omega = T_B^*\Omega$; products transfer in reversed order, as their adjoints require. Let Ω_r denote the shared component in e_r. The calibrated vector equations give

$$\Omega_1 = \sqrt{2}(w_{1,A} \otimes w_{1,B})\Omega_0, \qquad \Omega_2 = 2(v_A \otimes v_B)\Omega_0.$$

Since $a = e_2ae_0$, its transfers read $a_A\Omega_0 = a_B^*\Omega_2$ and $a_A^*\Omega_2 = a_B\Omega_0$. Substitute the second component formula and apply v_A^* or v_B^*. This yields both equations in (9). The root centralizer words b_0, b_0^*, c_0 give (10)–(11).

Magic units identify each of the other three directions in a bridge. Hence all nine sectors are local images of the same root component with coefficients 1, four copies of $\sqrt{2}$, and four copies of 2. This factors the entire state into a fixed finite scaffold state and the normalized root state. It is not a postselection argument. Lemma 3.2 gives (2). Every operation acts on original local supports and leaves the purifying environment alone, so the conclusion includes mixed states.

6.3 Simultaneous normal realization

Use the canonical Catalan operators and scaffold density

$$h_{\mathrm{sc}} = \frac{1}{25}\,\mathrm{diag}(h_{\mathrm{Cat}}, 2h_{\mathrm{Cat}}, 2h_{\mathrm{Cat}}, 2h_{\mathrm{Cat}}, 2h_{\mathrm{Cat}}, 4h_{\mathrm{Cat}}, 4h_{\mathrm{Cat}}, 4h_{\mathrm{Cat}}, 4h_{\mathrm{Cat}}). \tag{27}$$

Take identity transports along the bridges. Put $a = |e_2\rangle\langle e_0| \otimes A_{\mathrm{Cat}}$, with the canonical copies for b_r, c_r, p_r, q_r. The modular ratio makes a commute with (27); all nine equations hold.

We justify the unitary encodings simultaneously. The centralizer of an injective trace-class density is a product of finite matrix blocks, one for each positive eigenvalue. In each block the polar partial isometry of any contraction T extends to a unitary W. Then

$$U_\pm = W\big(|T| \pm i\sqrt{I - |T|^2}\big), \qquad T = (U_+ + U_-)/2.$$

Direct sums give unitaries in the same centralizer. Apply this to the amplified a and to the root contraction B. This does not extend the proper unamplified isometry A_{Cat} in a finite space. Use $2P - I, 2Q - I$ for the root reflections and

$$U_c = 2C - I + 2i\sqrt{C(I - C)}, \qquad C = (2I + U_c + U_c^*)/4.$$

Transport the same choices to the other corners and extend by identity outside them. This realizes all support and block equations.

Append a maximally entangled qutrit and use the graph formulas. Each proposed label commutes with the coarse FMS effects: the frame is preserved and each block unitary has identical transported flag and magic blocks. Refined effects are their actual joint products. Distinct labels use separate measurements. Thus all rows, FMS predicates, bindings and refinements have one simultaneous normal realization.

7 The rational Bell functional

For opposite-party binary effects E_i, F_i and rational coefficients $\alpha_0, \ldots, \alpha_m$, define

$$S = \alpha_0 I + \sum_i \alpha_i E_i, \qquad T = \alpha_0 I + \sum_i \alpha_i F_i, \qquad k = 1 + \tfrac{1}{2}\sum_i \alpha_i^2,$$
$$F_\alpha = ST + k\sum_i (E_i + F_i - 2E_iF_i). \tag{28}$$

This is a Bell expression after expanding ST: constants and marginals use a one-answer anchor. It contains no same-party products. For $\Delta_i = E_i - F_i$, cross-party commutation gives the exact identity

$$F_\alpha = \frac{S^2 + T^2}{2} + \sum_i \Delta_i^2 + \frac{1}{2}\sum_{i<j} (\alpha_j \Delta_i - \alpha_i \Delta_j)^2 + k\sum_i (E_i - E_i^2 + F_i - F_i^2). \tag{29}$$

Indeed $ST = (S^2 + T^2 - (S - T)^2)/2$; expansion of the last square and the elementary Lagrange identity yield (29). No same-party commutation or idempotency is assumed in this calculation. Every displayed summand is positive for effects. Equality gives perfect matching and the row identity $S = 0$ on the reducing state support by Section 2.

Define B as minus the sum of (28) for every one of the 4 529 core rows, one binary synchrony loss per binary symbol, and the ordinary losing-event probabilities of the conditional FMS predicates, refinements, activity and event bindings. Every loss has unit weight. There are 5 592 binary symbols, ten original FMS questions and twenty-two refined questions per party, in addition to the anchor. FMS questions have six answers including inactivity, and refinements have twelve. These finite rules specify the whole functional. Exact aggregation produces the advertised coefficient counts.

Proof of Theorem A. Equation (29) and nonnegative event probabilities prove $B \leq 0$ for arbitrary POVMs. At equality each loss vanishes separately. Support matching, the bridge and label lemmas, and the projection compiler therefore recover the full construction of Section 6. Lemma 3.2 gives the pure factor. The simultaneous realization there attains equality normally. Equation (8) and invariance of nonzero marginal spectra under isometries give infinite entropy on both sides.

For finite-dimensional nonattainment, the root has nonzero mass and $A^*A = I$ makes A unitary there. The range equation then forces $B = P = 0$. The initial equation gives $Q = I$, hence $C = I/2$. Covariance would say $A/2 = A/4$, a contradiction.

For approximation, let $a_n = \mathrm{Cat}_n / 4^n$, $Z_N = \sum_{n \le N} a_n$, $t_N = a_N / Z_N$, and use the canonical words with $n \le N$. Compress the word operators and make centralizer unitary completions on the finite scaffold as before. Every relation is exact except $A_N^* A_N - I = -\Pi_N$. The corresponding compiled row is

$$S_{\mathrm{iso}} = -(|1\rangle\langle 2| + |2\rangle\langle 1|) \otimes e_0 \otimes \Pi_N.$$

All event losses and synchrony losses vanish. Synchronization makes the row penalty equal to $\mathrm{Tr}(h_N S_{\mathrm{iso}}^2) = 2t_N/75$, the expectation of its square. Hence

$$B_N = -\frac{2t_N}{75} \longrightarrow 0, \qquad t_N \sim \frac{1}{2\sqrt{\pi} N^{3/2}}. \tag{30}$$

Finally the Gibbs entropy bound in natural units is $S(h) \ln 2 \le \beta \,\mathrm{Tr}(hH) + \ln \mathrm{Tr}\, e^{-\beta H}$. It follows by truncation, or the variational principle, also when the entropy is infinite. A lower bound on H makes its mean energy well-defined as an extended number. Finite mean energy would force finite entropy, giving the stated contradiction. □

7.1 Literal coefficient convention

The archive contains ordered question/answer lists, `core.json`, `format.json` and one sorted coefficient stream. Each sixteen-byte little-endian record is an unsigned 64-bit key followed by a signed 64-bit numerator. The coefficient is the numerator divided by 2048, with

$$\mathrm{key} = 2^{32}\mathrm{event}_A + \mathrm{event}_B, \qquad \mathrm{event} = 16\,\mathrm{question\ id} + \mathrm{answer\ index}.$$

Unlisted coefficients are zero. Indices start at zero. The anchor makes constants and marginals literal cross-party probabilities. No implicit affine substitution or distribution factor is needed.

8 Rational rows as Boolean winning conditions

The Bell penalties have signed coefficients. Reclassifying their negative terms as losses would not enforce their zero sets. We instead implement each rational row as a subspace orthogonality test.

Theorem B (rational-row compiler). *Let a finite strategy specification name finitely many synchronized binary projections and finitely many rational linear identities among them. There is a finite extension by Boolean tests with the following properties.*

Soundness: *in every perfect normal spatial strategy for the extended tests, the identities hold as operator identities for the named support-restricted projections on each local state support.*

Completeness: *any normal spatial strategy with these synchronized projections, satisfying the identities on its supports, extends to satisfy all the added tests perfectly after adjoining a finite maximally entangled state. The old measurements act trivially on the new factor. The rest of the old strategy, including nonsynchronous questions, is retained; if its old tests were perfect, the whole extension is perfect.*

8.1 A certified scalar matrix algebra

Choose $d_{\max} = 2^{2r}$ large enough for every row construction below. Use r synchronized magic squares, binding each context's cell marginals to synchronized binary cells. The Pauli-pair calculation in Lemma 4.1 gives two anticommuting pairs per square. Four-answer joint

questions for all pairs of axes from distinct squares force cross-square commutation. Hence their algebra is $M_{d_{\max}} \otimes I$ on the support. Synchronization makes the marginal $I_{d_{\max}}/d_{\max} \otimes h$.

Add joint questions between every named old projection and every scalar axis. Their sharp marginals force the old projections into the commutant: $E_i = I_{d_{\max}} \otimes e_i$. This step concerns only synchronized projections, never an FMS sign. Add global Z and global X measurements with all bit outcomes, binding each bit marginal to its axis. They are the actual joint spectral measurements.

For a row dimension $d = 2^n \le d_{\max}$ use the first n qubits; the rest are spectators. For each nonzero $j \in \mathbb{F}_2^n$, choose a basis $\ell_1, \dots, \ell_{n-1}$ of $j^\perp$. Add a joint context for $X^j, Z^{\ell_1}, \dots, Z^{\ell_{n-1}}$, binding its marginals to the corresponding global parity events. These commute because $\ell \cdot j = 0$. The zero-Z-syndrome outcomes with X^j sign $\pm$ are

$$R_j^{\pm} = \tfrac{1}{2}(|0\rangle \pm |j\rangle)(\langle 0| \pm \langle j|) \otimes I_{\text{spectator}}.$$

The reference $R_d = |+_d\rangle\langle +_d| \otimes I_{\text{spectator}}$ is the event that the first n global X bits are positive. It includes all spectator outcomes.

8.2 Signed reflections and column graphs

For one row $S = \alpha_0 I + \sum_i \alpha_i e_i$, let L clear all coefficient denominators and set

$$s_i = 2e_i - I, \quad c_i = L\alpha_i, \quad c_0 = 2L\alpha_0 + \sum_i L\alpha_i.$$

Then $2LS = c_0 I + \sum_i c_i s_i$. Expand this into signed reflections with integer repetitions, add a canceling $I, -I$ pair with $V_0 = I$, and pad by further such pairs to a power of four d. The list has even length because its coefficient sum is even. Thus

$$\sum_{j=0}^{d-1} V_j = 2LS, \qquad V_0 = I, \qquad V_j \in \{\pm I,\ \pm(2e_i - I)\}. \tag{31}$$

For each $j > 0$ use a six-answer joint question of e_i and the three-way scalar measurement $R_j^+, R_j^-, I - R_j^+ - R_j^-$. For $V_j = \pm I$, bind its binary bit to the anchor. Matching both marginals identifies all joint effects. Appropriate coarse events are the two-coordinate graph projections

$$G_j^{\pm} = \frac{1}{2}\begin{pmatrix} I & \pm V_j \\ \pm V_j & I \end{pmatrix}$$

on coordinates $0, j$, zero elsewhere. A negative reflection sign exchanges the two coarse events.

Add a synchronized d-answer question with projections D_t, indexed by $t \in \mathbb{F}_2^n$. Put $\chi_t(j) = (-1)^{t \cdot j}$. Forbid D_t together with G_j^- when $\chi_t(j) = 1$, or G_j^+ when $\chi_t(j) = -1$. The forbidden-event support rule gives actual orthogonality. A vector in $\operatorname{ran} D_t$ therefore has coordinates $v_j = \chi_t(j) V_j v_0$.

Define

$$J_t \xi = d^{-1/2}\big(\chi_t(j) V_j \xi\big)_{j=0}^{d-1}, \qquad P_t = J_t J_t^*.$$

Unitarity and character orthogonality give $J_t^* J_u = \delta_{tu} I$ and $\sum_t P_t = I$. Neither equation requires different V_j to commute. The kernel conditions give $D_t \le P_t$. Since both families sum to identity, $D_t = P_t$ for every t.

Finally forbid D_0 jointly with the opposite R_d. Then $D_0 R_d = 0$, equivalently $R_d J_0 = 0$. The column overlap is

$$J_0^* R_d J_0 = d^{-2}\Big(\sum_j V_j\Big)^*\Big(\sum_j V_j\Big).$$

It vanishes exactly when $S = 0$. This proves soundness on arbitrary normal supports.

For completeness append $\Phi_{d_{\max}}$, use the scalar Paulis and the actual joint products, and define the column partition by the J_t. Synchronized original projections commute with the marginal and transfer across a purification. Products transfer in reversed order. Define Bob's column effects by those transferred products, equivalently by transpose in canonical Schmidt coordinates. This order matters for noncommuting e_i. All column and synchrony tests hold. The row-zero test holds precisely because $S = 0$. These constructions act trivially on every old measurement's new factor and prove completeness, including for mixed original states.

8.3 The explicit game and proof of Theorem C

For the present 4 529 rows, $d = 16$ for 4 507 rows, 64 for twenty rows and 256 for two rows. Four magic squares certify eight scalar qubits. The old core has 5 590 projection variables; two additional activity symbols do not enter a row. There are 89 440 old-variable/axis joint questions, 333 scalar pair contexts, 69 375 slot joint questions and 4 529 column questions. Together with scalar cells, contexts, cross-square joints and global measurements these add 163 835 questions per party.

Retain the 6 916 old FMS/refinement/binding forbidden events and old binary synchrony tests. Add all the compiler rules. Accept any answer pair not forbidden by these rules and reject invalid answers. Choose questions independently and uniformly, so every pair has probability Q^{-2}, where $Q = 169\,460$. The predicate is Boolean; masks in `game.json` list the finite answer subsets used by each rule.

In a perfect strategy every pair has positive sampling probability, so each rejecting event is zero. Theorem B gives every old rational row. The restricted old strategy has $B = 0$ by (28), so Theorem A gives (2). Completeness extends the canonical normal realization by Φ_{256}. These separable infinite spaces identify locally with ℓ^2. Finite nonattainment follows from Theorem A.

The cutoff strategies also extend by Φ_{256}. Every new structural rule is exact; only one row-zero event fails. For a row,

$$p(D_0, R_d) = \frac{4L^2}{d^3} \operatorname{Tr}(h_{\mathrm{old}} S^2). \tag{32}$$

The maximally mixed scalar marginal supplies the third power of d, including spectator normalization. For the isometry row $L = 2, d = 16$ and $\operatorname{Tr}(h_{\mathrm{old}} S^2) = 2t_N/75$. Thus

$$1 - \omega_N = \frac{t_N}{9600 Q^2} \longrightarrow 0. \tag{33}$$

This completes the proof of Theorem C.

The known synchronous obstruction. Kim, Paulsen and Schafhauser prove that a synchronous game with a perfect spatial strategy has a perfect finite-dimensional strategy [20, Theorem 3.10 and Corollary 3.11]. On a normal support its synchronized projections reduce each positive finite-dimensional marginal eigenspace, where a maximally entangled strategy remains perfect. The present game retains nonsynchronous FMS and refined questions. Their transports change density eigenvalues. The full strategy cannot be restricted to a single marginal eigenspace. This observation explains the construction; it is not a new obstruction theorem.

9 Entropy near the optimum

Changes in version 1.1.0. The $\varepsilon^{-1/5}$ bound of v1.0.0 remains valid; its rate is a feature of radial truncation. Height cutoffs give a logarithmic upper bound, and Theorem D now proves

a matching-order lower bound for all normal spatial strategies (Appendices B-C; Appendix D adds the exact-algebra law and the extraction limit).

Define

$$E_B(\varepsilon) = \inf\{S(\rho_A) : B \geq -\varepsilon\}, \qquad E_G(\varepsilon) = \inf\{S(\rho_A) : \omega(G) \geq 1 - \varepsilon\}.$$

The infima range over normal spatial strategies. Purifying onto Bob preserves Alice's marginal and all probabilities, so the same infima are obtained with pure states. They need not be attained. The corresponding assertions hold with the parties exchanged.

Theorem D (near-optimal entropy). *For $X \in \{B, G\}$ there are constants $K'_X \geq 1$ and $\varepsilon_0 > 0$ such that, for $0 < \varepsilon \leq \varepsilon_0$, writing $L_\varepsilon = \log_2(1/\varepsilon)$,*

$$E_X(\varepsilon) \geq \frac{1}{18} L_\varepsilon - \frac{16}{3} \log_2(L_\varepsilon + 2) - 200 - \frac{2}{9} \log_2 K'_X,$$
$$E_X(\varepsilon) \leq \frac{2}{3} L_\varepsilon - \frac{2}{3} \log_2 L_\varepsilon + O_X(1).$$

Both inequalities hold for the infimum of either party's marginal entropy over all normal spatial strategies, including mixed states and POVMs. Consequently $E_X(\varepsilon) = \Theta(\log(1/\varepsilon))$.

Proof. Theorem B.1 gives local isometries and the exact domain restrictions needed by Theorem C.1, with every vector defect at most $K_X \varepsilon^{1/4}$. Set $K'_X = \max\{1, K_X\}$. For $K'_X \varepsilon^{1/4} \leq 1$, substitute $L = \frac{1}{4} L_\varepsilon - \log_2 K'_X$ in that rounding theorem and use $L + 2 \leq L_\varepsilon + 2$. The local isometries, pure local ancillas and zero-state padding preserve each original marginal spectrum. If $K'_X \varepsilon^{1/4} > 1$, the displayed lower bound is negative and follows from nonnegativity. The height-and-size construction below proves the upper bound. Infima preserve both bounds. □

The lower constants are explicit but large. The rounding bound is positive only at very small defect (its first positive integer logarithm is 1144). Under the unweakened substitution this already requires $\varepsilon \leq (K'_X)^{-4} 2^{-4576}$; the displayed simplified game bound may require still smaller ε. The leading constant for unrestricted strategies remains open between $1/18$ and $2/3$. Proposition B.6 also shows that this form of actual-state decoder cannot have exponent above $1/2$.

The retained radial result. Put

$$C_L = \frac{24}{25\sqrt{\pi}}, \quad C_S = \frac{64}{45\sqrt{\pi}}, \quad C = C_S C_L^{1/5} = 0.709793116354\ldots.$$

The scalar taper below still gives finite-dimensional strategies with $E_B(\varepsilon) \leq (C + o(1))\varepsilon^{-1/5}$ and $E_G(\varepsilon) \leq (CQ^{-2/5} + o(1))\varepsilon^{-1/5}$, where $Q = 169\,460$. These valid upper bounds are superseded by Theorem D. With the canonical measurements fixed, the power $1/5$ remains optimal among scalar spectral filters of the canonical marginal for both constructions; its proof is retained below.

9.1 Divergence at bounded entropy

Here is the compactness argument, included to identify precisely what it proves. Sort the Schmidt probabilities $p_1 \geq p_2 \geq \cdots$ of any pure state of entropy at most K. Since $p_j \leq 1/j$,

$$\sum_{j>d} p_j \leq \frac{K}{\log_2(d+1)}. \tag{34}$$

Compress each POVM to the (separable) state supports; this preserves all probabilities. If the Schmidt rank is finite, extend by a fixed deterministic answer on the complement in ℓ^2. After local basis changes write each state as $\sum_j \sqrt{p_j}|j,j\rangle$ in common copies of ℓ^2. Uniformly small tails yield a norm-convergent subsequence of these vectors with unit norm. Finite partial entropy sums and lower semicontinuity show that its entropy is at most K.

Bounded effect sets on a separable Hilbert space are sequentially weak-operator compact. Extract a common subsequence for the finitely many effects. Positivity and POVM sums survive the limit. On the first d Schmidt coordinates all probabilities involve only finitely many matrix entries. The tail bound then lets d tend to infinity, uniformly over the measurements. Thus entropy-at-most-K spatial correlations form a compact set in each finite scenario.

If a sequence had vanishing loss and bounded entropy, nearly minimizing pure strategies would have a limit attaining the optimum with finite entropy. Theorems A and C forbid this. Equivalently each finite entropy budget has a strictly positive loss gap. The argument gives no evaluated modulus for that gap.

9.2 Height cutoffs and the logarithmic upper bound

This subsection gives the height-and-size construction proving the upper bound in Theorem D.

Choose the unitary polar completions in Section 6 as follows. For B, cycle each chain $r, rB, \dots, rB^{h-1}$, where r ends in A and has height h, and fix the empty word. For the amplified a, interchange $e_0 \otimes w$ and $e_2 \otimes wA$, fixing the complement. The cycle preserves word size and maximum prefix height. The swap preserves the scaffold weight because $4 \cdot 4^{-(n+1)} = 4^{-n}$. Both are permitted centralizer polar completions. Transport the same choices along the two bridges.

For a prefix-closed, B-closed set S, write

$$Z_S = \sum_{w\in S} 4^{-n(w)}, \quad M_S = \sum_{w\in S} n(w)4^{-n(w)}, \quad b_S = \sum_{\substack{w\in S\\ wA\notin S}} 4^{-n(w)}, \quad q_S = b_S/Z_S.$$

The coefficient operator of the scaffold state restricted to S is diagonal, with entries

$$K_S(f,c,w) = \sqrt{d_c 4^{-n(w)}/(75Z_S)}, \qquad (d_c) = (1,2,2,2,2,4,4,4,4).$$

For a binary projection E, its synchronization loss is $\|EK_S - K_S E\|_2^2$. The B cycles reduce S. Only a swap whose parent is in the A boundary crosses its support. The finite word list contains no label mixing an a token and a B token; labels with two a tokens preserve histories. All core rows are still zero operator identities, and all old FMS and refinement losses vanish. Thus

$$\ell_B(S) = \sum_E \kappa_E \sigma_E, \qquad \kappa_E = 1 + \sum_{r:E\in r} \left(1 + \tfrac{1}{2}\sum_{F\neq I} \alpha_{r,F}^2\right).$$

The exact symbolic crossing calculation in the verification bundle gives 392 active symbols: 78 G, 78 H, 156 complements and 80 triangles. On these symbols the range of κ_E is $[7/2, 221/2]$. Summation gives

$$\ell_B(S) = \Lambda_B q_S, \quad \ell_G(S) = \Lambda_G q_S/Q^2, \quad \Lambda_B = \frac{50808529}{540000}, \quad \Lambda_G = \frac{14844029}{54000}. \tag{35}$$

For the game, each old symbol contributes $49\sigma_E$, each nonzero row slot contributes another $3\sigma_E$, and column synchronization and graph orthogonality supply the remaining terms. The old sign-event formula $\ell_G = \ell_B/Q^2$ is specific to radial filters and is not used here. The exact phase calculation checks arbitrary unit-modulus counter phases, not only selected heights.

The bundle records the exact rational crossing certificate, its independent implementation and the arbitrary-phase audit.

Let S_H consist of histories whose every prefix has height at most H. The mass at current height h is

$$m_h = \frac{H+1-h}{(H+1)2^h}, \qquad 0 \le h \le H.$$

Indeed $m_0 = 1$, $m_h = m_{h-1}/4 + m_{h+1}$ and $m_{H+1} = 0$; the repeated-root recurrence gives the displayed solution. Consequently

$$Z_H = \frac{2H + 2^{-H}}{H+1}, \quad b_H = \frac{2^{-H}}{H+1}, \quad q_H = \frac{1}{H2^{H+1}+1}. \tag{36}$$

For completeness the moment can be calculated without enumerating histories. Let $T_h(x)$ count continuations from height h and put $Z_H(x) = 1 + xT_1(x)$. Then

$$T_h = 1 + xT_{h+1} + \mathbf{1}_{h\ge 2}T_{h-1}, \qquad T_{H+1} = 0.$$

If $Y_h = xT_h'(x)$, differentiation gives the same linear system with inhomogeneous term xT_{h+1}. Solving at $x = 1/4$ gives

$$M_H = xZ_H'(x)\big|_{1/4} = \frac{2H(H-1)}{3(H+1)} + \frac{H(H+5)2^{-H}}{6(H+1)}.$$

The equations uniquely determine the solution because the finite-strip transfer matrix has spectral radius less than one. These formulas can also be verified by substitution in the two linear systems. Including the finite scaffold, the marginal entropy is

$$S_B(H) = \log_2 75 - \frac{8}{5} + \log_2 Z_H + \frac{2M_H}{Z_H}, \qquad S_G(H) = S_B(H) + 8. \tag{37}$$

Write $q = 2^{-H}$ and

$$\frac{M_H}{Z_H} = \frac{H-1}{3} + \delta_H, \quad \delta_H = \frac{q(H+1)(H+2)}{6(2H+q)}, \quad c_\infty = \log_2 150 - \frac{8}{5} - \frac{2}{3}.$$

The inequality $2^{-H}(H+1)^2(H+2) \le 6H$ follows at $H = 1$ by equality and then by comparing successive ratios. Hence $2\delta_H \le 1/(H+1)$, whereas

$$-\log_2(Z_H/2) \ge \frac{2-q}{2(H+1)\ln 2} > \frac{1}{H+1}.$$

It follows that $S_B(H) \le 2H/3 + c_\infty$ and $S_B(H) = 2H/3 + c_\infty + o(1)$.

Here is an integer-rounding check for the refined Bell constant. Choose the least H with $\Lambda_B q_H < \varepsilon$. If $\varepsilon \le \Lambda_B q_7$, then $H \ge 8$. Put $a = \log_2 \Lambda_B$, $x = H-1$, and $L = \log_2(1/(\Lambda_B q_{H-1}))$. The function $y - \log_2 y$ is increasing for $y > 1/\ln 2$, so the worst endpoint of the interval is its upper-loss endpoint. Furthermore

$$\frac{L}{x} = 1 + \frac{1 + \log_2 x - a + e_x}{x}, \qquad 0 < e_x = \log_2(1 + (x2^{x+1})^{-1}) < 0.001.$$

The maximum of $(\log_2 x + 1.001 - a)/x$ over positive real x is $1/(\ln 2\, 2^{a-1.001+1/\ln 2}) < 0.0114$. Therefore

$$S_B(H) - \frac{2}{3}L + \frac{2}{3}\log_2 L \le c_\infty + \frac{2}{3}a + \frac{2}{3}\log_2(1.0114) < 9.35.$$

With the fixed game normalization the same calculation changes only the additive constant. In particular the construction gives

$$E_B(\varepsilon) \le \frac{2}{3}\log_2(1/\varepsilon) - \frac{2}{3}\log_2\log_2(1/\varepsilon) + 9.35,$$
$$E_G(\varepsilon) \le \frac{2}{3}\log_2(1/\varepsilon) - \frac{2}{3}\log_2\log_2(1/\varepsilon) + O(1).$$

Equation (35) is an exact rational computation checked by two implementations; it affects only the constant in these upper bounds, not their order. The lower bound is proved by the decoding and rounding appendices.

Finite-dimensional implementation. The pure height cutoff has infinite Schmidt rank for every $H \ge 2$: all words $A(AB)^k$ survive. It must not be called a finite-dimensional strategy. Instead use

$$S_{H,N} = \{w : \max_{v \preceq w} \mathrm{ht}(v) \le H,\ n(w) \le N\}.$$

This is a finite, prefix-closed, B-closed set. At fixed H its mass and first moment converge to Z_H, M_H exponentially in N; the finite-strip transfer eigenvalue is $\cos^2(\pi/(H+1)) < 1$ for $H \ge 2$, and $H = 1$ is already finite. Its boundary ratio also converges to q_H. Compress each full measurement to the finite Schmidt support; positivity and the POVM sums are preserved, as are all probabilities on that state. The strict loss slack in the choice of H and the strict constant slack above allow a finite N. Explicitly, enumerate increasing N until the rational loss is below ε and the entropy from $Z_{H,N}, M_{H,N}$ is below the displayed bound. Convergence guarantees termination. The exact height-only loss formula is not asserted for this additional cutoff.

The explicit S_4 witness. Starting from S_4, forbid the A edge at surviving height-three words with $n(w) \ge 18$. At $u_0 = AAAABABABA$, which has seven A letters and height four, add the A child and all its continuations of height at most five. The added subtree uses that height-five rule rather than the pruning rule. Exact finite-strip summation gives

$$Z' = \frac{55409916813}{34359738368}, \quad M' = \frac{57550635573}{34359738368}, \quad b' = \frac{1288366711}{103079215104}.$$

Thus $q' = 1288366711/166229750439 < 1/129 = q_4$ and the history entropy decreases from 2.7668185404 to 2.7666935844 bits. Fixed scaffold factors add the same entropy to both states. This checked witness rules out exact optimality of pure height caps; it does not identify the unrestricted entropy frontier.

9.3 A spectral taper and its exact loss

In the canonical realization, let $r = 0, 1, 2$ denote the corner-weight exponent, with multiplicities $m_0 = 1, m_1 = m_2 = 4$. Including the qutrit, each marginal eigenvalue is $2^r 4^{-n}/150$. Define

$$J|\mathrm{frame}, r, w\rangle = (n(w) - r/2)|\mathrm{frame}, r, w\rangle.$$

Every synchronized core projection commutes with J, because J is affine in the logarithm of the canonical marginal. In particular the centralizer unitary completions preserve it. Take

$$f_N(j) = \begin{cases} 1 & j \le 0, \\ (1 - (j/N)^2)_+ & j > 0, \end{cases} \qquad \psi_N^f = \frac{f_N(J_A)\psi_*}{\sqrt{Z_N^f}}.$$

Here ψ_* is the full canonical state. The filter has finite Schmidt rank, supported in $n \le N$. Compressing the original measurements to that support gives valid finite POVMs with exactly the same probabilities; the support need not reduce the FMS signs. Filtering bare word size would fail to preserve centralizer tests, which is why the shifts $r/2$ are retained.

Write $a_n = \mathrm{Cat}_n / 4^n$ and $f_{n,r} = f_N(n - r/2)$. Exactly,

$$Z_N^f = \frac{1}{50} \sum_{n \ge 0} a_n (f_{n,0}^2 + 8 f_{n,1}^2 + 16 f_{n,2}^2) \longrightarrow 1. \tag{38}$$

Core rows are zero operator identities, and their projections remain synchronized after filtering. All activity, flag, retained-event and label bindings remain exact. Each refined measurement remains a joint product with its single label. Only wrong sign/sign events in original or duplicated FMS pairs can lose.

Put

$$D_1(N) = \sum_n a_n [f_N(n) - f_N(n - 1/2)]^2,$$
$$D_2(N) = \sum_n a_n [f_N(n - 1/2) - f_N(n - 1)]^2.$$

In bridge g, put $f_f = f_{n,g-1}$ and $f_m = f_{n,g}$, the filter values on its flag and magic corners. On a fixed word and frame, a forbidden sign pair in bridge g has amplitude

$$\frac{\sqrt{2^{g-1}}\, 2^{-n} (f_f - f_m)}{2\sqrt{150 Z_N^f}}.$$

Its probability, after summing the frame and words, is $2^{g-1} D_g / (200 Z_N^f)$. There are two wrong signs per pair. In each bridge there are four original pairs, twenty-two duplicated Alice pairs and twenty-two duplicated Bob pairs. Summing refined labels gives the coarse event. Consequently the *literal* Bell loss is

$$\ell_B(N) = -B(\psi_N^f) = \frac{12}{25} \frac{D_1(N) + 2D_2(N)}{Z_N^f}. \tag{39}$$

The serialized direct-event audit counts 368 potentially nonzero fine sign records; their labels sum to these 192 coarse events across both bridges. The derivation uses only scalar values of the filter on the J-spectrum. Thus (39) holds for every nonzero Borel scalar filter $f(J)$ with $\mathrm{Tr}(h_* |f(J)|^2) < \infty$, using its corresponding normalization Z and difference sums D_1, D_2; for complex filters the squared differences are $|f_f - f_m|^2$.

For $F(x) = (1 - x^2)_+$ and $a_n \sim n^{-3/2}/\sqrt{\pi}$, half-step difference sums give

$$D_g(N) \sim \frac{N^{-5/2}}{4\sqrt{\pi}} \int_0^1 x^{-3/2} F'(x)^2 \, dx = \frac{2}{3\sqrt{\pi}} N^{-5/2}.$$

To justify the Riemann sum, first restrict to $[\eta, 1]$. The omitted lower interval is bounded by a constant times $\eta^{3/2} N^{-5/2}$, since $F'(x) = -2x$. The finitely many endpoint terms are $O(N^{-4})$ near zero and $O(N^{-7/2})$ at the upper edge. Therefore

$$\ell_B(N) \sim C_L N^{-5/2}.$$

For example $Z_N^f \ge 1/2$ and each half-step difference is at most n/N^2, which also give a nonasymptotic bound $\ell_B(N) \le 5N^{-5/2}$ for $N \ge 1$.

The filtered eigenvalues give, with $0 \log 0 = 0$,

$$S_N = \log_2(150 Z_N^f) + \frac{1}{50 Z_N^f} \sum_n a_n \sum_{r=0}^{2} m_r 2^r f_{n,r}^2 [2n - r - 2 \log_2 f_{n,r}]. \tag{40}$$

Only the $2n$ term contributes to the leading order. All other terms are uniformly bounded using $\sum_n a_n = 2$ and boundedness of $u^2 |\log u|$. Since $Z_N^f = 1 + O(N^{-1/2})$,

$$S_N \sim \frac{\sqrt{N}}{\sqrt{\pi}} \int_0^1 x^{-1/2} (1 - x^2)^2 \, dx = C_S \sqrt{N}.$$

Choosing an integer $N = (C_L/\varepsilon)^{2/5}(1 + o(1))$ with sufficient positive slack proves the Bell upper bound. The $o(1)$ can decay slowly; the leading constant alone is not a finite-loss error estimate. The game extension adds eight bits. Its new row identities remain exact; only the retained old FMS sign events lose. Thus $\ell_G(N) = \ell_B(N)/Q^2$, proving the game constant. Formula (33) is for a different cutoff family and must not be used for these filtered states.

9.4 Restricted lower bound for all spectral filters

Normalize $\mathrm{Tr}(h_* |f(J)|^2) = 1$ and set $f_k = f(k/2)$ for $k = -2, -1, 0, \ldots$. Absolute values preserve entropy and normalization and can only decrease the difference energy, so assume $f_k \geq 0$. The shell masses and degeneracies are

$$\begin{aligned} \mu_{2m} &= (a_m + 16 a_{m+1})/50, & d_{2m} &= 3(\mathrm{Cat}_m + 4\,\mathrm{Cat}_{m+1}), \quad m \geq 0, \\ \mu_{2m-1} &= 8 a_m / 50, & d_{2m-1} &= 12\,\mathrm{Cat}_m, \quad m \geq 0, \\ \mu_{-2} &= 16/50, & d_{-2} &= 12. \end{aligned}$$

Thus $\sum_k \mu_k f_k^2 = 1$ and

$$S = H((\mu_k f_k^2)_k) + \sum_k \mu_k f_k^2 \log_2 d_k \geq \tfrac{1}{2} \sum_{k \geq 0} k \mu_k f_k^2. \tag{41}$$

The elementary bounds $\mathrm{Cat}_m \geq 2^{m-1}$ for $m \geq 1$ and $a_m \geq [2(m+1)^{3/2}]^{-1}$ imply $\log_2 d_k \geq k/2$ and $\mu_k \geq [100(k+3)^{3/2}]^{-1}$ for $k \geq 0$.

For a normalized filter, (39) becomes $\ell = (12/25) D$, where

$$D = \sum_{n \geq 0} a_n [(f_{2n} - f_{2n-1})^2 + 2(f_{2n-1} - f_{2n-2})^2].$$

This is the energy of a connected weighted half-line. The sum of inverse conductances from -2 to L is at most $2(L+3)^{5/2}$. Cauchy–Schwarz along the path gives

$$|f_k - f_{-2}| \leq \sqrt{2D(L+3)^{5/2}}, \qquad -2 \leq k \leq L.$$

For small D choose $L = \lfloor (32D)^{-2/5} - 3 \rfloor$, so the right side is at most $1/4$. If $f_{-2} \geq 1/2$, then $f_k \geq 1/4$ through L. Equation (41) bounds the entropy below by

$$\frac{1}{3200} \sum_{k=1}^{L} \frac{k}{(k+3)^{3/2}} \geq \frac{\sqrt{L}}{25600} \quad (L \geq 6).$$

If $f_{-2} < 1/2$, the low-shell amplitudes are at most $3/4$, so at least $7/16$ of the normalized mass lies above L. Then $S \geq 7L/32$. Both cases give $S \geq c \ell^{-1/5}$ for an absolute positive c.

For complex filters apply this argument to their absolute values; their energy is no larger. For filters $f(J) \otimes I$ of the game's canonical marginal, the same compiler extension gives $\ell_G = \ell_B/Q^2$ and $S_G = S_B + 8$. Together with the taper this proves restricted optimality for both B and G and proves the retained restricted radial statement.

The height upper bound rules out every positive unrestricted lower power. The matching-order lower bound instead uses the actual-state decoding and rounding appendices; it makes no assumption that a near-optimal strategy is a scalar spectral deformation.

10 Discussion

The finite game forces a normal pure factor with a critical Catalan tail. At word size n there are Cat_n Schmidt probabilities equal to $4^{-n}/2$. Their total mass is of order $n^{-3/2}$, while their entropy contribution is of order $n^{-1/2}$. The masses sum to one and the entropy diverges. A finite auxiliary system implements the test but does not cause the divergence. Forthcoming work gives a second finite game, built from a context-free grammar with critically weighted counts of order $n^{-5/4}$, whose canonical radial entropy exponent is $1/3$ instead of $1/5$. These are radial-family exponents; height cutoffs show that the latter is not the unrestricted rate for the present game.

The I_{3322} contrast. Coladangelo's attaining state [6, Definition 10.12 and Theorem 10.14] has finite entropy. This is a consequence of its construction, not an entropy claim made in that paper. Here is the tail argument. Its Schmidt vector λ satisfies a bounded Jacobi equation $J_\infty(c)\lambda = q_*\lambda$, with monotone labels of total variation at most two and $q_* > 1/4$. The half-line version of the estimate in its Lemma 10.10 bounds the tail restriction by $1/4 + \mathrm{TV}_N/2$. Choose N so this lies below q_*. The resolvent equation on that half-line has only a boundary source. The Combes–Thomas exponential-weight estimate [9], using bounded off-diagonal entries, gives $\sum_i e^{\mu|i|}\lambda_i^2 < \infty$ for some $\mu > 0$, on both tails. Gibbs' inequality then gives $H((\lambda_i^2)_i) < \infty$. Thus dimension nonattainment need not force entropy divergence. We do not need a claim about every I_{3322} optimum.

The sharp leading coefficient for unrestricted strategies remains open between $1/18$ and $2/3$. The exact-local-algebra coefficient is $2/3$ in the squared-mirror convention, as proved in Appendix D; that hypothesis is stronger than approximate game play. Improving decoding or row repair could narrow the unrestricted gap. Smaller question and answer sets are also desirable. Any reduction must preserve actual products and common calibrated transports.

The theorem concerns normal states on tensor products. We do not classify nonnormal states or nonspatial commuting strategies. Full-state self-testing has a different quantitative limitation: Proposition D.3 rules out every positive Hölder modulus for extraction of the whole Catalan state. Minimal alphabets and the optimal unrestricted leading constant remain open. These boundaries leave the distinction among dimension, entropy and type III structure visible.

How this work was done

The authors selected the questions and proof standards, directed the research, and take responsibility for every claim. AI systems from more than one developer assisted with exploration, derivation, programming and drafting.

The mathematical evidence consists of the proofs in this paper and the explicitly cited external theorems. The ancillary programs provide reproducible finite checks. The proofs include the coarse-cover construction in Appendix B and the isometric extension on the physical complement in Appendix C. The two Bell evaluators and two game evaluators use

separate implementations; their provenance and scope are stated in Appendix A. Historical checks of the retained radial result include exact scalar values and constants and a literal full-functional evaluation on filtered states. The latter reuses the cross-checked Bell measurement builder but independently constructs the filtered state, contracts the coefficients and compares the result with (39).

Finite exact evaluations test the exported predicates, coefficient tables, projection identities and cutoff boundary formulas. They do not prove infinite-dimensional attainment, universal entropy divergence, or a robustness theorem by sampling. The asymptotic conclusions rest on the written proofs. This preprint has not been peer reviewed.

A Reproducibility

The accompanying `verification/` directory contains both archives, two independent Bell evaluators, two game evaluators, the literal verifier checks, scalar near-optimality checks, recorded outputs and fresh execution logs. Its README gives commands, dependencies, provenance and limitations. `SHA256SUMS` identifies every bundled file apart from itself. Paths in public files are relative.

The Bell producer-side evaluator reads every literal coefficient and uses exact arithmetic in $\mathbb{Q}(i, \sqrt{2}, \sqrt{3})$. The review evaluator reconstructs the measurements independently in $\mathbb{Q}(\zeta_{24})$, using a different field representation. Both give

$$B_0 = -2/75, \qquad B_1 = -2/375, \qquad B_2 = -2/825.$$

Each finds only the isometry row failing, the predicted full boundary matrix, and zero direct FMS/refinement loss.

The producer-side game checker reuses the producer-side Bell evaluator's matrix kernel. It is independent of the game producer, but its old measurements are not a second independent implementation. The game review supplies that implementation: it independently rebuilds every old projection, FMS/refined measurement and forbidden event, then reconstructs the new gadgets from predicate data. Both game evaluators give the exact losses

N	Local dimension	$1 - \omega_N$
0	6 912	1/275680239360000
1	13 824	1/1378401196800000
2	27 648	1/3032482632960000

These agree with (33). The public verifier is checked on 10 076 128 column-graph and spectator answer cases, together with the complete pair-rule index. Independent brute-force and changed-column controls accompany the reduced game evaluator. The checks evaluate rule classes and exact structural zeros, not a dense enumeration of every answer pair in the full question square.

The scalar checker audits the literal direct-event classes, performs thirty-two exact rank-one contractions, checks rational taper losses, and evaluates positive scalar sums through $N = 10^6$. A second engine for (39) contracts all 73 769 literal coefficients on filtered states at word cutoffs $M = 1, 2$, including the tapers and mixed-sign rational filters. It reuses the cross-checked Bell measurement builder; its state construction, contraction, aggregation and comparison are separate. The scripts and recorded and fresh outputs are bundled with hashes. Floating-point asymptotic illustrations do not replace the analytic limits or establish a universal entropy lower rate without the written argument.

The functional archive SHA-256 is

```
bb5c0bf32e9a7175bef73d9b156a5c6f3
e3a0487462c90a9138e82f809927aaf.
```

The game archive SHA-256 is

```
f29237d3920afe1b4ff9a3749b6893416
9d8109bbaa0fa19dbbcc34f334824d2.
```

The line breaks above split each hash after thirty-two hexadecimal characters. Neither archive was regenerated for packaging. No proof-assistant certification is claimed.

Version 1.1.0 checks. The portable driver `verification/v11/run_review_checks.py` runs the retained fifteen checks and eight appendix checks in a new unique output directory. It resolves input archives relative to the extracted bundle, limits BLAS threads, and fails on a nonzero exit or a failed numerical/exact assertion. The new checks cover normal magic-square extraction, mixed and pure sign mirrors, cap/threshold/polar/cover inequalities, independent path enumeration, exact rational strip solutions through height 80, perturbed band-and-row models, and cutoff arithmetic. The bundle checksums record the released scripts and outputs. Finite checks test formulas and implementations; the dimension-independent conclusions rest on the proofs.

B Polynomial decoding on the actual state

B.1 Statement, conventions, and scope

We use the finite tests of Sections 4–8. The graph, placement, parameter, cover, and polynomial rows are those specified there. Constants may depend on this fixed finite list, its coefficients, and its question distribution. They never depend on Hilbert-space dimension, a Schmidt eigenvalue, or a state-sector mass.

Theorem B.1 (Actual-state decoding)**.** *There is a constant K such that every normal spatial strategy for $\mathcal{B}$ of loss ε admits local isometries V_L, V_R and operators A_s, B_s, C_s, P_s, Q_s on their target spaces, $s = L, R$, satisfying*

$$\|A_s\|, \|B_s\| \le 1, \qquad 0 \le C_s \le I, \qquad P_s^2 = P_s = P_s^*, \quad Q_s^2 = Q_s = Q_s^*.$$

For every purification Ω, put $\widehat{\Omega} = (V_L \otimes V_R \otimes I_E)\Omega$. Each of

$$\begin{aligned} &A_s^*A_s - I, \quad A_sA_s^* + B_sB_s^* + P_s - I, \quad B_s^*B_s + P_s + Q_s - I, \\ &C_sP_s - P_s, \quad C_sQ_s - Q_s/2, \quad C_sA_s - A_sC_s/2, \quad C_sB_s - 2B_sC_s, \quad P_sQ_s \end{aligned}$$

has vector norm at most $K\varepsilon^{1/4}$ on $\widehat{\Omega}$. The same holds for the five mirrors

$$\tfrac{1}{2}A_L - A_R^*, \quad A_L^* - \tfrac{1}{2}A_R, \quad B_L - B_R^*, \quad B_L^* - B_R, \quad C_L - C_R. \tag{42}$$

The same conclusion, with a different fixed K and the same exponent $1/4$, holds for the game G.

The proof retains the scalar M_3 frame in the root multiplicity space. It does not claim a quantitative tensor identification of every original label on all 27 frame/corner blocks. Those stronger identifications are unnecessary for Theorem B.1. The final operators are one common root table amplified to the nine scaffold sectors, evaluated on the *actual* transformed state.

The isometries and decoded operators are functions only of the measurement operators and fixed padding choices; they do not depend on the choice of purification. The local state

spaces can be assumed separable for the proof: take the closed span of the local state supports under all words in the finitely many measurements and their adjoints. This is a separable reducing space. Arbitrary zero-state complements can instead be retained and padded to a common cardinal dimension. Neither choice changes any estimate.

First dilate the finitely many local POVMs simultaneously to PVMs with one common local embedding and pure local ancillas. All tested probabilities and marginal spectra are preserved. For $\mathcal{B}$, the SOS identity in the construction gives, with $r = C_0\sqrt{\varepsilon}$, the following data: every rational row has vector residual at most r on each party; each synchronized binary projection has mismatch norm at most r; every forbidden-event amplitude is at most r. Fixed finite sums of events merely change C_0.

Throughout, $O(r^a)$ means a bound by a fixed constant times r^a. The proof supplies a finite recurrence for all such constants. It suffices to treat small r; increasing K covers the remaining losses by bounded trivial choices of the decoded operators.

B.2 A calculus that supplies every suffix mirror

Lemma B.2 (Word mirrors)**.** *Suppose local operators x_α have already proved opposite-party mirrors y_α with*

$$\|(x_{\alpha,L} - y_{\alpha,R})\Omega\| \le e_\alpha, \qquad \|x_\alpha\| \le M_\alpha, \qquad \|y_\alpha\| \le N_\alpha.$$

For a word $w = x_1 \cdots x_k$, its mirror is $w^\flat = y_k \cdots y_1$, and

$$\left\|(w - w^\flat)\Omega\right\| \le \sum_{j=1}^{k} \left(\prod_{i<j} M_i\right)\left(\prod_{i>j} N_i\right) e_j. \tag{43}$$

Consequently, for a bounded local operator or local-space map R,

$$\|Rw\Omega\| \le \left\|w^\flat\right\| \|R\Omega\| + \|R\| \left\|(w - w^\flat)\Omega\right\|. \tag{44}$$

Proof. Replace the rightmost left letter by its opposite mirror and commute that mirror past all remaining left letters. Repeat, working right to left. This gives (43). In (44), replace $w\Omega$ by $w^\flat\Omega$ and commute the opposite-party product past R. This argument works equally for rectangular maps R between local spaces. □

For raw synchronized projections p_α, take $y_\alpha = p_{\alpha,R}$, $M_\alpha = N_\alpha = 1$, and $e_\alpha = r$. If $p = \sum_w c_w w$ is a fixed raw polynomial, define

$$a(p) = \sum_w |c_w|, \qquad \ell(p) = \sum_w |c_w|\,|w|.$$

Then

$$\left\|p^\flat\right\| \le a(p), \quad \left\|(p - p^\flat)\Omega\right\| \le \ell(p)r, \quad \|Rp\Omega\| \le a(p)\,\|R\Omega\| + \|R\|\,\ell(p)r. \tag{45}$$

For real-coefficient raw polynomials, $p^\flat = p_R^*$. This formula supplies the mirror for every graph word, every matrix unit, every proper-prefix word, every parameter polynomial, and every suffix in the raw calculations below; no extracted frame or calibrated sign is needed.

Write $p \simeq q$ if both $p - q$ and its adjoint have $O(\sqrt{r})$ vector norm. For fixed raw polynomials this is a congruence under addition, multiplication, and adjoint. The initial seeds below are self-adjoint rows and *both ordered* projection overlaps. We never infer an adjoint estimate from the norm of an arbitrary residual alone. A term aRb with $\|R\Omega\| \le c_R\sqrt{r}$ is bounded, for $r \le 1$, by

$$\|aRb\Omega\| \le \left[a(a)a(b)c_R + a(a)a(R)\ell(b)\right]\sqrt{r}. \tag{46}$$

Summing (46) over an expanded finite algebraic derivation computes its constant.

Lemma B.3 (Projection-partition estimate)**.** *For synchronized projections $P_1, \dots, P_n$ with $\|(\sum_i P_i - I)\Omega\| \le \eta$,*

$$\sum_{i \neq j} \|P_i P_j \Omega\|^2 \le \eta^2 + \eta + 2n(n-1)r. \tag{47}$$

Proof. For $\phi(X) = \langle \Omega, X\Omega \rangle$ and any bounded local X, synchronization gives $|\phi(XP) - \phi(PX)| \le 2\|X\| r$: move the endpoint P to the opposite party in each expression and commute it past X. Thus $\phi(P_j P_i P_j)$ differs from $\phi(P_j P_i)$ by at most $2r$. Put $R = \sum_i P_i - I$. Summing gives $\sum_{i \neq j} \phi(P_i P_j) = \phi(R^2) + \phi(R)$, which proves the assertion. □

B.3 Raw graphs, triangles, words, and prescribed parameters

The complete partitions needed in the graph presentation are

$$(F_1, F_2, F_3), \quad (G, K_G, F_k), \quad (H, K_H, F_k),$$
$$(E_i, N_i, I - F_i), \quad (E_1, N_1, E_2, N_2, E_3, N_3),$$
$$(J_{11}, J_{10}, J_{01}, J_{00}, F_k).$$

Here E_i, N_i denote the projection *symbols* in the parameter-binding rules, before identifying their products. Their sum rows have residual at most $5r$, and each list has at most six terms. Lemma B.3 bounds every ordered overlap by $10\sqrt{r}$. Hence support, orthogonality, and all their adjoints are available in the raw congruence.

B.3.1 Graphs

For one graph on i, j, put

$$S = F_i + F_j, \quad Z = F_i - F_j, \quad X = 2G - S, \quad Y = 2H - S.$$

The partition seeds give

$$S^2 \simeq S, \quad Z^2 \simeq S, \quad SX \simeq XS \simeq X, \quad SY \simeq YS \simeq Y, \quad X^2 \simeq Y^2 \simeq S.$$

For example $G(I - S) \simeq 0$ follows from the partition containing G, K_G, F_k and the F cover, so $X^2 = 4G - 2GS - 2SG + S^2 \simeq S$. The row $5H + S = 3F_i + 4G$ gives $Y \simeq (3Z + 4X)/5$. Squaring and subtracting yields $\{Z, X\} \simeq 0$. Its i, i and j, j sandwiches give $F_i X F_i \simeq F_j X F_j \simeq 0$. Therefore, with

$$t_{ij} = F_i X F_j, \qquad t_{ji} = t_{ij}^*,$$
$$X \simeq t_{ij} + t_{ji}, \qquad t_{ij} t_{ji} \simeq F_i, \qquad t_{ji} t_{ij} \simeq F_j. \tag{48}$$

These statements apply to every labelled graph independently. All sandwiches and squares use (45). Also $\|t_{ij}\| \le 2$, since $-2I \le 2G - F_i - F_j \le 2I$.

B.3.2 Triangles and placements

For a triangle (U, V, W), put

$$K = I + t_{12}^U + t_{21}^U + t_{23}^V + t_{32}^V + t_{13}^W + t_{31}^W.$$

Its row gives $3D \simeq K$. Since $D^2 = D$, $K^2 \simeq 3K$. The $1, 3$ sandwich has only the diagonal contributions $2t_{13}^W$ and the path $t_{12}^U t_{23}^V$, hence

$$t_{12}^U t_{23}^V \simeq t_{13}^W. \tag{49}$$

The $3,1$ sandwich gives the adjoint assertion separately. The identity triangle gives the reference matrix-unit laws, with $t_{ii} = F_i$. The placement triangles $(U, 1, U)$ and $(1, U, U)$ identify the same label in all positions.

Define precisely the raw global label

$$U_{11}^{\mathrm{raw}} = t_{12}^{U} t_{21}, \qquad U^{\mathrm{raw}} = \sum_{i=1}^{3} t_{i1} U_{11}^{\mathrm{raw}} t_{1i}. \tag{50}$$

Matrix-unit multiplication and (49) give

$$\begin{gathered} (U^{\mathrm{raw}})^* U^{\mathrm{raw}} \simeq I, \quad U^{\mathrm{raw}} (U^{\mathrm{raw}})^* \simeq I, \quad [U^{\mathrm{raw}}, t_{ij}] \simeq 0, \\ t_{ij}^{U} \simeq U^{\mathrm{raw}} t_{ij} \simeq t_{ij} U^{\mathrm{raw}}, \qquad U^{\mathrm{raw}} V^{\mathrm{raw}} \simeq W^{\mathrm{raw}} \end{gathered} \tag{51}$$

whenever the corresponding product triangle is present. Inverse triangles give the inverse/adjoint labels. Induction over the specified proper-prefix construction proves (51) for every word in the finite list. This is an induction over actual products of the same tokens, not a fresh interpretation for each word. The elementary bound $\|U^{\mathrm{raw}}\| \le 4 + 16 + 16 = 36$ holds before any repair.

Here are the reductions behind these assertions. Write $u = U_{11}^{\mathrm{raw}}$. Then

$$u^* u = t_{12} (t_{12}^{U})^* t_{12}^{U} t_{21} \simeq F_1, \qquad u u^* \simeq F_1.$$

Multiplying the sums in (50) cancels $t_{1i} t_{j1} \simeq \delta_{ij} F_1$ and yields both unitary identities. Also $U^{\mathrm{raw}} t_{ij} \simeq t_{i1} u t_{1j} \simeq t_{ij} U^{\mathrm{raw}}$, and $U^{\mathrm{raw}} t_{12} \simeq u t_{12} \simeq t_{12}^{U}$. The placement triangles give $t_{13}^{U} \simeq t_{12}^{U} t_{23}$ and $t_{23}^{U} \simeq t_{21} t_{12}^{U} t_{23}$, proving the other placements and their adjoints. For a product triangle, substitute these placements in (49) to obtain $(U^{\mathrm{raw}} V^{\mathrm{raw}} - W^{\mathrm{raw}}) t_{13} \simeq 0$. Multiply by t_{31} and transport the resulting F_1 corner through t_{i1}, t_{1i}; commutation with the reference units gives the global product equation.

Sandwich a rational graph row $2 \sum_w q_w G_{12}(U_w) - (\sum_w q_w)(F_1 + F_2) = 0$ by F_1, F_2, multiply by t_{21}, and transport to the three diagonal corners with t_{i1}, t_{1i}. It gives $\sum_w q_w U_w^{\mathrm{raw}} \simeq 0$. Substitute the previously proved word identities. This proves every prescribed polynomial input, and its adjoint, at $O(\sqrt{r})$.

B.3.3 The parameters are the specified physical events

For a prescribed projection E, the six-part partition and the rows $\sum_i E_i = E$ imply

$$E_i \simeq E F_i \simeq F_i E.$$

The four J projections form a partition of $F_i + F_j$. Their two marginal rows give

$$(E_i + E_j) G_{ij}(1) \simeq J_{11} \simeq G_{ij}(1)(E_i + E_j),$$

$$J_{00} \simeq (S - E_i - E_j)(S - G_{ij}(1)).$$

Consequently $[E, F_i] \simeq [E, G_{ij}(1)] \simeq [E, t_{ij}] \simeq 0$, and

$$2 G_{ij}(s_E) - S \simeq \big(2(E_i + E_j) - S\big)\big(2 G_{ij}(1) - S\big).$$

Taking its off-diagonal sandwich and using (50) proves

$$t_{ij}^{s_E} \simeq (2E - I) t_{ij}, \qquad s_E^{\mathrm{raw}} \simeq 2E - I. \tag{52}$$

Thus all thirteen physical parameter events, including the magic axes and corners, occur in the same raw algebra. The entire raw stage is $O(\sqrt{r})$, stronger than $O(r^{1/4})$.

B.4 Normal row–column magic-square rigidity

We give the needed arbitrary-Hilbert-space statement directly. The finite-sum representation method is the same general mechanism as state-dependent Gowers–Hatami stability [35, 15] and the state-dependent stability argument in [7, Lemma 4.7]; the proof here starts with row–column measurements and does not invoke its finite-dimensional equation–variable self-testing theorem as a black box.

Lemma B.4 (Normal magic-square lemma)**.** *For projective row–column magic-square measurements on a unit vector $\Omega \in \mathcal{H}_L \otimes \mathcal{H}_R \otimes \mathcal{H}_E$, write M_{ij}, N_{ij} for the cell reflections and $h = \max_{ij} \|(M_{ij} - N_{ij})\Omega\|$. There are local isometries, trivial on E, extracting the standard real magic-square observables P_{ij} such that*

$$\left\|\widehat{\Omega} - \Phi_4 \otimes \eta\right\| \le 1250h, \qquad \|(V_s M_{ij} - (P_{ij} \otimes I)V_s)\Omega\| \le 632h,$$

with N_{ij} on Bob. A word of length ℓ has intertwining error at most $[632\ell + \ell(\ell-1)]h$. The POVM version follows by simultaneous local Naimark dilation. The unnormalized version has the same bounds in terms of unnormalized mismatch norms, with no inverse state mass.

Proof. Every Alice word in the nine reflections has the reverse Bob word as a norm-one mirror, with error at most its length times h. Write Alice's square as

$$\begin{pmatrix} a & b & u \\ c & d & v \\ e & f & w \end{pmatrix}.$$

The row equations hold exactly: $u = ab$, $v = cd$, $w = ef$, and $[a, b] = [c, d] = [e, f] = 0$. Transfer to Bob's exact column contexts to get

$$\|(e - ac)\Omega\|, \|(f - bd)\Omega\|, \|(w + uv)\Omega\| \le 3h, \quad \|[a, c]\Omega\|, \|[b, d]\Omega\| \le 4h.$$

For instance $ac\Omega$ transfers to $N_{21}N_{11}\Omega = N_{31}\Omega$ with error $2h$, and N_{31} transfers back with error h.

The expansion $(ac - e)bd + e(bd - f)$ and the already proved mirror of bd give $\|(acbd - ef)\Omega\| \le 10h$; the reversed expression gives $\|(bdac - fe)\Omega\| \le 10h$. Therefore

$$D = acbd + abcd, \quad C = acbd - bdac, \qquad \|D\Omega\| \le 13h, \quad \|C\Omega\| \le 20h.$$

Exact row commutations give $D = a(cb + bc)d$ and $D - C = b(ad + da)c$. Cancel the left unitaries and remove the rightmost d or c using its norm-one mirror. Thus $\|\{b, c\}\Omega\| \le 15h$ and $\|\{a, d\}\Omega\| \le 35h$. Transfer these degree-two identities to Bob. On both parties, the six Pauli commuting/anticommuting relations have error at most $40h$.

Let Γ be the order-32 group of bit tuples (t, p, q, r, s) with multiplication

$$(t, p, q, r, s)(t', p', q', r', s') = (t + t' + sp' + rq',\ p + p',\ q + q',\ r + r',\ s + s'),$$

with addition modulo two. Its normal forms are $J^t a^p b^q c^r d^s$. Define actual unitaries

$$f(t, p, q, r, s) = (-1)^t a^p b^q c^r d^s.$$

A product of two normal forms requires at most six swaps to sort: three for the second a, two for the second b, one for the second c. The right suffix of a swap has length at most six, so its previously derived mirror bounds the swap error by $40h + 12h$. Consequently

$$\|(f(g)f(k) - f(gk))\Omega\| \le 312h. \tag{53}$$

Squared letters cancel exactly; $f(gJ) = -f(g)$ exactly.

Define the actual isometry

$$V_s\xi = 32^{-1/2} \sum_{g\in\Gamma} |g\rangle \otimes f_s(g)\xi.$$

For the right regular representation $R(k)|g\rangle = |gk^{-1}\rangle$, orthogonality of the register and (53) give $\|(R(k)V_s - V_s f_s(k))\Omega\| \le 312h$. The image of V_s lies exactly in the central-minus subspace, since $R(J)V_s = -V_s$. That part of the group algebra is $M_4(\mathbb{C})$: take $a = Z \otimes I$, $b = I \otimes Z$, $c = I \otimes X$, $d = X \otimes I$; their sixteen monomials are a Hilbert–Schmidt orthogonal matrix basis. A fixed finite-register unitary identifies the central-minus regular representation with these matrices tensored with a multiplicity space.

After both maps, the four matching-generator defects are at most $625h$. The four stabilizers $P(g)_L P(g)_R$ commute, and their common $+1$ space is exactly $\mathbb{C}\Phi_4$. For $Q = |\Phi_4\rangle\langle\Phi_4| \otimes I$,

$$\left\|(I-Q)\widehat{\Omega}\right\|^2 \le \tfrac{1}{4}\sum_g \left\|(I - P(g)_L P(g)_R)\widehat{\Omega}\right\|^2 \le (625h)^2.$$

Normalize the comparison junk vector in $Q\widehat{\Omega}$; the distance is at most $1250h$. If this bound is at least two, an arbitrary unit junk vector suffices. This normalization is not a physical operation.

The nine cell group elements are $a, b, ab, c, d, cd, ac, bd, -abcd$. Alice's cells differ from their normal forms by at most $3h$. Bob's differ by at most $320h$: use the mirror, the Alice comparison, and (53) to replace $f(g)^*$ by $f(g)$ for these involutory group elements. Adding $312h$ gives $632h$. Telescope words with the already proved raw reverse-word mirrors to get the stated length bound. Every outcome projector is a fixed polynomial in its context's cells. The fixed real unitary $H_2 \otimes H_2$, with H_2 the Hadamard matrix, converts this Pauli square to the magic square of Section 4, preserves Φ_4, and gives the specified overlaps $\langle c_x, r_y\rangle = (-1)^{xy}/\sqrt{2}$. For an unnormalized vector, apply the normalized statement and multiply by its norm; the mass cancels. The zero vector is immediate. □

B.5 One actual pair of isometries and the two comparison blocks

Set $u = \sqrt{r}$. All constructions in this section have error $O(u)$.

B.5.1 Mirrors before compression

In each original FMS row or column, let T_{ij} be its signed nonflag cell sum, zero on flag and inactive answers. A checked pair gives $\|(T_{ij,L} - T_{ij,R})\Omega\| = O(r)$: its squared value is zero on every accepted answer pair and bounded by four on every rejected pair. The nonflag projectors have the same estimate. Nonflag projectors of distinct contexts in the same bridge agree on the state by comparison through one opposite context.

Every fine nonflag outcome D_a already has a bounded mirror: if α is its context's nonflag projection, then

$$D_a = \alpha \prod_{j=0}^{2} (I + a_j T_{ij})/2.$$

Use the opposite nonflag projector and the reversed opposite cell factors. Each factor is a contraction, so the mirror has norm at most one and error $O(r)$. Flag and inactive projections have their direct bound mirrors. Sanitizing flag/inactive answers to one fixed valid magic triple adds finitely many such terms, still with fixed bounded mirrors. This observation is needed for *output* leakage in the next step.

B.5.2 Round only three coarse sectors

Use one bridge-1 context on each party. Let F_0 be its flag projection, F_1 its nonflag projection, and $P = F_0 + F_1$ its exact noninactive projection. The flag binding identifies F_0 with the prescribed root event at $O(r)$. Let $Q = F_2$ be bridge 2's nonflag projection (Alice's R_0, Bob's C_0). All three projections have already established bounded opposite mirrors. The polynomial cover, reduced by the raw parameter identities, gives

$$R = P + Q - I = F_0 + F_1 + F_2 - I, \qquad \|R\Omega\| = O(u), \qquad \|(F_{j,L} - F_{j,R})\Omega\| = O(r).$$

The cover is a polynomial input, not a linear partition row; its error here is $O(u)$. The exact identities

$$PQ = P(P + Q - I) = PR, \qquad QP = Q(P + Q - I) = QR$$

give $\|PQ\Omega\|, \|QP\Omega\| = O(u)$ directly. Since $F_0, F_1 \le P$, the two orders of each overlap with Q are $O(u)$: $F_iQ = F_iPQ$, and $QF_i = QPF_i$; in the latter expression carry the suffix F_i by its previously proved bounded mirror. Also $F_0F_1 = F_1F_0 = 0$ in this context. Thus every ordered overlap $F_kF_j\Omega$, $k \neq j$, is $O(u)$.

Define $T\xi = \sum_{j=0}^{2} |j\rangle \otimes F_j\xi$. Extend its polar partial isometry to an isometry V by local zero-state padding. Since $T^*T = \sum_j F_j$, spectral calculus gives

$$\|(V - T)\Omega\| \le \|(I - T^*T)\Omega\| = O(u).$$

For the exact coordinate projections π_j,

$$\|(\pi_jT - TF_j)\Omega\|^2 = \sum_{k \neq j} \|F_kF_j\Omega\|^2 = O(u^2).$$

The other terms in $\pi_jV - VF_j$ are $O(u)$; the term $(V - T)F_j\Omega$ uses the already available mirror of F_j. Consequently $\|(\pi_jV - VF_j)\Omega\| = O(u)$.

Apply these local isometries and call the actual state Ω_1. The exact matched-sector projection is $P_{\mathrm{mat}} = \sum_j \pi_{j,L}\pi_{j,R}$, and

$$\|(I - P_{\mathrm{mat}})\Omega_1\|^2 = \tfrac{1}{2} \sum_j \|(\pi_{j,L} - \pi_{j,R})\Omega_1\|^2 = O(r).$$

Its components $\Psi_j = \pi_{j,L}\pi_{j,R}\Omega_1$ are comparison vectors only. The physical state is never postselected.

B.5.3 Compress, dilate, and extract each magic sector

Transport every original measurement by the isometries and extend it as a PVM on the zero-state complement. For a row's nonflag projection α, $\|(\alpha - F_g)\Omega\| = O(r)$. If a tested nonflag pair has $D\alpha = D$, then

$$D_LE_RF_{g,L}\Omega = D_LE_R\Omega + D_LE_R(F_{g,L} - \alpha_L)\Omega = O(r).$$

This uses opposite-party commutation and bounded prefixes, not a local suffix rule for a bipartite residual. Flag/inactive amplitudes on $F_g\Omega$ are $O(r)$ by the same identity with $D\alpha = 0$. On Ψ_g, these amplitudes are therefore $O(u)$. Compress each local measurement to π_g, relabel flag/inactive answers to one fixed valid triple, and perform a common local Naimark dilation *inside that sector*. The unnormalized magic-game loss is $O(u^2)$. On Ψ_g each cell mismatch satisfies $h^2 = 4\,\mathbb{P}(\text{cell disagreement})$, with both quantities unnormalized.

Here is the required action comparison. Each sanitized outcome projection or cell reflection X commutes with its own context's α and has the bounded mirror just proved. Thus

$[F_g, X]\Omega = [F_g - \alpha, X]\Omega = O(r)$ by Lemma B.2. Apply the coarse intertwining also to $X\Omega$, using that previously established fine-outcome mirror of X. This gives $[\pi_g, \widehat{X}]\Omega_1 = O(u)$, hence $\left\|(I - \pi_g)\widehat{X}\Psi_g\right\| = O(u)$. For an original projection D, its compression $Q = \pi_g D \pi_g$, and a Naimark projection $\widetilde{D}$ with embedding J,

$$\left\|\widetilde{D}J\xi - JQ\xi\right\|^2 = \left\langle \xi, (Q - Q^2)\xi \right\rangle = \|(I - \pi_g)D\xi\|^2, \qquad \xi \in \pi_g\mathcal{H}. \tag{54}$$

The reflection version is $\|\xi\|^2 - \|\pi_g X\xi\|^2$. Thus the action comparison is $O(u)$, with *no new square root*.

Apply Lemma B.4 in its unnormalized form. Its error is $O(u)$ independently of $\|\Psi_g\|$; for a zero sector choose arbitrary embeddings. Combine the two sector isometries and the root embedding by local direct sum. The resulting *one pair* of physical isometries gives an actual state $\widehat{\Omega}$ and a reference vector

$$\Psi = \Psi_0 + \Phi_4 \otimes \eta_1 + \Phi_4 \otimes \eta_2, \qquad \left\|\widehat{\Omega} - \Psi\right\| = O(u). \tag{55}$$

All junk vectors may contain E, which no map has touched. Pad each magic multiplicity space K_g coherently across all four corners $m_{g,ab}$, using the same added summand in each corner, so that the exact magic matrix units extend on the padding. Original corner and positive-axis events are within $O(u)$, in state action, of the specified exact magic projectors. This follows from (54), outcome rigidity, and their binary event bindings. Outside the relevant coarse sector, an outcome $D_a \le \alpha$ satisfies $D_a(I - F_g)\Omega = D_a(\alpha - F_g)\Omega = O(r)$. Replacing F_g by π_g and its state component by Ψ_g costs $O(u)$, proving the required full-state estimate. On the opposite party use the original binary mirror and the exact real-projector matching in Φ_4.

In particular the flag of bridge 2 is the *same exact* $m_{1,00}$ corner of bridge 1. We now have nine exact corner projections partitioning the local spaces. No scalar M_3 was extracted. After every local isometry the old measurements and binary projections are conjugated and extended as PVMs on the zero-state complement. Their previously proved raw polynomial relations transfer to Ψ with error $O(u)$ by boundedness and (55); this does not assume that Ψ lies in the old isometry image, and those relations are not re-derived from weaker row estimates. Exact corner and magic matrix-unit actions have their exact opposite mirrors on Ψ.

B.6 Calibrated signs and shared labels without coefficient inversion

Work on the reference vector (55). Calibration against the retained row/column outcomes identifies each raw sign sum S_0 with its exact comparison support $S = f + c_x$ (Alice), or $T = f + r_y$ (Bob), with error $O(u)$. Indeed retained triple events match the corresponding opposite triples in vector norm, their sums identify the sign support, and the preceding section identifies those triples with the standard magic projections. The same argument applies to every refined duplicate.

For one bridge, identify the selected direction multiplicities. On the cut $L \mid RE$, the comparison blocks are

$$W_{xy} = \operatorname{diag}(F, \sigma_{xy} D/\sqrt{2}), \quad \sigma_{xy} = (-1)^{xy}, \qquad K = \operatorname{diag}(F, D). \tag{56}$$

Here F, D are arbitrary rectangular Hilbert–Schmidt coefficient operators. They are not assumed positive, injective, or invertible. Every right operator acts on R tensored with I_E. The selected direction changes are actual finite magic matrix units.

B.6.1 Compression and sharpness are linear in vector error

Let raw PVM signs have $S_0 = P_+ + P_-$ and $X = P_+ - P_-$, so $X^2 = S_0$. Write $X_c = SXS$ and define complete binary effects on S by

$$E_a = (S + aX_c)/2 = SP_aS + K_A/2, \qquad K_A = S - SS_0S \geq 0.$$

If all support and original forbidden amplitudes are bounded by $\delta_{\text{sign}} = Cu$, then

$$\langle \Psi, K_A \Psi \rangle = \|(I - S_0)S\Psi\|^2 \leq \delta_{\text{sign}}^2.$$

Define F_b, K_B, T_0, T, Y_c similarly. For a forbidden sign pair,

$$\|P_aSQ_bT\Psi\| \leq \|P_aQ_b\Psi\| + 2\delta_{\text{sign}} \leq 3\delta_{\text{sign}}.$$

To verify the middle term, use $P_aQ_b = P_aQ_bS_0T_0$ and $ST - S_0T_0 = S(T - T_0) + (S - S_0)T_0$; the last suffix is opposite-party and commutes. The completion terms then give $p_{\text{bad}} \leq (41/2)\delta_{\text{sign}}^2$ for the two wrong-parity pairs. Consequently the exact identity

$$\begin{aligned} 4p_{\text{bad}} = {} & \left\| X_cW - \sigma W Y_c^T \right\|_2^2 + \left\| (I - X_c^2)^{1/2} W \right\|_2^2 \\ & + \left\| W(I - (Y_c^T)^2)^{1/2} \right\|_2^2 \end{aligned} \tag{57}$$

controls all three norms by $\sqrt{82}\,\delta_{\text{sign}}$.

The actual raw sign also has linear output leakage, because

$$SX(I - S)XS = SS_0S - X_c^2 \leq S - X_c^2.$$

The weights of K and W differ by at most $\sqrt{2}$, so $\|(I - S)XS\Psi\| \leq \sqrt{164}\,\delta_{\text{sign}}$. Input leakage satisfies $\|X(I - S)\Psi\| \leq \|S_0(S_0 - S)\Psi\| \leq \delta_{\text{sign}}$. Therefore

$$\|(X - X_c)\Psi\| \leq (1 + \sqrt{164})\delta_{\text{sign}}. \tag{58}$$

This proof uses no sign mirror before establishing it.

B.6.2 A bounded raw-sign mirror precedes label transport

On the matched reference vector the exact sign-support identity is $S_L\Psi = S_R''\Psi$, where S_R'' is the opposite selected support rotated to the same direction by the exact magic units. For a positive comparison pair put $L = \text{diag}(1, \sqrt{2})$, so $H = KL^{-1}$. If $E = X_cH - HY_c^T$, then

$$X_cK - K(LY_cL^{-1})^T = EL. \tag{59}$$

Rotate Bob's selected r direction to the corresponding c direction using the exact magic units. The resulting Bob-only mirror of X_c is LY_cL^{-1}, of norm at most $\sqrt{2}$, extended by zero outside this sign support. Combining (59) with (58) gives a bounded mirror for the *raw* sign X, with error $O(u)$. The symmetric construction gives Bob's raw-sign mirror. Only the fixed scalar L is inverted.

B.6.3 The same opposite questions fix every refinement

A refined PVM has raw coarse sign X' and label marginal Z'. They commute exactly. Against the unchanged opposite sign its comparison equation has the same H, so

$$\left\| (X_c - X_c')H \right\|_2 = O(u).$$

The fixed weight comparison and (58) give $\|(X - X')\Psi\| = O(u)$. Tested label matching gives $\|(Z - Z')\Psi\| = O(u)$, and Z' already has the norm-one opposite mirror Z_R.

Put $D_0 = X - X'$ and $Q_0 = Z - Z'$. The exact identity

$$[X, Z] = [D_0, Z'] + [X, Q_0]$$

has controlled suffixes: Z' uses its tested mirror, and X uses (59). Explicitly,

$$\|[D_0, Z']\Psi\| \le 2\,\|D_0\Psi\| + 2\delta_{\text{sign}}, \quad \|[X, Q_0]\Psi\| \le (1 + \sqrt{2})\,\|Q_0\Psi\| + 2e_X.$$

Thus $[X, Z]\Psi = O(u)$ for every refined label, before assuming any frame preservation.

B.6.4 Calibrate and repair the same two transports

Put $J_L H = H J_R$ and $W_{00} = W_{01} = W_{10} = H$, $W_{11} = J_L H$. Let $E_{xy} = X_x W_{xy} - (-1)^{xy} W_{xy} Y_y^T$. Direct expansion gives

$$\{X_0, J_L\}H = E_{11} + (E_{00} - E_{10})J_R + J_L E_{01},$$
$$H\{Y_0^T, J_R\} = E_{11} - E_{10}J_R + J_L(E_{01} - E_{00}).$$

The signs in these equations are the compressed signs; (58) compares them with the raw ones. Both weighted anticommutators are $O(u)$. If $X_0 = \left(\begin{smallmatrix} a & \beta^* \\ \beta & b \end{smallmatrix}\right)$, then

$$\begin{aligned} &\left\|(I - \beta^*\beta)^{1/2} F\right\|_2^2 + \tfrac{1}{2}\left\|(I - \beta\beta^*)^{1/2} D\right\|_2^2 \\ &\quad = \left\|(I - X_0^2)^{1/2} H\right\|_2^2 + \|\mathrm{diag}(X_0)H\|_2^2 = O(u^2). \end{aligned} \tag{60}$$

For Bob first use the fixed partial unitary $\sqrt{2}\,\Pi_{c_0}\Pi_{r_0}$ to identify his selected corner. After common zero-state padding, extend the polar partial isometry of each β to a unitary $w : f\mathcal{H} \to c_0\mathcal{H}$. The source and range spaces can all be padded to the same infinite dimension, so the two bridges share the same padded e_1 space. For a contraction β,

$$\|(w - \beta)\xi\| \le \|(I - \beta^*\beta)\xi\|, \qquad \|(w^* - \beta^*)\zeta\| \le \|(I - \beta\beta^*)\zeta\|.$$

These follow from $|1 - s| \le |1 - s^2|$ on $[0, 1]$ and the polar decomposition; kernel padding has the same scalar value at $s = 0$. Equation (60) controls both estimates by $O(u)$ on the corresponding state components. No spectral inverse is used.

The off-diagonal blocks of $X_0 H - H Y_0^T = O(u)$ give

$$w_L\Psi_f = 2^{-1/2} w_R^*\Psi_c + O(u), \qquad w_L^*\Psi_c = 2^{1/2} w_R\Psi_f + O(u). \tag{61}$$

These are also the global mirrors for w, w^*, since their exact sources are f, c. Applying w_L to the second equation proves

$$\Psi_c = \sqrt{2}(w_L \otimes w_R \otimes I_E)\Psi_f + O(u). \tag{62}$$

Every map in this argument is on one physical party. Rectangular F, D suffice throughout.

B.7 Transport the raw labels; leave the frame in the root

The six frame labels refined in each bridge are $F_1, F_2, F_3, G_{12}(1), G_{13}(1), G_{23}(1)$. Section B.6 gives their commutators with the original sign X_0 at $O(u)$. Each t_{ij} is a fixed polynomial in those labels; Leibniz expansion and their raw suffix mirrors give $[X_0, t_{ij}]\Psi = O(u)$.

For a transported label $v_{g,k}$, its raw global representative (50) is a polynomial in the frame projections and the *refined* graph projection $G_{12}(v_{g,k})$. Therefore

$$[X_0, v_{g,k}^{\mathrm{raw}}]\Psi = O(u). \tag{63}$$

The adjoint commutator follows either by the same graph polynomial calculation or by $[X, U^*] = -U^*[X, U]U^*$ modulo the already proved two-sided unitary defects. All suffixes in the latter formula are raw except the one X, whose mirror was established first.

The raw block equation is

$$v_{g,k}^{\mathrm{raw}} \simeq I - (e_{g-1} + M_g) + e_{g-1}u_{g-1,k}^{\mathrm{raw}} + \sum_{a,b} R_{ab}e_g u_{g,k}^{\mathrm{raw}} R_{ab}^*. \tag{64}$$

Here replace the prescribed raw corner and axis events by the exact comparison ones. Their actions differ by $O(u)$ and both families have already proved bounded mirrors, so the replacement preserves $O(u)$. The raw support equation $(U - I)(I - e) = 0$ and its adjoint, together with two-sided raw unitarity, give both $Ue \simeq eU$ and $U(I - e) \simeq I - e$.

Compress (63) between the exact magic corner e_g and flag e_{g-1}. In (64), only the flag block and the selected magic block survive, so, for the unrepaired off-diagonal sign block β,

$$\beta u_{g-1,k}^{\mathrm{raw}} - u_{g,k}^{\mathrm{raw}}\beta = O(u)$$

on Ψ. Replacing β by the *same calibrated* w_g uses the raw suffix mirror of $u_{g-1,k}^{\mathrm{raw}}$; its other occurrence is a bounded prefix. The adjoint calculations give the reverse transport and the transport of adjoint labels. Thus

$$w_g u_{g-1,k}^{\mathrm{raw}} \simeq u_{g,k}^{\mathrm{raw}} w_g, \qquad w_g^* u_{g,k}^{\mathrm{raw}} \simeq u_{g-1,k}^{\mathrm{raw}} w_g^*. \tag{65}$$

Here and below the congruence includes adjoints, proved with the corresponding adjoint-label calculation. The order is raw frame/words, comparison geometry, sign mirrors, shared-label commutators, then label transport. No scalar frame is used to decode labels before these steps.

Let $v = w_2 w_1$. It is an exact unitary from the padded e_0 corner to e_2, and the two preceding mirrors give

$$v_L\Psi = \tfrac{1}{2}v_R^*\Psi + O(u), \qquad v_L^*\Psi = 2v_R\Psi + O(u), \qquad \Psi_2 = 2(v_L \otimes v_R)\Psi_0 + O(u). \tag{66}$$

The product mirror is obtained in the order prescribed by Lemma B.2; it is $w_{1,R}^* w_{2,R}^*/2$. This supplies every suffix containing a calibrated transport before root generators are formed.

B.8 Bounded root generators and all five mirrors

Until the final repair, use the raw common token polynomials. The core rows, with the exact comparison corner replacements, give

$$\begin{aligned}
&a \simeq e_2 a e_0, \quad a^*a \simeq e_0, \quad aa^* + b_2 b_2^* + p_2 \simeq e_2,\\
&b_0^* b_0 \simeq e_0 - p_0 - q_0, \quad p_0 q_0 \simeq 0, \quad c_0 p_0 \simeq p_0, \quad c_0 q_0 \simeq q_0/2,\\
&c_2 a \simeq a c_0/2, \qquad c_0 b_0 \simeq 2 b_0 c_0.
\end{aligned}$$

The two root reflection equations also give $p_0^2 \simeq p_0 \simeq p_0^*$ and the analogous q_0 relations. All tokens have fixed norm bounds. For example $U - U^* = U^*(U^2 - I) + (I - U^*U)U$ proves approximate self-adjointness of a raw root reflection, with a previously controlled suffix.

Define provisionally $A_0 = v^*a$, $B_0 = b_0$, $C_0 = c_0$, $P_0 = p_0$, $Q_0 = q_0$ on the root, inserting the exact source/range corner compressions. They satisfy the local Catalan polynomials at $O(u)$ on the unnormalized Ψ_0. For clarity the only transported calculations are

$$\begin{aligned} A_0^*A_0 &= a^*vv^*a \simeq e_0, \\ A_0A_0^* &= v^*aa^*v \simeq e_0 - v^*b_2b_2^*v - v^*p_2v \simeq e_0 - b_0b_0^* - p_0, \\ C_0A_0 - \tfrac{1}{2}A_0C_0 &= (c_0v^* - v^*c_2)a + v^*(c_2a - \tfrac{1}{2}ac_0) \simeq 0. \end{aligned}$$

Use (65) also for adjoints to replace $v^*b_2b_2^*v$. Each right suffix is either a raw word or v, v^* followed by raw words; (45), (66), and Lemma B.2 give all of their mirrors. The remaining root relations are the displayed core rows.

B.8.1 Exact domain restrictions without an exponent loss

Extend every provisional root operator by zero off e_0. Since Ψ has exactly matched corner sectors, its action on Ψ equals its action on Ψ_0. A product mirror of such an operator can be compressed to the opposite root corner with no increase in norm or error. Thus the global raw/transport suffix estimates already apply to every repair below on the unnormalized root vector; they do not depend on the final root mirrors, which are proved afterward. For every raw unitary token U, let $\widetilde{U}$ be its polar contraction cap: if $U = q|U|$, set $\widetilde{U} = q\min(|U|, I)$. Then

$$\left\|(\widetilde{U} - U)\Psi\right\| \le \|(U^*U - I)\Psi\|, \qquad \left\|(\widetilde{U}^* - U^*)\Psi\right\| \le \|(UU^* - I)\Psi\|. \tag{67}$$

This is the scalar bound $(s-1)_+ \le |s^2 - 1|$, applied on each polar side. No small singular value is divided by. Define

$$a = e_2(\widetilde{U}_{a+} + \widetilde{U}_{a-})e_0/2, \quad B = e_0(\widetilde{U}_{0,b+} + \widetilde{U}_{0,b-})e_0/2,$$

$$A = v^*a, \qquad C = e_0(2I + \widetilde{U}_{0,c} + \widetilde{U}_{0,c}^*)e_0/4.$$

These have the required exact contraction and positivity properties. For the projection P, Hermitianize and compress the raw p_0 to $h_p = e_0(p_0 + p_0^*)e_0/2$. Its idempotency defect on Ψ_0 is $O(u)$, by the preceding root calculus. Set $P = \mathbf{1}_{[1/2,\infty)}(h_p)$ on the root, and define Q identically from q_0. The scalar inequality

$$|\mathbf{1}_{[1/2,\infty)}(t) - t| \le 2|t^2 - t| \quad (t \in \mathbb{R}) \tag{68}$$

gives $O(u)$ direct and adjoint closeness. These projections need not commute exactly; the required PQ defect is transferred from the explicit core row $p_0q_0 \simeq 0$.

All repairs preserve every fixed polynomial residual: telescope with repaired bounded prefixes and *original* suffixes. The latter already have their mirrors, including (66). This argument also supplies bounded mirrors for each repaired token from its raw mirror and its direct/adjoint closeness. Hence the domain repair costs no square root.

B.8.2 The root mirrors

Raw real-coefficient graph words have adjoint mirrors. Their repaired, exactly corner-compressed versions therefore satisfy

$$a_L\Psi_0 = a_R^*\Psi_2 + O(u), \qquad a_L^*\Psi_2 = a_R\Psi_0 + O(u).$$

Substitute (66). Apply v_L^* in the first equation and v_R^* in the second. Since operators on opposite parties commute and $v_s^*v_s = e_0$, this gives

$$A_L\Psi_0 = 2A_R^*\Psi_0 + O(u), \qquad 2A_L^*\Psi_0 = A_R\Psi_0 + O(u).$$

The two B mirrors and the C mirror follow directly from their raw adjoint mirrors and repair closeness. Thus all five equations (42) hold on Ψ_0 with error $O(u)$. This argument uses neither a Schmidt inverse nor an entropy comparison with the reference state.

B.9 Extend the root table to the actual full state

Lemma B.5 (Nine-sector extension)**.** *Let exact local sector decompositions have unitaries $T_{s,j} : \mathcal{H}_{s,0} \to \mathcal{H}_{s,j}$, $j = 0, \dots, 8$, with $T_{s,0} = I$. Suppose the actual vector Ξ satisfies*

$$\left\| \Xi - \sum_{j=0}^{8} c_j (T_{L,j} \otimes T_{R,j}) \Psi_0 \right\| \le \eta, \qquad c = (1, \sqrt{2}, \sqrt{2}, \sqrt{2}, \sqrt{2}, 2, 2, 2, 2).$$

The local block-unitaries $W_s = \sum_j |j\rangle \otimes T_{s,j}^*$ map to $\mathbb{C}^9 \otimes \mathcal{H}_{s,0}$. Amplify any root table by I_9. For each bounded local or mirror residual R obtained as the I_9 amplification of a fixed root polynomial (or of a fixed root mirror polynomial),

$$\|R(W_L \otimes W_R \otimes I_E)\Xi\| \le 5 \|R\Psi_0\| + \|R\| \, \eta. \tag{69}$$

Proof. The comparison vector becomes $(\sum_j c_j |jj\rangle) \otimes \Psi_0$. Apply the residual, use $\sum_j c_j^2 = 25$, and bound the difference vector by its norm times $\|R\|$. □

In the present construction take $T_0 = I$, the four first-magic transports $t_{1,ab} w_1$, and the four second-magic transports $t_{2,ab} v$. The magic units are exact in the comparison representation; pad their multiplicities coherently with the preceding polar repairs. Equation (62) and its two-bridge version give the hypothesis of Lemma B.5 with $\eta = O(u)$, including the $O(u)$ actual/reference discrepancy in (55). The other three directions in each magic block are exact local images of its selected component. This uses only transport of *state components*, not transport of every decoded label to those directions.

All repaired root generators satisfy exact domain restrictions, so their I_9 amplifications do as well. Every residual has a fixed operator-norm bound, and the root vector residuals are $O(u)$. Equation (69) proves the Bell part of Theorem B.1, since $u = \sqrt{r} = O(\varepsilon^{1/4})$. The composition of the physical local maps is one pair of isometries, trivial on E. The scalar frame remains inside the root multiplicity space and contributes no postselection factor.

B.10 The Boolean compiler has a linear loss reduction

Here r_G bounds all compiler mismatch norms and forbidden-event amplitudes. The uniformly sampled fixed game gives $r_G = C_G \sqrt{\varepsilon_G}$. We prove each original rational row has residual $O(r_G)$, not merely $O(\sqrt{r_G})$.

B.10.1 Raw scalar matrix units and physical slot events

Every synchronized old binary projection, scalar binary projection, and column projection D_t has its raw opposite mirror. A measured joint-context marginal matched to one of these projections differs from it by $O(r_G)$ and has the same opposite mirror. Thus context products can be replaced by raw products at $O(r_G)$ using Lemma B.2. Context commutation and parity transfer at that same order. Invalid-answer mass, if included in a common alphabet, can first be assigned to a fixed valid answer; its amplitude is $O(r_G)$.

The four-corner magic-square algebra from Section B.4, now applied to synchronized scalar cells, gives the Pauli relations at $O(r_G)$. Cross-square joint contexts give all cross-square axis commutators at $O(r_G)$. For a row dimension $d_{\text{row}} = 2^n \le 256$, use the fixed polynomials

$$e = \prod_{\ell=1}^{n} (I + Z_\ell)/2, \qquad E_{ij} = X^i e X^j$$

with a fixed order for each product. Pauli reductions give, with $O(r_G)$ direct and adjoint error,

$$E_{ij}E_{k\ell} \simeq \delta_{jk}E_{i\ell}, \quad E_{ij}^* \simeq E_{ji}, \quad \sum_i E_{ii} \simeq I.$$

In this section only, $\simeq$ denotes $O(r_G)$ rather than $O(\sqrt{r})$. Each finite reduction has the reverse raw-word suffix mirror. Old-projection/axis joint contexts give commutation of every old reflection with the scalar units at the same order. No scalar tensor factor needs to be extracted.

The global bit and parity contexts identify the actual scalar events with

$$R_{d_{\mathrm{row}}} \simeq R = \frac{1}{d_{\mathrm{row}}} \sum_{i,k} E_{ik}, \qquad R_j^{\pm} \simeq \tfrac{1}{2}(E_{00} + E_{jj} \pm E_{0j} \pm E_{j0}).$$

For $R_j^{\pm}$, expand the joint spectral factors of X^j and the basis of $j^{\perp}$ and reduce Pauli monomials. The zero syndrome leaves exactly coordinates $0, j$, and the X^j sign supplies the off-diagonal sign. Replacing the matched measured marginals by the raw axes has $O(r_G)$ error. The six-answer slot PVM has exactly commuting marginals. Therefore its physical coarse graph is

$$G_j^{\sigma} \simeq \overline{G}_j^{\sigma} = \tfrac{1}{2}(E_{00} + E_{jj} + \sigma(E_{0j} + E_{j0})V_j),$$

where each V_j is an *exact* old signed reflection, $V_j^2 = I$, and $V_0 = I$. Their sum is exactly $\sum_j V_j = 2LS$ for the original row S.

B.10.2 Identify the column projections in the tested orientation

For $\chi_t(j) = (-1)^{t \cdot j}$ define

$$T_t = \sum_j \chi_t(j) E_{j0} V_j, \qquad P_t = d_{\mathrm{row}}^{-1} T_t T_t^*.$$

Finite matrix-unit multiplication gives

$$T_t^* T_u \simeq d_{\mathrm{row}} \delta_{tu} E_{00}, \quad P_t P_u \simeq \delta_{tu} P_t, \quad \sum_t P_t \simeq I.$$

Only $V_j^2 = I$ is used; different V_j need not commute.

A forbidden event $D_{t,L} G_{j,R}^{-\chi_t(j)}$ gives Alice's left kernel $D_t \overline{G}_j^{-\chi_t(j)} \Omega = O(r_G)$ using the graph's derived raw mirror, and Bob's right kernel $\overline{G}_j^{-\chi_t(j)} D_t \Omega = O(r_G)$ using synchronization of D_t. Both conclusions follow from one tested orientation. In the kernel and support calculations in this paragraph, $\simeq$ denotes only the displayed one-sided vector estimate. No adjoint estimate for a kernel is assumed. The two-sided raw polynomial identities used as coefficients remain available as before. Put $C_j = E_{jj} - \chi_t(j) E_{j0} V_j$ and $C_0 = 0$. Polynomial reduction gives

$$C_j \simeq 2E_{jj}\overline{G}_j^{-\chi_t(j)}, \quad C_j^* \simeq 2\overline{G}_j^{-\chi_t(j)} E_{jj},$$
$$\sum_j C_j \simeq I - T_t E_{00}, \qquad \sum_j C_j^* \simeq I - E_{00} T_t^*.$$

Thus on Alice $D_t \simeq D_t E_{00} T_t^* \simeq D_t P_t$, and on Bob $D_t \simeq T_t E_{00} D_t \simeq P_t D_t$. The exact column PVM completeness then gives respectively

$$P_t = \sum_u D_u P_t \simeq \sum_u D_u P_u P_t \simeq D_t, \quad P_t = \sum_u P_t D_u \simeq \sum_u P_t P_u D_u \simeq D_t.$$

Every suffix, including D_u, is raw and synchronized, so this costs no square root. At the conclusion $D_t \simeq P_t$, the adjoint estimate is also available, since D_t and $P_t = d_{\mathrm{row}}^{-1} T_t T_t^*$ are exactly self-adjoint.

B.10.3 The anchored vector identity

The final forbidden event gives $P_0R\Omega = O(r_G)$ on Alice and $RP_0\Omega = O(r_G)$ on Bob. With $W = \sum_j V_j = 2LS$, the anchor $V_0 = I$ yields the *two vector polynomial identities*

$$E_{00}P_0RE_{00} \simeq d_{\mathrm{row}}^{-2}E_{00}W, \qquad E_{00}RP_0E_{00} \simeq d_{\mathrm{row}}^{-2}E_{00}W. \tag{70}$$

Indeed $P_0 \simeq d_{\mathrm{row}}^{-1}\sum_{i,j}E_{ij}V_iV_j$ and $R = d_{\mathrm{row}}^{-1}\sum_{k,\ell}E_{k\ell}$; the 00 corner of their product is $\sum_j V_0V_j/d_{\mathrm{row}}^2$, and the reversed product gives $\sum_i V_iV_0/d_{\mathrm{row}}^2$. Transporting this corner through the fixed raw scalar units gives

$$W \simeq d_{\mathrm{row}}^2\sum_k E_{k0}[E_{00}P_0RE_{00}]E_{0k}$$

on Alice, with RP_0 on Bob. The right suffix $E_{00}E_{0k}$ has its already proved raw mirror. Hence $\|S_s\Omega\| \le Cr_G$ on both parties. Only fixed integers d_{row}, L are divided by.

The SOS expression for the restricted old strategy now has row-square terms, mismatch-square terms, and old forbidden probabilities all $O(r_G^2)$. Naimark-dilated projections have zero sharpness defect. Summing the fixed list proves

$$\mathrm{loss}_{\mathcal{B}} \le C\varepsilon_G. \tag{71}$$

Apply the Bell decoder. This proves the game part of Theorem B.1 with the same exponent $1/4$.

B.11 Dependency, exponent, and suffix ledgers

The quantitative orders below apply with $r = C\sqrt{\varepsilon}$ for either input, using Section B.10 first for the game.

Stage	Vector order	Bell/game exponent
SOS or compiler rows; raw synchronization and event amplitudes	r	$1/2$
Projection overlaps; all raw graph/word/parameter identities	$\sqrt{r}$	$1/4$
Actual coarse-sector isometry; normal magic comparison	$\sqrt{r}$	$1/4$
Sign compression, sharpness, bounded mirrors, calibration	$\sqrt{r}$	$1/4$
Shared refinements and common raw-label transport	$\sqrt{r}$	$1/4$
Root token cap, projection threshold, local relations, mirrors	$\sqrt{r}$	$1/4$
Nine-sector extension on the actual state	$\sqrt{r}$	$1/4$

For completeness, the following table is the entire suffix registry. Arbitrary products use exactly the reverse-product rule (43); linear combinations use the same coefficients. Thus it covers every suffix in the proof rather than an unspecified collection of permitted replacements.

Local symbol or expression	Previously derived opposite mirror	First available
Raw binary projection	Opposite copy, norm one	SOS/tests
Raw polynomial p	Reversed raw polynomial $p^\flat$, norm $\le a(p)$	Section B.2
Original FMS fine outcome or cell	Reversed product of matched coarse cells/nonflag projectors	Section B.5
Exact comparison corner or magic unit	Opposite transpose/adjoint on Ψ, norm one	Section B.5
Raw original or refined sign	Rotated LY_cL^{-1}, norm $\le \sqrt{2}$	Section B.6
Refined label marginal Z'	Tested opposite binary Z_R, norm one	Its label test
w_g; w_g^*	$w_{g,R}^*/\sqrt{2}$; $\sqrt{2}w_{g,R}$	Section B.6
v; v^*	$v_R^*/2$; $2v_R$	Section B.7
Capped raw token and its adjoint	Capped opposite adjoint/token, norm one	Section B.8
Repaired root P, Q, C, B, B^*	Opposite P, Q, C, B^*, B, norm one	Section B.8
Root A; A^*	$2A_R^*$; $A_R/2$	Section B.8
Compiler scalar word or column symbol	Reversed raw synchronized word	Section B.10

Rows added later are never used to justify earlier steps. When a comparison vector replaces the actual vector, both residual and mirror errors gain only their fixed operator-norm multiples of the vector discrepancy. The final use of the comparison vector is (69); entropy is never transferred by continuity.

B.12 Consequences and the sharp ceiling on a decoder exponent

Proposition B.6 (No actual-state mixed-strategy exponent exceeds 1/2)**.** *A decoder with the local A^*A relation and both A mirrors, acting trivially on a purification environment, cannot have a uniform residual exponent $\gamma > 1/2$. This already holds for finite-dimensional mixed near-optimal strategies.*

Proof. On a unit product vector $\zeta = a \otimes b$, suppose these three residuals have norm at most δ_{prod}. Put $\alpha = \langle a, A_L a\rangle$ and $\beta = \langle b, A_R b\rangle$. Taking expectations in the mirrors gives

$$|\alpha/2 - \overline{\beta}| \le \delta_{\text{prod}}, \qquad |\alpha - \overline{\beta}/2| \le \delta_{\text{prod}},$$

so $|\alpha| \le 2\delta_{\text{prod}}$. Project the first mirror onto $a^\perp \otimes \mathcal{H}_R$: it gives $\|(I - |a\rangle\langle a|)A_L a\| \le 2\delta_{\text{prod}}$. Thus $\|A_L a\|^2 \le 8\delta_{\text{prod}}^2$, while $A_L^* A_L \simeq I$ gives $\|A_L a\|^2 \ge 1 - \delta_{\text{prod}}$. Therefore

$$\delta_{\text{prod}} \ge c_* = (\sqrt{33} - 1)/16.$$

Mix a finite cutoff strategy of loss $\eta \le \varepsilon^2$ with a fixed deterministic product strategy of loss $\ell > 0$, using orthogonal local flags and block-diagonal measurements. Set $p = (\varepsilon - \eta)/(\ell - \eta)$ for the product branch. The loss is exactly ε and $p \sim \varepsilon/\ell$. Purify with an orthogonal environment flag. Local isometries preserve the product form of that branch, and residuals cannot cancel across the untouched environment flag. A global residual bound δ therefore gives a product-branch bound $\delta_{\text{prod}} \le \delta/\sqrt{p}$. Hence $\delta \ge c_*\sqrt{p} = \Omega(\sqrt{\varepsilon})$. □

The constructed exponent is 1/4. Exponent 1/2 would yield lower coefficient 1/9 through Appendix C; its attainability remains open.

C State-weighted Catalan rounding

C.1 Statement and conventions

All logarithms and entropies are in base two. The logarithmic variable $L_\delta = \log_2(1/\delta)$ is written L only inside this appendix; subscripts L, R always denote parties. The band count N in the banding subsection and the history cutoff N in the renewal lemma are independent parameters, denoted N_{band} and N_{hist} when used together in the assembly. The band error η is renamed u before the renewal comparison error is introduced. Let ρ be a normal state on $\mathcal{H}_L \otimes \mathcal{H}_R$ and fix a purification $\Omega \in \mathcal{H}_L \otimes \mathcal{H}_R \otimes \mathcal{E}$. For a local operator X_s, its state-weighted norm is $\|X_s\Omega\|$. All operators and physical isometries below act trivially on $\mathcal{E}$. Local Hilbert spaces, multiplicity spaces, and $\mathcal{E}$ may be infinite dimensional. Entropy means marginal von Neumann entropy; for mixed states we do not identify it with a mixed-state entanglement measure.

Suppose on each party $s = L, R$ that

$$\|A_s\|, \|B_s\| \le 1, \quad 0 \le C_s \le I, \quad P_s^2 = P_s = P_s^*, \quad Q_s^2 = Q_s = Q_s^*.$$

Assume that each of the following local residuals has norm at most δ on Ω:

$$\begin{gathered} A_s^*A_s - I, \quad A_sA_s^* + B_sB_s^* + P_s - I, \quad B_s^*B_s + P_s + Q_s - I, \\ C_sP_s - P_s, \quad C_sQ_s - Q_s/2, \quad C_sA_s - A_sC_s/2, \quad C_sB_s - 2B_sC_s, \quad P_sQ_s. \end{gathered} \tag{72}$$

The five mirror residuals are

$$A_L/2 - A_R^*, \quad A_L^* - A_R/2, \quad B_L - B_R^*, \quad B_L^* - B_R, \quad C_L - C_R, \tag{73}$$

and each likewise has vector norm at most δ. Tensor identities are suppressed.

Theorem C.1 (Catalan rounding)**.** *Under these assumptions, for $0 < \delta \le 1$ and $L = \log_2(1/\delta)$,*

$$S(\rho_s) \ge \frac{2}{9}L - \frac{16}{3}\log_2(L+2) - 200, \qquad s = L, R. \tag{74}$$

The bound is uniform over all dimensions, auxiliary spaces, multiplicities, and normal mixed states. If all defects vanish, both marginal entropies are infinite.

The proof is constructive at the level of local zero-state extensions and local isometries. Comparison vectors are used only to estimate a finite-register conditional entropy; they are never substituted for the physical state. The proof of the theorem is completed in Section C.5. Every lemma needed for it is proved below. The Catalan construction supplies the notation; no external quantitative rounding theorem is invoked.

The bounded-mirror rule. If $\|(X_L - Y_R)\Omega\| \le e$, then for every bounded left-local map R_L,

$$\|R_LX_L\Omega\| \le \|Y_R\|\|R_L\Omega\| + \|R_L\|e.$$

This follows by inserting $Y_R\Omega$ and commuting opposite-party operators. The mirror must already have been established before this rule is applied. A one-sided vector error is never propagated through an arbitrary same-party suffix.

C.2 Spectral banding with explicit constants

All Hilbert spaces below may be infinite dimensional. An environment is suppressed from the notation and every operator acts trivially on it. Let Ω be a unit vector. Assume that A_s, B_s are contractions, $0 \le C_s \le I$, and P_s, Q_s are projections, for $s = L, R$. Assume that all the local residuals in (72) have norm at most δ on Ω, as do the five mirror residuals in (73). Products on opposite parties commute. Put

$$R_{A,s} = A_s^* A_s - I, \quad R_{B,s} = B_s^* B_s + P_s + Q_s - I, \quad R_{\mathrm{r},s} = A_s A_s^* + B_s B_s^* + P_s - I.$$

Their operator norms are at most $1, 2, 2$, respectively.

C.2.1 Mirrors needed before any localization

The given mirrors imply

$$\|(P_L - P_R)\Omega\| \le 6\delta, \tag{75}$$

$$\|(Q_L - Q_R)\Omega\| \le 10\delta. \tag{B1}$$

Indeed

$$A_L A_L^* \Omega - A_R A_R^* \Omega = A_L(A_L^* - A_R/2)\Omega + \tfrac{1}{2} A_R(A_L - 2A_R^*)\Omega,$$

whose norm is at most 2δ. For the other two products the explicit expansions are

$$B_L B_L^* - B_R B_R^* = B_L(B_L^* - B_R) + B_R(B_L - B_R^*),$$

$$B_L^* B_L - B_R^* B_R = B_L^*(B_L - B_R^*) + B_R^*(B_L^* - B_R).$$

Each has vector error at most 2δ. Subtract the two range residuals to obtain the first inequality in (B1), and then subtract the two source residuals for B to obtain the second. No state-dependent error has been multiplied on the right in this argument.

For $T \in \{A, A^*, B, B^*\}$ let

$$k_A = k_{B^*} = \tfrac{1}{2}, \qquad k_{A^*} = k_B = 2.$$

On either party, with $\bar{s}$ denoting the other party,

$$\|(C_s - k_T C_{\bar{s}})T_s\Omega\| \le c_T\delta, \qquad (c_A, c_{A^*}, c_B, c_{B^*}) = (3/2, 7/2, 3, 5/2). \tag{B2}$$

For A and B this follows directly from their local covariance relation and the C mirror. For example,

$$(C_L - C_R/2)A_L\Omega = (C_L A_L - A_L C_L/2)\Omega + \tfrac{1}{2} A_L(C_L - C_R)\Omega.$$

For A^* first replace $A_L^*\Omega$ by $A_R\Omega/2$, at error δ. The commuting operator $C_L - 2C_R$ has norm at most 2; the remaining expression is

$$\tfrac{1}{2} A_R(C_L - C_R)\Omega - (C_R A_R - A_R C_R/2)\Omega.$$

This proves the constant $7/2$. For B^* replace $B_L^*\Omega$ by $B_R\Omega$. Now $\|C_L - C_R/2\| \le 1$, and the remaining expression is

$$B_R(C_L - C_R)\Omega - \tfrac{1}{2}(C_R B_R - 2B_R C_R)\Omega.$$

The other party is identical. Thus every spectral comparison below is between commuting opposite-party operators.

C.2.2 The common random partition

Fix $N \geq 4$. For $t \in [2/3, 3/4]$, define a partition of $[0, \infty)$ by

$$I_0 = [t, \infty), \quad I_h = [t2^{-h}, t2^{-(h-1)}) \ (1 \leq h < N), \quad I_N = [0, t2^{-(N-1)}).$$

Write $\Pi_h(X) = 1_{I_h}(X)$, $E_h^s = \Pi_h(C_s)$ and

$$\eta = 40\, 2^{N/2} \delta^{1/2}. \tag{B3}$$

Assume for the remainder of this section that $\eta \leq 1/128$. For commuting positive operators X, Y and a vector v, their joint spectral measure gives

$$\mathbb{E}_t \sum_{h=0}^{N} \|(\Pi_h(X) - \Pi_h(Y))v\|^2 \leq 48\, 2^N \|v\| \|(X - Y)v\|. \tag{B4}$$

To see this, a scalar threshold $t2^{-j}$ separates x, y for a set of t of length at most $2^j|x - y|$. The sum of squared indicator differences is twice the indicator that the two points lie in different bands. Divide by the interval length $1/12$, sum over $0 \leq j < N$, and use Cauchy–Schwarz in the joint spectral measure. In fact the same proof gives $24(2^N - 1)$ in place of $48\, 2^N$.

Apply (B4) to (C_L, C_R, Ω) and to the eight triples $(C_s, k_T C_{\bar{s}}, T_s\Omega)$. By (B2), the expected sum of all nine nonnegative quantities is at most

$$(48 + 8 \cdot 192) 2^N \delta = 1584\, 2^N \delta \leq \eta^2.$$

Fix a value of t at which this sum is at most η^2. Let

$$M = \sum_{h=0}^{N} E_h^L E_h^R, \qquad \Phi = M\Omega, \qquad \Phi_h = E_h^L E_h^R \Omega.$$

These vectors are intentionally not normalized. They satisfy

$$E_h^L \Phi = E_h^R \Phi = \Phi_h, \qquad e := \|\Omega - \Phi\| \leq \eta, \qquad q := \|\Phi\| \geq \sqrt{1 - \eta^2}. \tag{B5}$$

The equality underlying (B5) is $1 - q^2 = \frac{1}{2} \sum_h \|(E_h^L - E_h^R)\Omega\|^2$. In particular the terminal band remains in Φ throughout all the following algebraic estimates.

C.2.3 Localization without a new square root

The local relations $(C - I)P\Omega = O(\delta)$ and $(C - I/2)Q\Omega = O(\delta)$ imply

$$\|(I - E_0^s)P_s\Omega\| \leq 4\delta, \qquad \|(I - E_1^s)Q_s\Omega\| \leq 8\delta. \tag{B6}$$

For Q, the distance of $1/2$ from the complement of $[t/2, t)$ is at least $1/8$. The constant 6 would not be justified uniformly for this interval of thresholds. Using the already established mirrors (B1), matching (B5), and then (B6), one obtains

$$\left(\sum_{h \geq 1} \|P_s \Phi_h\|^2 \right)^{1/2} = \|(I - E_0^{\bar{s}})P_s\Phi\| \leq 11\eta, \tag{76}$$

$$\left(\sum_{h \neq 1} \|Q_s \Phi_h\|^2 \right)^{1/2} = \|(I - E_1^{\bar{s}})Q_s\Phi\| \leq 19\eta. \tag{B7}$$

For example the first right side is bounded by $6\delta + 4\delta + e$. This is precisely where a prior P, Q mirror avoids an unnecessary square-root loss.

Every local residual R_s satisfies the useful aggregate estimate

$$\left(\sum_h \|R_s\Phi_h\|^2\right)^{1/2} = \|R_s\Phi\| \le \delta + \|R_s\|e. \tag{B8}$$

The equality holds because $\Phi_h = E_h^{\bar{s}}\Phi$, and the opposite-party projections commute with R_s. Thus the source and range residuals have bounds $2\eta, 3\eta, 3\eta$ on the matched vector. No terminal mass appears in (B8).

The spectral comparisons chosen above also imply the following four aggregate leakage bounds, on each party:

$$\left(\sum_{h=0}^{N-2} \|(I - E_{h+1})A\Phi_h\|^2\right)^{1/2} \le 2\eta, \tag{77}$$

$$\left(\sum_{h=1}^{N-1} \|(I - E_{h-1})B\Phi_h\|^2\right)^{1/2} \le 2\eta, \tag{78}$$

$$\left(\sum_{h=1}^{N-1} \|(I - E_{h-1})A^*\Phi_h\|^2\right)^{1/2} \le 2\eta, \tag{79}$$

$$\left(\sum_{h=0}^{N-2} \|(I - E_{h+1})B^*\Phi_h\|^2\right)^{1/2} \le 2\eta. \tag{B9}$$

For completeness, replace Φ_h by $E_h^{\bar{s}}\Phi$, commute this projection past the generator, and observe that the unwanted joint spectral block is contained in the mismatch event for $(C_s, k_T C_{\bar{s}})$. The sum over h is orthogonal on the opposite party. Its norm on $T_s\Omega$ is at most η, and replacing Ω by Φ costs at most e. The shift at $h = 0$ for $k_T = 1/2$ is valid because $2t > 1$; the top two cases excluded in (B9) are not asserted.

The forbidden bottom actions have bounds

$$\|B_s\Phi_0\| \le 10\eta, \qquad \|A_s^*\Phi_0\| \le 12\eta. \tag{B10}$$

Indeed on $E_0^{\bar{s}}$, the commuting operator $C_s - 2C_{\bar{s}}$ has absolute value at least $2t - 1 \ge 1/3$. Apply (B2) and then replace Ω by Φ.

C.2.4 The exact input to the incoming-row repair lemma

For $1 \le v \le N - 2$ define the incoming edges

$$x_{A,v} = E_v A E_{v-1}, \qquad x_{B,v} = E_v B E_{v+1}.$$

The following estimates hold separately on each party, with all sums taken over this entire collection of rows:

$$\left(\sum_v \|(x_{A,v}^* x_{A,v} - E_{v-1})\Phi_{v-1}\|^2\right)^{1/2} \le 4\eta, \tag{80}$$

$$\left(\sum_v \|(x_{B,v}^* x_{B,v} - E_{v+1})\Phi_{v+1}\|^2\right)^{1/2} \le 35\eta, \tag{81}$$

$$\left(\sum_v \|(x_{A,v}x_{A,v}^* + x_{B,v}x_{B,v}^* - E_v)\Phi_v\|^2\right)^{1/2} \le 18\eta. \tag{B11}$$

Here and below an edge on each party is denoted by the same letter. For the first line, expand

$$(x^*_{A,v}x_{A,v} - E_{v-1})\Phi_{v-1} = E_{v-1}(A^*A - I)\Phi_{v-1} - E_{v-1}A^*(I - E_v)A\Phi_{v-1}$$

and use (B8)–(B9). The second line has the additional terms $P\Phi_{v+1}, Q\Phi_{v+1}$, bounded by (B7); its constants are $3 + 11 + 19 + 2 = 35$. For the third line compress the range residual by E_v, then subtract $E_vA(I - E_{v-1})A^*\Phi_v$ and $E_vB(I - E_{v+1})B^*\Phi_v$, and use (B7)–(B9). The constants are $3 + 11 + 2 + 2 = 18$.

Both oriented edge mirrors also hold in aggregate. For example,

$$x_{A,v,L}\Phi_{v-1} - 2x^*_{A,v,R}\Phi_v = E^L_v E^R_{v-1}(A_L - 2A^*_R)\Phi.$$

The projections for distinct v are orthogonal. Consequently the aggregate direct A mirror error is at most 5η on each party, so in particular m^R_e for an A edge has bound 5η. The adjoint orientation is bounded by $(5/2)\eta$; including it again below is a harmless overestimate. Each B orientation costs at most 3η. The same statements hold after interchanging parties. Combining the squares of all source, range and oriented mirror errors in (B11), on both parties, gives at most

$$2(4^2 + 35^2 + 18^2)\eta^2 + 2(5^2 + (5/2)^2)\eta^2 + 4 \cdot 3^2\eta^2 = \tfrac{6457}{2}\eta^2 < 57^2\eta^2.$$

After replacing Φ by the normalized matched vector Φ/q, the aggregate row-repair input is therefore bounded by

$$d \le 64\eta \le 1. \tag{B12}$$

This bound has no factor depending on N. In particular it contains neither the terminal mass nor its square root.

C.2.5 The unused bottom block

The edge $B_1 : 1 \to 0$ is absent from the retained incoming rows. The renewal operator has no incoming edge at level zero, and the optional exact extension sets $\widehat{B} = 0$ on levels $0, 1$. Thus the unused bottom block is vacuous; no deletion error enters the argument.

C.2.6 A cumulative-tail recurrence with kernel mass four

Define, for $0 \le j < N$,

$$F^s_j = 1_{[0,t2^{-j})}(C_s) = \sum_{h=j+1}^{N} E^s_h, \qquad \alpha_j = \|F^L_j\Phi\|^2.$$

Set $F^s_{-1} = I$. Endpoint conventions use the same half-open partition as above. For $0 \le j \le N - 2$, matching, the original mirrors on Φ, and the selected spectral comparisons give

$$\|A^*_L F^L_j\Phi - \tfrac{1}{2}A_R F^L_{j-1}\Phi\| \le \tfrac{7}{2}\eta, \tag{82}$$

$$\|B^*_L F^L_j\Phi - B_R F^L_{j+1}\Phi\| \le 5\eta. \tag{B16}$$

For the first line, commute F^R_j past A^*_L; its mirror costs $\delta + (3/2)e \le (5/2)\eta$. The remaining threshold comparison is

$$F_j(C_R)A_R\Phi - F_j(C_L/2)A_R\Phi,$$

and costs at most 2η, multiplied by $1/2$. The second line is the same argument with $B^*_L \simeq B_R$ and $F_j(C_R)$ versus $F_j(2C_L) = F_{j+1}(C_L)$. A threshold mismatch is a subset of the corresponding

partition mismatch, so these comparisons are covered by the already selected t. There is no new random choice.

The range equation on $F_j^L\Phi$ costs at most 3η, by commuting the opposite-party tail projection. Its P term is at most $121\eta^2$. The squared-norm comparison in the first line of (B16) costs at most $(3/2)(7/2)\eta$, and the A source equation contributes at most $\eta/2$. For the second line the squared-norm comparison costs at most 10η. The B source equation contributes 3η, and its P, Q terms contribute at most $(11^2 + 19^2)\eta^2$. Thus, for $1 \le j \le N-2$,

$$|\alpha_j - \tfrac{1}{4}\alpha_{j-1} - \alpha_{j+1}| \le \tfrac{87}{4}\eta + 603\eta^2 \le 640\eta. \tag{B17}$$

The same formula at $j = 0$ uses $\|\Phi\|^2$ in place of α_{-1}, but is not needed.

Since $0 \le \alpha_1 \le \alpha_0 \le 1$, solving the scalar recurrence gives, for $0 \le k < N$,

$$\alpha_k \le (k+1)2^{-k} + 2560\eta. \tag{B18}$$

Indeed the homogeneous solution is $[\alpha_0 + (2\alpha_1 - \alpha_0)k]2^{-k}$, bounded above by $(k+1)2^{-k}$. The impulse kernel is $(n+1)2^{-n}$, and its total mass is exactly

$$\sum_{n=0}^{\infty}(n+1)2^{-n} = 4.$$

Finally M commutes with F_k^L, so the two vectors $F_k^L\Phi$ and $F_k^L(\Omega - \Phi)$ are orthogonal. Consequently the original state satisfies

$$\|F_k^L\Omega\|^2 \le (k+1)2^{-k} + 2561\eta, \qquad 0 \le k < N. \tag{B19}$$

Matching gives the same bound on the other party. The large constants are explicit and dimension independent; they were chosen for transparent bookkeeping rather than optimization.

Dependency ledger. The order is: original local equations and five mirrors; product mirrors and then P, Q mirrors; commuting generator-image covariance; one common random partition; a matched vector retaining its terminal band; spectral localization of P, Q; aggregate local defects and edge leakages; source, range and both edge-mirror inputs; the unused bottom block. The cumulative tail recurrence uses only the items preceding row repair. No estimate in the row-repair input uses bottom-block handling, tail truncation, a repaired operator, or any entropy comparison.

C.3 Repair of incoming rows

Lemma C.2 (State-weighted repair of incoming rows)**.** *Let $\mathcal{G} = (V, E)$ be a finite directed graph with in- and out-degree at most $D \ge 1$. For $s = L, R$ let $\mathcal{H}^s = \bigoplus_{v\in V}\mathcal{H}_v^s$, and let*

$$\Phi = \sum_{v\in V}\Phi_v, \qquad \Phi_v \in \mathcal{H}_v^L \otimes \mathcal{H}_v^R \otimes \mathcal{K}, \qquad \sum_v \|\Phi_v\|^2 = 1.$$

Every row to be repaired is assumed to have at least one incoming edge. For each edge e, let $x_e^s : \mathcal{H}_{s(e)}^s \to \mathcal{H}_{t(e)}^s$ have norm at most $M_{\mathrm{op}} \ge 1$, and let $0 < \lambda_e \le \Lambda$, where $\Lambda \ge 1$. Put

$$g_e^s = (x_e^s)^* x_e^s, \qquad r_v^s = \sum_{t(e)=v} x_e^s (x_e^s)^* - I_{\mathcal{H}_v^s}, \qquad m_e^s = x_e^s \Phi_{s(e)} - \lambda_e (x_e^{\bar{s}})^* \Phi_{t(e)}.$$

The sums over v below range over the rows to be repaired, and all edges considered end in those rows. Suppose

$$d^2 \ge \sum_{s=L,R}\left\{\sum_e \|(g_e^s - I)\Phi_{s(e)}\|^2 + \sum_v \|r_v^s\Phi_v\|^2 + \sum_e \|m_e^s\|^2\right\}, \qquad 0 \le d \le 1.$$

After adjoining zero-state local summands, there are edge operators u_e^s such that every incoming row $U_v^s = (u_e^s)_{t(e)=v}$ is unitary from the direct sum of its source spaces onto the enlarged target space. In particular,

$$(u_e^s)^* u_f^s = \delta_{ef} I \quad (t(e) = t(f)), \qquad \sum_{t(e)=v} u_e^s (u_e^s)^* = I.$$

Set $\mathfrak{b} = DM_{\mathrm{op}}^2 + 1$ and

$$K = 2\big(\Lambda^2 M_{\mathrm{op}}^2 D + 2\mathfrak{b}\Lambda M_{\mathrm{op}}\sqrt{D} + M_{\mathrm{op}}^2\sqrt{D} + \mathfrak{b} + 1\big).$$

Then

$$\left(\sum_{s,e} \|(u_e^s - x_e^s)\Phi_{s(e)}\|^2\right)^{1/2} \le \sqrt{K}\,\sqrt{d},$$
$$\left(\sum_{s,e} \|((u_e^s)^* - (x_e^s)^*)\Phi_{t(e)}\|^2\right)^{1/2} \le d.$$

Consequently the aggregate repaired mirror error is at most $\sqrt{K}\sqrt{d} + (1+\Lambda)d$. The padding acts trivially on Φ and its purifying environment, so it changes neither marginal spectrum.

Proof. Fix a party and suppress its superscript. Write $R_v = (x_e)_{t(e)=v}$ and let j_e be the canonical inclusion of the eth source space in the row domain. For $\phi = \Phi_{s(e)}$ and $v = t(e)$, expansion of r_v gives the exact cross-Gram identity

$$\sum_{\substack{t(f)=v \\ f\neq e}} \|x_f^* x_e \phi\|^2 = \langle x_e\phi, r_v x_e \phi\rangle + \langle \phi, g_e(I - g_e)\phi\rangle. \tag{RG}$$

This identity does not require x_e to be a contraction. Its left side is nonnegative, although the two terms on the right need not be.

Let $q_e = \|m_e\|$, and write $x_e\Phi_{s(e)} = w_e + m_e$, where $w_e = \lambda_e(x_e^{\bar{s}})^*\Phi_v$. Since $\|r_v\| \le \mathfrak{b}$,

$$|\langle x_e\Phi_{s(e)}, r_v x_e \Phi_{s(e)}\rangle| \le |\langle w_e, r_v w_e\rangle| + 2\mathfrak{b}\Lambda M_{\mathrm{op}}\|\Phi_v\| q_e + \mathfrak{b} q_e^2.$$

Operators on opposite parties commute, whence

$$|\langle w_e, r_v w_e\rangle| = \lambda_e^2 |\langle \Phi_v, x_e^{\bar{s}}(x_e^{\bar{s}})^* r_v \Phi_v\rangle| \le \Lambda^2 M_{\mathrm{op}}^2 \|\Phi_v\|\,\|r_v\Phi_v\|.$$

This is the only propagation through a suffix in the argument: it uses the already assumed opposite-party edge mirror, followed by exact inter-party commutation. Summing over edges, the degree bound and Cauchy–Schwarz give

$$\sum_e |\langle x_e\Phi_{s(e)}, r_{t(e)} x_e \Phi_{s(e)}\rangle| \le \Lambda^2 M_{\mathrm{op}}^2 D\, d + 2\mathfrak{b}\Lambda M_{\mathrm{op}}\sqrt{D}\, d + \mathfrak{b} d^2.$$

Likewise,

$$\sum_e |\langle \Phi_{s(e)}, g_e(I-g_e)\Phi_{s(e)}\rangle| \le M_{\mathrm{op}}^2 \sum_e \|\Phi_{s(e)}\|\,\|(I-g_e)\Phi_{s(e)}\| \le M_{\mathrm{op}}^2\sqrt{D}\, d.$$

Thus (RG) bounds the sum of all off-diagonal Gram errors for this party. The diagonal entries of $(R_v^* R_v - I) j_e \Phi_{s(e)}$ are exactly $(g_e - I)\Phi_{s(e)}$. Adding their squared norms, and then adding both parties, proves

$$\sum_{s,e} \|((R_{t(e)}^s)^* R_{t(e)}^s - I) j_e \Phi_{s(e)}\|^2 \le Kd. \tag{GG}$$

For completeness, the required unitary polar extension is available simultaneously at all vertices. Choose an infinite cardinal κ at least as large as the dimensions of every local vertex space, and adjoin a copy of $\ell^2(\kappa)$ to each vertex space, placing zero state on every new summand. Extend each original x_e^s by zero. For a nonempty incoming row both the kernel and the cokernel of the enlarged row have dimension κ. Its polar partial isometry therefore extends to a unitary U_v^s from its enlarged row domain onto its enlarged codomain. Choices at different vertices are independent: each edge is a column of exactly one incoming row. In separable spaces one may take $\kappa = \aleph_0$.

Write $R = U|R|$ for such a polar extension. For every ζ, spectral calculus and $|1-t| \le |1-t^2|$ for $t \ge 0$ give

$$\|(U - R)\zeta\| = \|(I - |R|)\zeta\| \le \|(I - R^*R)\zeta\|.$$

Since $RR^* = U|R|^2U^*$, the same argument gives

$$\|(U^* - R^*)\eta\| \le \|(I - RR^*)\eta\|.$$

These inequalities remain valid after tensoring with all spectator spaces. Applying the first to $j_e\Phi_{s(e)}$ and using (GG) proves the direct estimate. For the adjoint estimate, orthogonality of the row-domain summands gives

$$\sum_{t(e)=v} \|(u_e^* - x_e^*)\Phi_v\|^2 = \|(U_v^* - R_v^*)\Phi_v\|^2 \le \|r_v\Phi_v\|^2.$$

Summing proves its asserted constant 1. Finally, each new mirror is the sum of its direct repair error, its old mirror error, and λ_e times the opposite-party adjoint repair error. Minkowski's inequality gives the last claim. □

Remark. An empty incoming row cannot be made unitary merely by padding the existing source spaces: its domain is zero. Root vertices must be excluded from the row lemma or supplied with an explicit source edge. No inverse bound on λ_e is needed for the displayed orientation; it is needed only when the input mirrors are supplied in the opposite normalization.

C.4 Finite-strip renewal and its entropy

Lemma C.3 (Finite-strip renewal)**.** *Let $H \ge 2$. On each party let $\mathcal{H} = \bigoplus_{h=0}^{H} \mathcal{H}_h$. Let $a_h : \mathcal{H}_h \to \mathcal{H}_{h+1}$, $0 \le h \le H-2$, and $b_h : \mathcal{H}_h \to \mathcal{H}_{h-1}$, $2 \le h \le H$, be isometries. Assume that the ranges of a_{h-1} and b_{h+1} are orthogonal whenever both enter the same vertex. Write $a = \sum a_h$, $b = \sum b_h$, and, on the matched space $\mathcal{M} = \bigoplus_h (\mathcal{H}_h^L \otimes \mathcal{H}_h^R)$,*

$$T = \tfrac{1}{2} a_L \otimes a_R + b_L \otimes b_R.$$

The identity on an arbitrary purifying environment is suppressed. The retained local direct sums may sit in larger physical spaces with arbitrary additional orthogonal summands; all local isometries constructed below are extended to those summands. Then

$$\|(I - T)^{-1}\| \le 4H^2 2^{H/2}.$$

If $\xi_s \in \mathcal{H}_s^L \otimes \mathcal{H}_s^R \otimes \mathcal{E}$, $s \in \{0, H\}$, and $y = (I-T)^{-1}(\xi_0 + \xi_H) \ne 0$, there are local isometries, acting trivially on $\mathcal{E}$, and a finite register F on the left with

$$\log_2 \dim F \le N + 2, \qquad N = 2H^2 \lceil 8 \log_2(2H) \rceil,$$

such that the normalized state $\widehat{y} = y/\|y\|$ satisfies

$$H(F \mid J)_{\widehat{y}} \ge \tfrac{2}{3} H - \tfrac{8}{3}.$$

Here J is the remaining left system and conditional entropy is understood in its finite-F extension, so no finiteness assumption on the multiplicity entropy is needed.

Proof. Put $D = \sum_{h=0}^{H} 2^{h/2} E_h$ on $\mathcal{M}$. The two allowed edge coefficients of $U = DTD^{-1}$ are both $1/\sqrt{2}$. Distinct directed paths of equal length have orthogonal ranges: cancel equal final edges, using their isometry property, until distinct final edges have orthogonal ranges. Consequently $\|U^n\|^2$ is bounded by the largest survival probability through time n of the fair nearest-neighbor walk on the strip, killed when it attempts a missing edge. There are no incoming edges at $0, H$. At an internal starting vertex h the mean killing time is $h(H-h)$. At either source it is $1 + (H-1)/2$. Thus every mean is at most $H^2/2$. For $\ell_{\mathrm{blk}} = 2H^2$, Markov's inequality and the Markov property give

$$\|U^n\|^2 \le 4^{-\lfloor n/\ell_{\mathrm{blk}} \rfloor}.$$

In particular the Neumann series converges in operator norm and

$$\|(I-T)^{-1}\| \le 2^{H/2} \sum_{n \ge 0} \|U^n\| \le 2^{H/2} \ell_{\mathrm{blk}} \sum_{j \ge 0} 2^{-j} = 4H^2 2^{H/2}.$$

All path ranges originating at the two sources are mutually orthogonal, even for different path lengths. Indeed, after cancellation of common final edges, either two distinct incoming edges remain, or one path has ended at a source while the other would have to enter that source. The latter is impossible because both sources have no incoming retained edge. The same argument applies on both parties. A legal path w having $n_A(w)$ up steps contributes $2^{-n_A(w)} W_w^L \otimes W_w^R \xi_s$. Thus, up to local isometries, each source contributes a pure path register with probabilities

$$p_s(w) = 4^{-n_A(w)}/Z_s, \qquad Z_s = \sum_{w \text{ from } s} 4^{-n_A(w)},$$

tensored with the normalized source vector. The two source labels are locally distinguishable.

For the bottom source the strip is the height cutoff $n = H - 1$:

$$Z_0 = \frac{2n + 2^{-n}}{n+1}, \qquad \frac{M_0}{Z_0} = \frac{n-1}{3} + \frac{2^{-n}(n+1)(n+2)}{6(2n+2^{-n})}, \quad M_0 = \sum_w n_A(w) 4^{-n_A(w)}.$$

Here is an explicit derivation. If $z_h(x)$ is the unnormalized mass at height h when each up edge has squared weight x, then $z_0(x) = 1$, $z_H(x) = 0$, and $z_h(x) = x z_{h-1}(x) + z_{h+1}(x)$ for $1 \le h \le H-1$. At $x = 1/4$ the solution is $z_h = (H-h)/(H 2^h)$. Writing $m_h = x z_h'(x)|_{x=1/4}$ gives $m_0 = m_H = 0$ and $m_h = m_{h-1}/4 + m_{h+1} + z_{h-1}/4$. Its explicit solution is

$$m_h = \frac{h(H-h)(2H+3-h)}{6H2^h}.$$

Substitution verifies both recurrences, and uniqueness follows from the strictly subunit spectral radius of the finite strip. Summing z_h and m_h gives the displayed formulas for Z_0 and M_0/Z_0. In particular $S_0 = \log_2 Z_0 + 2M_0/Z_0 \ge 2H/3 - 2/3$. For $n \ge 4$ this follows from $Z_0 \ge 8/5$, $\log_2(8/5) > 2/3$, and $M_0/Z_0 \ge (n-1)/3$. For $H = 2, 3, 4$ the entropies respectively are

$$\log_2 5 - \tfrac{8}{5}, \qquad \log_2(17/12) + \tfrac{46}{51}, \qquad \log_2(49/32) + \tfrac{72}{49}.$$

The first exceeds $2/3$ because $5^{15} > 2^{34}$. For the next two use $\log_2 x \ge 20(x-1)/(7(x+1))$, which follows from $\ln x \ge 2(x-1)/(x+1)$ and $\ln 2 < 7/10$.

For the top source the unnormalized mass at height h is $z_h = (h/H) 2^{H-h}$ for $1 \le h \le H$, and zero at 0. This is the unique solution of $z_h = z_{h-1}/4 + z_{h+1}$, $z_0 = 0$, $z_H = 1$. Therefore

$$Z_H = \frac{2^{H+1} - H - 2}{H} \ge \frac{2^H}{H}.$$

Its path entropy is at least $\log_2 Z_H \geq H - \log_2 H \geq 2H/3 - 2/3$. The last inequality follows by checking $H = 2, 3, 4, 5$ and differentiating $H/3 + 2/3 - \log_2 H$ for $H \geq 5$; the $H = 5$ check is $5^3 < 2^7$.

It remains to use a finite register without assuming finite entropy of the source vectors. A length-k path from source s ending at height h has weight $4^{-n_A(w)} = 2^{s-h}2^{-k}$. Since $Z_s \geq 2^s/(H+1)$, the normalized total probability at length k is at most

$$(H+1)4^{-\lfloor k/\ell_{\mathrm{blk}} \rfloor}.$$

Set $J_0 = \lceil 8\log_2(2H) \rceil$ and $N = \ell_{\mathrm{blk}} J_0$. Because $\log_2(1/p_s(w)) \leq 2|w| + H + 1$, the entropy carried by paths longer than N is at most

$$(H+1)\ell_{\mathrm{blk}} 4^{-J_0} \left\{ 2\ell_{\mathrm{blk}} \left[\tfrac{4}{3}(J_0+1) + \tfrac{4}{9} \right] + \tfrac{4}{3}(H+1) \right\} < 1.$$

For example, use $H + 1 \leq 3H/2$, $J_0 + 2 \leq 10H$, and $4^{-J_0} \leq (2H)^{-16}$ to bound this by $162H^6/(2H)^{16} < 1$. If q_s is the short-path probability, the entropy chain rule therefore gives

$$q_s H(p_s \mid |w| \leq N) \geq S_s - 2.$$

On the left, map each short path space isometrically to its binary word in F, together with its source label, a short flag, and its source vector in J. Map each long path to one extra symbol $\perp$ in F and retain its word and source information in J. Map the entire orthogonal complement isometrically into $|\perp\rangle_F \otimes J_{\mathrm{extra}}$, where J_{extra} is a separate copy of that complement inside J. This is a genuine local isometry on the full physical space, including any actual-state mass outside the path spaces. There are fewer than 2^{N+2} register symbols. Right path orthogonality makes the reduced left state block diagonal in the path and source labels. Its finite conditional entropy is exactly

$$H(F \mid J)_{\widehat{y}} = \sum_{s \in \{0,H\}} \frac{Z_s \|\xi_s\|^2}{\|y\|^2} q_s H(p_s \mid |w| \leq N),$$

which proves the asserted bound. □

Lemma C.4 (Entropy comparison with the actual state)**.** *Under the preceding assumptions let Ω be normalized in the retained spaces or in larger physical local spaces containing them as orthogonal summands, with $\|\Omega - y\| \leq \kappa \leq 1/2$. Then*

$$S(\rho_L) \geq \frac{2}{3}H - \frac{14}{3} - 4\kappa(N+2).$$

Proof. Normalization gives $\|\Omega - \widehat{y}\| \leq 2\kappa$, hence the trace distance of the reduced states after the same local isometry is at most $t = 2\kappa$. The Alicki–Fannes continuity method [1] gives the following inequality in the finite-register form of [36, Lemma 2], with the infinite-dimensional conditioning-system extension of [21]; we include its proof. For a finite register F of dimension d and an arbitrary conditioning space J,

$$|H(F \mid J)_\rho - H(F \mid J)_\sigma| \leq 2t \log_2 d + (1+t) h_2\left(\frac{t}{1+t}\right).$$

For completeness, write the Jordan decomposition of $\rho - \sigma$ as $t(\tau_+ - \tau_-)$ and use the two decompositions

$$\omega = \frac{\rho + t\tau_-}{1+t} = \frac{\sigma + t\tau_+}{1+t}.$$

Conditional entropy of a binary mixture exceeds the averaged conditional entropy by a quantity between zero and the binary mixing entropy. Subtracting the two decompositions and using

$|H(F \mid J)| \le \log_2 d$ proves the inequality. The argument, expressed using conditional mutual information with a classical binary flag, is valid for infinite-dimensional J.

If $S(\rho_L) = \infty$ there is nothing to prove. Otherwise the isometry preserves $S(\rho_L) = S(FJ)$ and Araki–Lieb gives $S(J) \le S(FJ) + \log_2 d < \infty$. Thus $S(\rho_L) \ge H(F \mid J)$. Use $\log_2 d \le N+2$, $t \le 2\kappa \le 1$, and the bound $(1+t)h_2(t/(1+t)) \le 2$ to conclude. □

C.5 Assembly and explicit logarithmic remainder

Proof of Theorem C.1. It suffices to prove the left marginal bound. The assumptions are symmetric under exchange of the parties, with the two displayed A mirrors exchanged up to sign.

Put $L = \log_2(1/\delta)$ and

$$H = \left\lfloor \frac{L}{3} - 8\log_2(L+2) - 128 \right\rfloor. \tag{83}$$

First suppose $H \ge 64$. Use the spectral-banding section with

$$N_{\mathrm{band}} = H + 1, \qquad u = 40\, 2^{(H+1)/2}\delta^{1/2}.$$

The letter u denotes the quantity called η in that section; it is distinct from the eventual renewal comparison error. The choice (83) implies $u \le 2^{-60}$, since

$$u \le 40\sqrt{2}\, 2^{-L/3}(L+2)^{-4}2^{-64} < 2^{-58}(L+2)^{-4}$$

and $H \ge 64$ implies $L \ge 192$. In particular all the banding smallness conditions hold.

Write $\Phi^{\mathrm{all}} = M\Omega = \sum_{h=0}^{H+1} \Phi_h$ for the matched, unnormalized vector retaining the terminal band. Define only now

$$\Psi = \sum_{h=0}^{H} \Phi_h, \qquad q = \|\Psi\|, \qquad \phi = \Psi/q = \sum_{h=0}^{H} \phi_h.$$

The projection defining Ψ is orthogonal. Equations (B5) and (B18) give

$$1 - q^2 \le (H+1)2^{-H} + 2561u =: v. \tag{84}$$

Our parameter choices give $v < 1/4$, hence $q > 1/2$, and

$$\|\Omega - \phi\| \le 2\sqrt{v}. \tag{85}$$

This is a comparison of vectors, not a postselection operation. The original terminal local spaces are retained as orthogonal complements throughout: the actual vector still contains its terminal and unmatched components. Each later local isometry is extended to these complements, and every padding inclusion embeds the entire actual vector.

Repair precisely the incoming rows with targets $1, \dots, H-1$. Their edges are

$$A_h : h \longrightarrow h+1 \ (0 \le h \le H-2), \qquad B_h : h \longrightarrow h-1 \ (2 \le h \le H).$$

All source and target bands are genuine bands; the terminal band $H+1$ is absent from these rows. The estimates (B11) and the projected mirrors concern the individual vectors Φ_h and are unchanged by removing the terminal component. Their aggregate before normalization is less than $57u$. After dividing by $q > 1/2$, the row-repair input therefore satisfies

$$d \le 128u \le 1. \tag{86}$$

The square root of the terminal mass has not entered d.

For $D = 2$, $M_{\mathrm{op}} = 1$ and $\Lambda = 2$, Lemma C.2 gives

$$K_* = 2(12 + 13\sqrt{2}), \qquad L_* = 2K_* + 18 = 66 + 52\sqrt{2} < 140.$$

Let a_h, b_h denote the repaired columns, and write

$$a = \sum_{h=0}^{H-2} a_h, \qquad b = \sum_{h=2}^{H} b_h, \qquad T = \tfrac{1}{2} a_L \otimes a_R + b_L \otimes b_R.$$

Every incoming interior row is unitary, and the aggregate repaired edge-mirror norm is at most

$$\sqrt{K_*}\sqrt{d} + 3d \le \sqrt{L_* d}.$$

The two sources are the *whole* matched components ϕ_0 and ϕ_H.

We spell out the renewal residual. On each interior level h the exact left row identity gives

$$\begin{aligned}\phi_h - \tfrac{1}{2}(a_{h-1,L} \otimes a_{h-1,R})\phi_{h-1} - (b_{h+1,L} \otimes b_{h+1,R})\phi_{h+1} &= a_{h-1,L}(a^*_{h-1,L}\phi_h - \tfrac{1}{2} a_{h-1,R}\phi_{h-1}) \\ &\quad + b_{h+1,L}(b^*_{h+1,L}\phi_h - b_{h+1,R}\phi_{h+1}).\end{aligned}$$

The two summands have orthogonal left ranges, and different target levels are orthogonal. The bracketed terms are the already repaired mirrors in the opposite orientation, with an extra factor $1/2$ on A. Consequently

$$\|\phi - T\phi - \phi_0 - \phi_H\| \le \sqrt{L_* d}.$$

Set $y = (I - T)^{-1}(\phi_0 + \phi_H)$. Lemma C.3 yields

$$\begin{aligned}\|\phi - y\| &\le 4H^2 2^{H/2}\sqrt{L_* d} \\ &\le 4096 H^2 2^{3H/4}\delta^{1/4}.\end{aligned} \tag{87}$$

For the numerical inequality use $d \le 128u$, $L_* < 140$, and $u = 40\sqrt{2}\, 2^{H/2}\delta^{1/2}$; the resulting prefactor $4\sqrt{128 \cdot 140 \cdot 40\sqrt{2}}$ is less than 4096. Combining (85)–(87), and absorbing the $\sqrt{u}$ term, gives

$$\|\Omega - y\| \le 8192 H^2 2^{3H/4}\delta^{1/4} + 2\sqrt{H+1}\, 2^{-H/2} =: \zeta. \tag{88}$$

For the absorption, $2\sqrt{2561u}$ has prefactor less than 762 in front of $2^{H/4}\delta^{1/4}$, which is dominated by the unused $4096 H^2 2^{3H/4}\delta^{1/4}$.

Let

$$N_{\mathrm{hist}} = 2H^2\lceil 8\log_2(2H)\rceil.$$

For integers $H \ge 2$, $\log_2(2H) \le H$, so $N_{\mathrm{hist}} + 2 \le 18H^3$. The first term of (88), after multiplication by $N_{\mathrm{hist}} + 2$, is at most

$$18\, 2^{-83}\frac{H^5}{(L+2)^6} < \frac{1}{32},$$

using (83) and $H \le L + 2$. The second is bounded by

$$36H^3\sqrt{H+1}\, 2^{-H/2} < \frac{1}{32} \qquad (H \ge 64).$$

For the latter, the expression decreases for $H \ge 64$: the logarithmic derivative is $3/H + 1/[2(H+1)] - (\ln 2)/2 < 0$. At $H = 64$ its value is $36\sqrt{65}/2^{14} < 1/32$. We have proved

$$\zeta(N_{\mathrm{hist}} + 2) < 1/16, \qquad \zeta < 1/2.$$

In particular $y \neq 0$. Lemma C.4 gives

$$S(\rho_L) \geq \frac{2}{3}H - \frac{14}{3} - \frac{1}{4} = \frac{2}{3}H - \frac{59}{12}.$$

Since the floor in (83) loses at most one,

$$S(\rho_L) \geq \frac{2}{9}L - \frac{16}{3}\log_2(L+2) - 91.$$

This is stronger than (74) in the range just considered.

If the integer in (83) is less than 64, then $L/3 - 8\log_2(L+2) - 128 < 64$. The right side of (74) equals

$$\frac{2}{3}\left(\frac{L}{3} - 8\log_2(L+2) - 128\right) - \frac{344}{3} < 0.$$

It is therefore bounded by $S(\rho_L) \geq 0$. Finally, if the defects vanish, the hypotheses hold for every positive δ, and the right side of (74) tends to infinity as $\delta \downarrow 0$. □

C.6 A full exact extension, with a simpler top boundary

We now construct an exact extension. After the top masses have been bounded, arbitrary unitary completion has the required squared-error order. The top-source convention of Section C.4 remains in force.

Lemma C.5 (Exact extension from repaired finite rows)**.** *Let $H \geq 2$. On each party let $\mathcal{H}^s = \bigoplus_{h=0}^{H} \mathcal{H}_h^s$, and let $\Phi = \sum_{h=0}^{H} \Phi_h$ be a unit matched vector, with $\mu_h = \|\Phi_h\|^2$. Suppose the finite graph with edges*

$$A_h : h \longrightarrow h+1 \quad (0 \leq h \leq H-2), \qquad B_h : h \longrightarrow h-1 \quad (2 \leq h \leq H)$$

has contractions satisfying the hypotheses of Lemma C.2 with aggregate defect $d \leq 1$ and mirror weights 2 for A_h and 1 for B_h. Thus precisely the rows with targets $1, \ldots, H-1$ are repaired. Put

$$K_* = 2(12 + 13\sqrt{2}), \qquad L_* = 2K_* + 18.$$

After local zero-state padding there are contractions $\widehat{A}^s, \widehat{B}^s$, a positive contraction $\widehat{C}^s$, and projections $\widehat{P}^s, \widehat{Q}^s$ obeying every local Catalan relation exactly as operator identities. Their five squared mirror defects on Φ have sum at most

$$\Delta_\Phi \leq L_* d + \mu_{H-1} + 7\mu_H. \tag{EX}$$

If Ω is any unit vector in the same original local spaces, possibly including additional local orthogonal summands, with $\|\Omega - \Phi\| \leq \eta$, the same operators, on the unchanged embedded Ω, satisfy

$$\Delta_\Omega \leq 2L_* d + 2\mu_{H-1} + 14\mu_H + 27\eta^2. \tag{EX'}$$

In particular no postselection or replacement of the physical state is made in this extension.

Proof. Apply Lemma C.2 with $D = 2, M_{\mathrm{op}} = 1, \Lambda = 2$. The displayed K_* is its constant. Each interior incoming row

$$[\widehat{A}_{h-1}^s, \widehat{B}_{h+1}^s] : \widetilde{\mathcal{H}}_{h-1}^s \oplus \widetilde{\mathcal{H}}_{h+1}^s \longrightarrow \widetilde{\mathcal{H}}_h^s, \qquad 1 \leq h \leq H-1,$$

is unitary. Make all padded vertex spaces have the same infinite Hilbert dimension κ. Add vertex spaces $\widetilde{\mathcal{H}}_h^s \cong \ell^2(\kappa)$ for every $h > H$. If the actual vector has additional original local orthogonal summands, embed those isometrically into level $H+1$, choosing κ large enough

also for them. All remaining new summands have zero actual-state component. Every level above H has zero comparison-state component in Φ. For every $h \geq H$, choose an arbitrary unitary row with the same displayed domain and codomain. Each column is assigned exactly once, by its target row, so these choices are compatible. Define the global operators by their columns and put

$$\widehat{P}^s = E_0^s, \quad \widehat{Q}^s = E_1^s, \quad \widehat{C}^s = \sum_{h \geq 0} 2^{-h} E_h^s,$$

where E_h^s projects onto the enlarged level-h space. Set $\widehat{B}^s = 0$ on levels $0, 1$. Orthogonality of the vertex spaces and unitarity of every incoming row give

$$(\widehat{A}^s)^* \widehat{A}^s = I, \quad \widehat{A}^s (\widehat{A}^s)^* + \widehat{B}^s (\widehat{B}^s)^* = I - E_0^s, \quad (\widehat{B}^s)^* \widehat{B}^s = I - E_0^s - E_1^s.$$

The counter relations and $\widehat{P}\widehat{Q} = 0$ follow directly from the source and target levels. All operator identities hold on the entire enlarged space, not merely on the state.

The sum of squared repaired edge-mirror errors in the finite graph is at most

$$(\sqrt{K_*}\sqrt{d} + 3d)^2 \leq (2K_* + 18)d = L_* d.$$

For an A edge the corresponding original mirror has an additional factor $1/2$, so this bound also bounds its contribution to the five original mirrors. Different edge indices belong to orthogonal pairs of local levels, hence their squared contributions add without cross terms.

Only three new edges can contribute on Φ: A_{H-1}, A_H, and B_{H+1}. The two oriented original mirrors at A_{H-1} contribute at most $\mu_{H-1} + 4\mu_H$, by

$$\|\tfrac{1}{2}\widehat{A}^s_{H-1}\Phi_{H-1} - (\widehat{A}^{\bar{s}}_{H-1})^*\Phi_H\|^2 \leq \tfrac{1}{2}\mu_{H-1} + 2\mu_H.$$

The two mirrors at A_H contribute $\mu_H/2$, and those at B_{H+1} contribute at most $2\mu_H$. Their sum is at most $\mu_{H-1} + 7\mu_H$. The C mirror vanishes exactly on the matched vector. This proves (EX).

Let $M_1, \ldots, M_5$ be the five global mirror operators. Their norms are at most $3/2, 3/2, 2, 2, 1$, respectively; the final bound uses that the two counter operators commute and have spectrum in $[0, 1]$. Consequently $\sum_j \|M_j\|^2 \leq 27/2$. Minkowski followed by $(x + y)^2 \leq 2x^2 + 2y^2$ gives

$$\sum_j \|M_j\Omega\|^2 \leq 2\sum_j \|M_j\Phi\|^2 + 27\|\Omega - \Phi\|^2,$$

proving (EX'). The only maps applied to Ω are inclusions into local zero-state summands. They preserve its marginal spectra and act as identity on the purifying environment. □

Remark (Bounding the next-to-top mass). If a discarded top contraction $X^s : \mathcal{H}^s_{H-1} \to \mathcal{H}^s_H$ satisfies

$$\|((X^L)^* X^L - I)\Phi_{H-1}\| \leq e, \qquad \|X^L\Phi_{H-1} - 2(X^R)^*\Phi_H\| \leq e, \qquad 0 \leq e \leq 1,$$

then

$$\mu_{H-1} \leq 4\mu_H + 6e.$$

Indeed $\mu_{H-1} \leq \|X^L\Phi_{H-1}\|^2 + e \leq (2\sqrt{\mu_H} + e)^2 + e \leq 4\mu_H + 6e$. Thus (EX') becomes

$$\Delta_\Omega \leq 2L_* d + 22\mu_H + 12e + 27\eta^2.$$

The completion alters the top-boundary edges after their state mass has been bounded. It preserves the desired order $\Delta = O(d + \mu_H + e + \eta^2)$ and therefore the same leading exponent.

Corollary C.6 (Quantitative exact extension)**.** *With the spectral bands and notation of the assembly, for $H \geq 64$ and $u = 40\,2^{(H+1)/2}\delta^{1/2} \leq 2^{-60}$, the original state admits a local zero-state embedding and exact Catalan operators whose five squared mirror errors have sum*

$$\Delta \leq 2^{21}(u + H2^{-H}).$$

The embedded state's marginal spectra are unchanged.

Proof. For the normalized comparison vector ϕ, put $\mu_h = \|\phi_h\|^2$ and use $d \leq 128u$ and $\|\Omega - \phi\|^2 \leq 4v$ in (EX'). The cumulative-tail bound and $q^{-2} \leq 4$ give

$$\mu_{H-1} + \mu_H \leq 4\big((H-1)2^{-(H-2)} + 2560u\big) \leq 16H2^{-H} + 10240u.$$

Thus (EX') is at most

$$2(140)(128)u + 14(16H2^{-H} + 10240u) + 108((H+1)2^{-H} + 2561u).$$

For $H \geq 64$, this is bounded by $2^{21}(u + H2^{-H})$. All inclusions are local and the state is unchanged. □

C.7 Compatibility with the decoder and the game corollary

Theorem B.1 asserts exactly the domain restrictions and residual list (72)–(73), with $\delta = K\varepsilon^{1/4}$ on

$$\widehat{\Omega} = (V_L \otimes V_R \otimes I_{\mathcal{E}})\Omega.$$

The local maps are isometries and hence preserve all nonzero marginal eigenvalues and hence entropy; dilations and padding may add zero eigenvalues.

The nine sectors. The final decoded operators have the form $I_9 \otimes A$, and similarly for B, C, P, Q. Tensoring with I_9 preserves every norm restriction and polynomial relation. The decoder's comparison vector has coefficients

$$(1, \sqrt{2}, \sqrt{2}, \sqrt{2}, \sqrt{2}, 2, 2, 2, 2), \qquad \sum_j c_j^2 = 25.$$

Its extension lemma bounds each residual on the *actual* transformed state by five times its root-vector error plus the residual operator norm times the comparison-vector error. Thus the amplification is part of the decoder constant K; no corner is postselected here, and no factor $1/25$ or $1/75$ multiplies the entropy coefficient. The scalar M_3 frame remains in the root multiplicity space, which the rounding theorem treats without a dimension bound.

Mixed states. The theorem uses the actual purification throughout. The environment is not absorbed into a physical party or acted upon. The normalized matched vector and the renewal vector are proof comparisons only. The final entropy estimate concerns the original marginal ρ_L, because the finite-register isometry preserves its nonzero spectrum. It follows equally for ρ_R by symmetry. This proves compatibility with the decoder's mixed-state convention without an entropy-continuity assumption on the comparison junk state.

Corollary C.7 (Game and Bell lower bounds; proved from the decoding input)**.** *Use Theorem B.1, with constants $K_{\mathcal{B}}$ and K_G, for the fixed Bell functional and game. Put $K'_X = \max\{1, K_X\}$ for $X \in \{\mathcal{B}, G\}$. If $0 < \varepsilon \leq 1$ and $L_\varepsilon = \log_2(1/\varepsilon)$, then*

$$E_X(\varepsilon) \geq \frac{1}{18}L_\varepsilon - \frac{16}{3}\log_2(L_\varepsilon + 2) - 200 - \frac{2}{9}\log_2 K'_X.$$

Here E_X is the infimum of the left marginal entropy over normal spatial strategies of loss at most ε. The same statement holds for the right marginal entropy. With the height-and-size cutoff construction, this gives

$$E_{\mathcal{B}}(\varepsilon),\ E_G(\varepsilon) = \Theta(\log(1/\varepsilon)).$$

This is an order-of-growth law; it does not identify a sharp leading constant.

Proof. For $K'_X\varepsilon^{1/4} \le 1$, substitute $L = \frac{1}{4}L_\varepsilon - \log_2 K'_X$ in Theorem C.1 and use $L+2 \le L_\varepsilon+2$. Isometries preserve the original marginal entropy. If $K'_X\varepsilon^{1/4} > 1$, the displayed lower bound is negative, so nonnegativity of entropy suffices. Taking infima proves the lower bound. The height-and-size construction of Section 9.2 gives

$$E_X(\varepsilon) \le \frac{2}{3}\log_2(1/\varepsilon) - \frac{2}{3}\log_2\log_2(1/\varepsilon) + O_X(1),$$

which proves the last assertion. The decoder is proved in Appendix B; the cutoff crossing identities are independently checked in the finite verification bundle. □

C.8 Dependency order and boundary conventions

Stage	Estimates already available before any use
P, Q mirrors	Original A, A^*, B, B^* mirrors and local source/range equations; all products expanded with bounded left prefixes.
Generator-image spectra	Original C covariance; adjoint images first use the original opposite-party adjoint mirror.
Common partition	Only commuting pairs $(C_s, kC_{\bar{s}})$, on the specified generator-image vectors.
Localization and rows	The fixed partition, matched vector including its terminal band, P, Q mirrors, and aggregate local defects.
Unused bottom block	The edge $B_1 : 1 \to 0$ is absent from both renewal and row repair; its deletion costs nothing.
Unitary row repair	Both oriented edge mirrors are proved before the range residual is moved past an edge in the cross-Gram estimate.
Tail truncation	Per-component interior-row data have already been bounded without terminal mass. Truncation is solely a comparison.
Renewal	Exact repaired incoming rows and their repaired mirrors. Neither boundary has an incoming retained edge.
Entropy comparison	Orthogonal source-generated path spaces, exponential path-length tail, finite register, and conditional-entropy continuity.

Entropy comparison in unbounded dimension. A small vector distance between Ω and a matched or renewed vector does not control their full marginal entropy in unbounded dimension. Replacing the last section by such an assertion would invalidate the theorem. The proof instead builds a register of explicitly bounded dimension, retains the entire remaining system as the conditioner, and proves the conditional-entropy continuity inequality with no bound on that system's dimension. The inequality $S(FJ) \ge H(F \mid J)$ then concerns the actual state.

The top-source convention is essential. If the incoming edge A_{H-1} were retained while the whole height-H component remained a source, variable-length path spaces could overlap, and the orthogonality proof and stated Z_H would not apply. We remove that edge from T and use exactly

$$T = \tfrac{1}{2}a_L \otimes a_R + b_L \otimes b_R, \quad a = \sum_{h=0}^{H-2} a_h, \quad b = \sum_{h=2}^{H} b_h,$$

with source $\phi_0 + \phi_H$ throughout. The optional exact extension has a different purpose and never changes this renewal operator.

The order of row localization is also essential. If one first discards the terminal band and then transfers every local residual by the global norm distance to the truncated vector, the square root of the tail enters the row input and destroys the claimed rate. Equations (B8)–(B12) are proved with the terminal band present, and only individual retained components are subsequently normalized. This order is essential.

C.9 What is proved about possible improvements

Proved exponent accounting. If the vector residual before the renewal inverse has the form $CH^m 2^{bH}\varepsilon^a$, then the inverse contributes $2^{H/2}$ and polynomial factors. Taking

$$H = \frac{a}{b + 1/2}\log_2(1/\varepsilon) - O(\log\log(1/\varepsilon))$$

gives the leading entropy coefficient

$$\frac{4a}{3(1+2b)}.$$

Here spectral banding gives $u \asymp 2^{H/2}\delta^{1/2}$, row repair gives vector error $O(\sqrt{u})$, and thus $a = \gamma/4$, $b = 1/4$ when $\delta = K\varepsilon^\gamma$. The coefficient is $2\gamma/9$, yielding $1/18$ at $\gamma = 1/4$ and, conditionally, $1/9$ at $\gamma = 1/2$.

Heuristic improvement direction; no improved leading coefficient claimed. The square root in the direct row-repair estimate is the explicit rate loss after banding. A stronger estimate for these specific Catalan rows could improve the coefficient. For example, if their direct repair error were $O(d)$ with the same hypotheses and uniform constants, the same bookkeeping would give coefficient $\gamma/3$ at game level, or $1/3$ in $\log_2(1/\delta)$. This is only a sufficient hypothetical strengthening; the present proof does not establish it, either for Catalan rows or for arbitrary graphs. Logarithmic remainders and additive constants can be reduced by tuning the cutoff without changing the leading coefficient.

Decoder ceiling. Proposition B.6 proves $\gamma \le 1/2$ for actual-state mixed-strategy decoding. Attainment of this ceiling is open. A squared-mirror defect Δ must be distinguished from a vector defect $\delta = \sqrt{\Delta}$ when comparing leading constants.

C.10 The single finite-strip lemma supporting the rounding

Lemma C.8 (State-weighted Catalan strip)**.** *Fix $H \ge 2$. Let a unit vector $\phi = \sum_{h=0}^{H}\phi_h$ be matched in vertex spaces on the two parties, with arbitrary environment. Let contractions $x_{A,h} : h \to h+1$ for $0 \le h \le H-2$ and $x_{B,h} : h \to h-1$ for $2 \le h \le H$ be given on both parties. For every edge e, put $\lambda_e = 2$ for A and $\lambda_e = 1$ for B, and set*

$$g_e^s = (x_e^s)^* x_e^s, \quad r_v^s = \sum_{t(e)=v} x_e^s (x_e^s)^* - I_v, \quad m_e^s = x_e^s \phi_{s(e)} - \lambda_e (x_e^{\bar{s}})^* \phi_{t(e)}.$$

Suppose, summing over $s = L, R$, all retained edges, and interior targets $1 \le v \le H-1$, that

$$\sum_{s,e} \|(g_e^s - I)\phi_{s(e)}\|^2 + \sum_{s,v} \|r_v^s \phi_v\|^2 + \sum_{s,e} \|m_e^s\|^2 \le d^2 \le 1.$$

Let Ω be any actual unit vector in those spaces or in larger local spaces containing them as orthogonal summands, and put

$$K_* = 2(12 + 13\sqrt{2}), \quad \tau = \|\Omega - \phi\| + 4H^2 2^{H/2}(\sqrt{K_*}\sqrt{d} + 3d),$$

$$N = 2H^2\lceil 8\log_2(2H)\rceil.$$

If $\tau \le 1/2$, then

$$\boxed{S(\rho_L^{\Omega}) \ge \frac{2}{3}H - \frac{14}{3} - 4\tau(N+2).}$$

The statement is uniform in dimensions and multiplicities. Its proof uses local zero-state extensions and local isometries only; no operation acts on the environment or postselects the actual state.

Proof. Lemma C.2 gives exact incoming rows and aggregate mirror error at most $\sqrt{K_*}\sqrt{d}+3d$. The orthogonal-row expansion displayed in Section C.5 gives the renewal residual with this same bound. Lemma C.3 constructs the two-source renewal vector at distance at most τ from Ω. Lemma C.4 proves the displayed entropy inequality. Each of these lemmas was proved independently before the assembly, so this reduction is not circular. □

D Exact local algebras and extraction limits

This appendix separates two refinements from the approximate-relation theorem. The first assumes exact local relations and has a sharp leading coefficient. The second limits a full-state self-testing modulus. Neither refinement is needed for Theorem D.

D.1 The sharp exact-algebra law

Theorem D.1 (Exact local relations; three mirrors)**.** *Suppose both parties obey (4)–(6) as operator identities. On a normal purification Ω, set*

$$\Delta_3 = \|(A_L^* - A_R/2)\Omega\|^2 + \|(B_L^* - B_R)\Omega\|^2 + \|(C_L - C_R)\Omega\|^2.$$

There is an absolute constant C_0 such that, for sufficiently small $\Delta_3 > 0$ and either party s,

$$S(\rho_s) \ge \frac{2}{3}\log_2(1/\Delta_3) - 8\log_2\log_2(1/\Delta_3) - C_0.$$

In particular the same bound holds with Δ_3 replaced by the sum Δ of all five squared mirror defects. For that five-mirror convention the leading coefficient $2/3$ is sharp, uniformly over representations and normal states.

Proof. Put $\delta = \sqrt{\Delta_3}$ and $\ell = \log_2(1/\delta)$. The spectral argument in the first paragraph of the proof of Lemma 3.1 uses only the local identities: P, Q are orthogonal reducing spectral subprojections, B is a partial isometry, and AA^*, BB^*, P are an orthogonal partition. An interval in $(1/2, 1)$ would be transported isometrically by B above one. The same reasoning on $1_{\{1\}}(C) - P$ gives $E_0 = P$; on $1_{\{1/2\}}(C) - Q$ it gives $E_1 = Q$, since the image would lie in P and in the orthogonal range of B. Repeated upward transport excludes every interval $(2^{-h-1}, 2^{-h})$. Thus

$$I = E_\infty + \sum_{h\ge 0} E_h, \qquad E_h = 1_{\{2^{-h}\}}(C), \quad E_\infty = 1_{\{0\}}(C).$$

This step makes no assertion that the zero-counter mass vanishes.

For $k \ge 0$ put $F_k = 1_{[0,2^{-k}]}(C)$ and $t_k = \|F_k^L\Omega\|^2$, so $t_0 = 1$. For $k \ge 1$, the range partition and covariance give the exact identity

$$t_k = \|F_{k-1}^L A_L^*\Omega\|^2 + \|F_{k+1}^L B_L^*\Omega\|^2.$$

Replace the first adjoint by $A_R/2$ and the second by B_R. Their squared-norm errors are at most $3\delta/2$ and 2δ, respectively. Opposite-party commutation and the exact source relations give the first squared norm $t_{k-1}/4$ and the second

$$t_{k+1} - \|F^L_{k+1}(P_R + Q_R)\Omega\|^2.$$

On this last joint spectral support, $C_L \le 1/4$ and $C_R \ge 1/2$. The commuting spectral-gap estimate bounds its mass by $16\delta^2$. Hence, for $\delta \le 1$,

$$|t_k - t_{k-1}/4 - t_{k+1}| \le 20\delta.$$

The homogeneous solution from t_0, t_1 is $[t_0 + (2t_1 - t_0)k]2^{-k}$, at most $(k+1)2^{-k}$. The impulse kernel has total mass $\sum_{j\ge 0}(j+1)2^{-j} = 4$. Therefore

$$t_k \le (k+1)2^{-k} + 80\delta. \tag{89}$$

Only the three displayed mirrors have been used.

Let D_∞ project onto equal finite heights together with equal zero-counter sectors. If unequal counters have at least one height at most m, their distance is at least 2^{-m-1}. All remaining unequal pairs have left counter at most 2^{-m-1}. Thus

$$\|(I - D_\infty)\Omega\|^2 \le 4^{m+1}\delta^2 + (m+2)2^{-m-1} + 80\delta.$$

With $m = \lfloor 2\ell/3 \rfloor$, this is at most $c_1(\ell+1)\delta^{2/3}$ for an absolute c_1. For $D_H = \sum_{h=0}^{H} E^L_h E^R_h$ and $\Omega_H = D_H\Omega$, (89) consequently gives

$$\|\Omega - \Omega_H\| \le c_2\sqrt{\ell+1}\,\delta^{1/3} + c_2\sqrt{H+2}\,2^{-H/2}. \tag{90}$$

The estimate includes zero-counter mass and is independent of a smallest Schmidt value.

Retain $a = A\sum_{h=0}^{H-2} E_h$ and $b = B\sum_{h=2}^{H} E_h$. Their columns are isometries with orthogonal incoming ranges, and $aa^* + bb^* = \sum_{h=1}^{H-1} E_h$. On matched heights let

$$T_H = \tfrac{1}{2}a_L \otimes a_R + b_L \otimes b_R, \qquad \xi_j = E^L_j E^R_j \Omega \quad (j = 0, H).$$

There is no incoming retained edge at either boundary. Define

$$\Pi_+ = \sum_{h=1}^{H-1} E^L_{h-1}E^R_h, \qquad \Pi_- = \sum_{h=1}^{H-1} E^L_{h+1}E^R_h.$$

Covariance proves, as exact vector identities,

$$(a^*_L - a_R/2)\Omega_H = \Pi_+(A^*_L - A_R/2)\Omega, \quad (b^*_L - b_R)\Omega_H = \Pi_-(B^*_L - B_R)\Omega.$$

Indeed selecting the displayed output heights forces matched input heights in each term. Multiply these identities by a_L, b_L and use the interior row partition. Their ranges are orthogonal, so

$$(I - T_H)\Omega_H = \xi_0 + \xi_H + r_H, \qquad \|r_H\| \le \delta.$$

In particular the height-matching error has not passed through the inverse. Lemma C.3 gives

$$y_H = (I - T_H)^{-1}(\xi_0 + \xi_H), \qquad \|y_H - \Omega_H\| \le 4H^2 2^{H/2}\delta.$$

Combining with (90), and writing the resulting bound as ζ_H, yields

$$\zeta_H \le c_3\left(\sqrt{\ell+1}\,\delta^{1/3} + \sqrt{H+2}\,2^{-H/2} + H^2 2^{H/2}\delta\right).$$

Choose $H = \lfloor 2\ell - 12\log_2(\ell+2) - c_4 \rfloor$. For sufficiently large ℓ, $H \geq 2$ and $\zeta_H(N_{\text{hist}}+2)$ is bounded by an absolute constant: use $N_{\text{hist}} + 2 \leq 18H^3$ and $2^{H/2}\delta \leq 2^{-c_4/2}(\ell+2)^{-6}$. The first two terms decay exponentially in ℓ after multiplication by H^3. Also $\zeta_H < 1/2$ and $y_H \neq 0$ for sufficiently large ℓ.

The local finite-history isometry and Lemma C.4, applied on either side, now give

$$S(\rho_s) \geq \tfrac{2}{3}H - O(1) = \tfrac{4}{3}\ell - 8\log_2(\ell+2) - O(1).$$

The path construction is symmetric, so this gives both marginal bounds even though the renewal residual used only the left adjoint mirrors. Since $\log_2(1/\Delta_3) = 2\ell$, the stated remainder follows. If all three defects vanish, apply the same argument with every positive δ to obtain infinite entropy. Replacing Δ_3 by the larger Δ changes the lower bound only in the sufficiently small range, where its right side is increasing in $\log_2(1/\Delta)$.

For sharpness use the canonical height-H state on the unchanged exact local representation. B and C mirrors are exact. Each of the two A mirrors has squared norm $q_H/4$, since only an omitted A child contributes, with coefficient $1/2$. Thus $\Delta = q_H/2$. Equations (36) and (37), with the fixed scaffold term removed, give entropy $2H/3 + O(1)$ and $\log_2(1/\Delta) = H + \log_2 H + O(1)$. No larger uniform leading coefficient can hold. □

Corollary D.2 (The boundary-law class)**.** *For the prefix-closed, B-closed canonical restrictions of Section 9.2, the infimum entropy at boundary ratio at most q has leading term $(2/3)\log_2(1/q)$. The same leading coefficient holds for that class of strategies for B and G, with their fixed normalizations.*

Proof. On the unamplified word state, prefix closure and B closure make both B mirrors exact; C is diagonal and matched. Each A mirror has squared norm $q_S/4$, by the same omitted-child calculation. Hence $\Delta_S = q_S/2$ for every such set, including infinite sets of finite entropy. Theorem D.1 gives the lower leading term uniformly over the class. Height cutoffs give the upper term, and finite height-and-size cutoffs approximate them with strict loss slack. The scaffold adds a fixed entropy; (35) multiplies loss by a fixed positive constant. Neither changes a leading logarithmic coefficient. □

D.2 A limit on full-state extraction

Proposition D.3 (No faster full-state self-testing modulus)**.** *For sufficiently small $\eta > 0$, a pure state ϕ within vector distance η of $\psi_{\text{Cat}} \otimes \xi$, after local isometries and with arbitrary auxiliary vector ξ, has marginal entropy at least $c\eta^{-2}$ for an absolute $c > 0$. Consequently a uniform vector extraction modulus for all near-perfect strategies of B or G must satisfy*

$$\eta(\varepsilon) \geq c'\big(\log_2(1/\varepsilon)\big)^{-1/2}$$

for sufficiently small ε, with a fixed $c' > 0$. In particular no positive Hölder rate is possible.

Proof. Let p be the ordered Schmidt probabilities of ψ_{Cat}, and $d_N = \sum_{n=0}^{N} \text{Cat}_n$. The elementary Catalan bound $\text{Cat}_n\, 4^{-n} \geq [2(n+1)^{3/2}]^{-1}$ gives

$$\tau_p(d_N) := \sum_{i > d_N} p_i \geq \frac{1}{2\sqrt{N+2}}, \qquad \log_2 d_N \geq N - 1.$$

For the first inequality sum the bound over $n > N$ and compare with the integral of $x^{-3/2}/4$ from $N+2$ to infinity. For the second use $\text{Cat}_N \geq 2^{N-1}$, which follows inductively from the two endpoint terms in the Catalan convolution (with $N = 1$ checked directly).

Tensoring by an auxiliary Schmidt law cannot decrease an ordered tail. To see this, write $p \otimes q = \sum_j q_j (p \otimes e_j)$. The sum of the largest d entries is convex and is the same for each

vector $p \otimes e_j$, so it is at most the sum of the largest d entries of p. The conclusion also holds for countably many q_j by monotone approximation.

For any coefficient Hilbert–Schmidt operator, its distance to operators of rank at most d is the square root of its ordered tail. This follows by projecting onto the d largest singular directions for the upper bound, and by the variational characterization of the sum of the largest d squared singular values for the lower bound. Distance to a fixed set is 1-Lipschitz. Therefore, for the Schmidt law p_ϕ,

$$\sqrt{\tau_{p_\phi}(d)} \geq \sqrt{\tau_p(d)} - \eta.$$

Take $N = \lfloor (256\eta^4)^{-1} \rfloor$ and $\eta \leq 1/8$. Then $N + 2 \leq (128\eta^4)^{-1}$, so $\tau_p(d_N) \geq 4\eta^2$ and $\tau_{p_\phi}(d_N) \geq \eta^2$. Since the probabilities after position d_N are at most $1/d_N$,

$$S(\phi_L) \geq \tau_{p_\phi}(d_N) \log_2 d_N \geq \frac{1}{512\eta^2}.$$

This remains true with an untouched purifying environment in the auxiliary system, since the cut is the original left marginal against everything else.

Apply the estimate to the pure finite height-and-size strategies in Theorem D, whose entropy is at most $C_X \log_2(1/\varepsilon)$ for small ε. Local isometries preserve this entropy. A uniform extraction guarantee $\eta(\varepsilon)$ would imply $1/(512\eta(\varepsilon)^2) \leq C_X \log_2(1/\varepsilon)$, proving the modulus bound. If the modulus does not tend to zero the same lower asymptotic restriction is immediate. □

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